



MOTOROLA INC.
COMPONENT PRODUCTS DEPT.

Two-Phase Microprocessor Clocks Designed to drive the Motorola MC6800 MPU

The *Functional Module* approach to data communications hardware design significantly decreases the time between the "idea" stage and the marketable product.

A fundamental building block in a modular microcomputer system is the 2-phase clock oscillator used to drive the microprocessor. Motorola is uniquely qualified to provide this building block because of expertise in the three relevant fields: oscillator design, quartz crystal technology, and thick film hybrid integrated circuit manufacturing.

This one-of-a-kind expertise has created several clocks designed to drive Motorola's MC6800 Microprocessor. This plug-in unit contains the crystal, the oscillator circuit, the NMOS and TTL drivers, and the waveshaping and interface circuitry; all the components necessary to provide the critical non-overlapping 2-phase waveforms used by the MC6800 MPU.

FEATURES

Clock Module—Each clock module requires only a single 5 volt power supply. The NMOS outputs can drive highly capacitive loads ranging from 80 pf to 160 pf and meet all MPU input waveshape and timing requirements.

Each TTL output signal leads the ϕ_2 NMOS so that additional system device delays can be accommodated. All TTL outputs are buffered so they can drive 5 TTL devices and maintain all output specifications.

Each module is crystal-controlled and is compensated for variations in temperature, voltage, and load. The standard frequency of each model is 1 MHz; however, other frequencies between 250 kHz and 2.5 MHz can be ordered.

Reliability—Decreased Component Count—Thick film hybrids offer a reliability advantage that comes primarily from reduced component count and therefore reduced interconnections. Further, the single hermetic seal on the hybrid package reduces the failure rate whereas in a discrete design a separate sealing process with an associated failure rate is needed for each component.

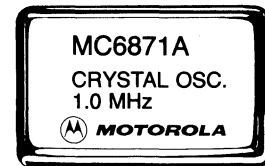
High Density Packaging—The hybrid MPU clock allows compact microcomputer design. It takes up only 1.34" x .840" space and has a seated height of .200".

Ruggedized Design—Maximum reliability at minimum cost is the result of combining three of Motorola's fields of experience: quartz crystal technology, clock oscillator design, and thick film hybrid integrated circuit manufacturing. Mass automated production techniques assure volume production. Gold plating of all crystals and Class 100 clean room processing testify that no short cuts are taken that might diminish reliability. Environmental testing proves the effectiveness of the rugged design for those applications in which shock and vibration are likely hazards.

Complete Process Control—Motorola is the only totally integrated manufacturer of quartz frequency control devices; full control of all processes from growing, sawing, lapping, and finishing quartz to combining it with other components into an electronic product—the MC6870A, MC6871A, and MC6871B MPU clocks.

Volume Production—Production facilities are oriented to mass automated production techniques. And, if required, capital for expansion is available to meet even greater requirements.

MC6870, MC6871 series



actual size

environmental specifications

Temperature Cycle: ± 5 ppm max., 0 to 120°C, 3 cycles, 2 hrs. max. each, 25 $\pm 2^\circ\text{C}$ ref.

Shock: 1000G's 0.35 millisecond, $\frac{1}{2}$ sine wave, 3 shocks each plane

Vibration: 10-55 Hz, .060" D.A.; 55-2000Hz, 35 G's. Duration Time—12 Hours

Humidity: 85% Rel. Humidity, @ $+85^\circ\text{C}$, 250 Hours

mechanical specifications

Gross Leak Test: All units 100% leak tested in de-ionized H_2O .

Hermetic Sealed Package: Mass spectrometer leak rate less than 2×10^{-8} atmos. cc/sec. of helium.

Seal Strength: 20 lbs. max. force perpendicular to top and bottom.

Pin Material: Phosphor bronze, $\frac{1}{4}$ hard, Grade A .00003" thick gold flash finish.

Bend Test: Will withstand maximum bend of 90° reference to base for 1 bend.

Marking Ink: Epoxy, heat cured.

Solvent Resistance: Isopropyl Alcohol Trichloroethane Freon TMC. No marking or seal destruction. Dipped 1 minute @ $+25^\circ\text{C}$ $\pm 5^\circ\text{C}$ in solvent.

Note: (1) Unit can be cleaned by only one type solvent listed.

Note: (2) Ultrasonic degreaser not to be used unless frequency and vibration of cleaner specified.

solderability specifications

Materials:

1.1 Solder: 60% tin and 40% lead.

1.2 Flux: The flux shall be 25 percent by weight of Grade WW rosin and 75 percent by weight of 99 percent isopropyl alcohol.

Procedure:

2.1 Solder Bath: The solder bath shall be maintained at $232 \pm 6^\circ\text{C}$.

2.2 Solderability: Dip the terminals into the flux to the depth that is to be soldered or to a maximum depth of .025" from the body of the oscillator. Keep them in the flux for at least 5 seconds. Withdraw them from the flux. Dip them immediately into the molten solder to the same depth. Keep them in the molten solder for 2 to 5 seconds. Withdraw them and allow the solder to cool in air.

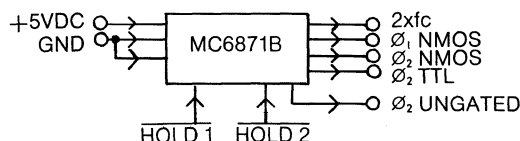
Requirements:

3.1 The terminals are considered solderable and acceptable for electrical connection purposes if 90 percent of the cold solder surface is uniform and free from breaks and pinholes. The other 10 percent of the cooled solder surface may show only pinholes, voids, or rough spots that are not concentrated in one area.

850 kHz to 2.5 MHz



alternate function microprocessor clock
250 kHz to 2.5 MHz



specifications

Rating	Symbol	Value	Unit
Supply Voltage	V_{cc}	$5.00 \pm 5\%$	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature	T_{stg}	-55 to +125	°C
Power Supply Drain (max.)	I_{sd}	100	mA

ELECTRICAL CHARACTERISTICS ($V_{CC} = 5.0 \pm 5\%$, $V_{SS} = 0$, $T_A = 0^\circ$ to 70°C , unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
Frequency					
Operating Frequency	f_c	.250		2.5	MHz
Frequency stability (inclusive of calibration tolerance at +25°C, operating temperature, input voltage change, load change, aging, shock and vibration)			$\pm .01$		%

NMOS Outputs at 1.0 MHz Operation***

NMOS Outputs at 0.3V dc level					
Pulse Width (meas. at $V_{cc} = -0.3V$ dc level)	$T_{\phi,H}$ $T_{\phi,L}$	430 450			ns ns
Logic Levels	V_{OLC} V_{OLC}	$V_{ss} - .1$ $V_{cc} - .3$	— —	$V_{ss} + .3$ $V_{cc} + .1$	Vdc Vdc
Rise and Fall Times	t_r t_f	5 5	12 12	50 50	ns ns
*Overshoot/Undershoot Logic "1" Logic "0"	V_{OS}	$V_{cc} - .5$ $V_{ss} - .5$		$V_{cc} + .5$ $V_{ss} + .5$	Vdc Vdc
Pulse duration of any overshoot or undershoot	T_{OS}			40	ns
Period @ 0.3V dc Level	$t_{\phi,c}$		1.00		us
Edge Timing @ $V_{cc} = 0.3V$ dc	T_x	940			ns
NMOS Relationship @ +0.5V dc	t_{d1} t_{d2}	0 0		8.0	us

TTL Outputs

In ref. to ϕ_2 , NMOS @ 0.3V dc					
ϕ_2 TTL @ 1.4V dc	T_A T_H	15 10	30 25	45 40	ns ns
ϕ_2 Ungated @ 1.4V dc	T_C T_J	30 20	50 40	70 60	ns ns
2xfc @ 1.4V dc	T_B	40	80	120	ns
Logic Levels	V_{OH} V_{OL}	2.4	3.2 .3		Vdc Vdc
Rise and Fall Times					
.4V and 2.4V	t_r			15	ns
2.4V and .4V	t_f			15	ns
Logic "0" Sink (/Gate)	I_{OL}			-1.6	mA
Logic "1" Source (/Gate)	I_{OH}			+4.0	mA
Current Output Shorted	I_{KS}	-18		-57	mA

Load

NMOS—Load Capacity ϕ_1, ϕ_2	C_{NMOS}	80	120	160	pf
TTL—No. of Loads				5	tfl
TTL—Load Capacity	C_{TTL}			50	pf

Logic Inputs ("0" Level applies HOLD)**

Holds ϕ_1 NMOS 'High', ϕ_2 NMOS 'Low', ϕ_2 TTL 'Low'	$\overline{\text{HOLD}} 1$	-2		+4	Vdc
Holds ϕ_1 NMOS 'Low', ϕ_2 NMOS 'High', ϕ_2 TTL 'High'	$\overline{\text{HOLD}} 2$	-2		+4	Vdc

* Into specified test load

**Must be externally held at "1" level (2.4V min., 5.0V max.) if not used

***Apply the following parameters for frequencies other than 1 MHz:

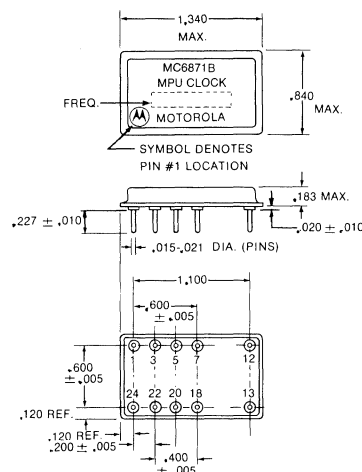
$T\phi_1 H = 0.5$ (P-140) ns

$$T_{\phi_2 H} = 0.5 \text{ (P-100) ns}$$

$T_x = (P-60) \text{ ns}$
where $P = \text{degrees}$

where P = desired period of operation in nanoseconds

DIMENSIONS



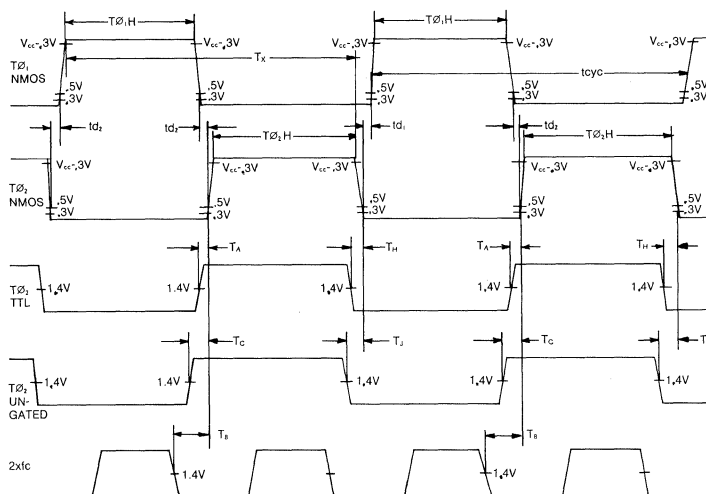
PIN	CONNECTION
1	GND
3	$\varnothing_1$ TTL UNGATED
5	$\varnothing_2$ TTL
7	V_{cc} (+5VDC)
12	$\varnothing_3$ NMOS
13	$\varnothing_4$ NMOS
18	GND
20	$\overline{\text{HOLD 1}}$
22	$\overline{\text{HOLD 2}}$
24	2xfc

Note: 4xfc available on request

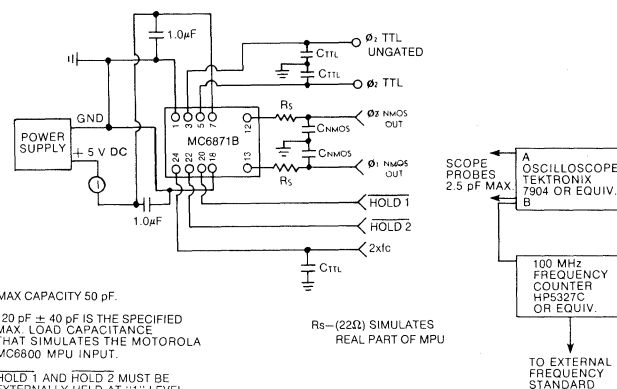
Note: All dimensions are in inches

WAVEFORM TIMING.

ALL TIME IN NANOSECONDS



TEST DIAGRAM



C_{TTL} — MAX CAPACITY 50 pF.

C_{NMOS} = 120 pF ± 40 pF IS THE SPECIFIED
MAX. LOAD CAPACITANCE
THAT SIMULATES THE MOTOROLA
MC6800 MPU INPUT.

Rs—(22Ω) SIMULATES
REAL PART OF MPU

*HOLD 1 AND HOLD 2 MUST BE EXTERNALLY HELD AT "1" LEVEL (2.4VDC MIN., 5.0VDC MAX.) WHEN NOT USED

TO EXTERNAL
FREQUENCY
STANDARD





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Product Preview

QUAD THREE-STATE BUS TRANSCEIVER

This quad three-state bus transceiver features both excellent MOS or MPU compatibility, due to its high impedance PNP transistor input, and high-speed operation made possible by the use of Schottky diode clamping. Both the -40 mA driver and -16 mA receiver outputs are short-circuit protected and employ three-state enabling inputs.

The device is useful as a bus extender in systems employing the M6800 family for other comparable MPU devices. The maximum input current of 200 μ A at any of the device input pins assures proper operation despite the limited drive capability of the MPU chip. The inputs are also protected with Schottky-barrier diode clamps to suppress excessive undershoot voltages.

Propagation delay times for the driver portion are 16 ns typical while the receiver portion runs 6 ns for t_{pHL} and 13 ns for t_{pLH} . The MC8T26 is identical to the NE8T26 and it operates from a single +5 V supply.

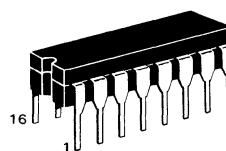
- High Impedance Inputs
- Single Power Supply
- High Speed Schottky Technology
- Three-State Drivers and Receivers
- Compatible With M6800 Family Microprocessor

MC6880

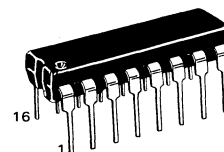
MC8T26

QUAD THREE-STATE BUS
TRANSCEIVER WITH HIGH
IMPEDANCE PNP INPUTS

SILICON MONOLITHIC
INTEGRATED CIRCUITS

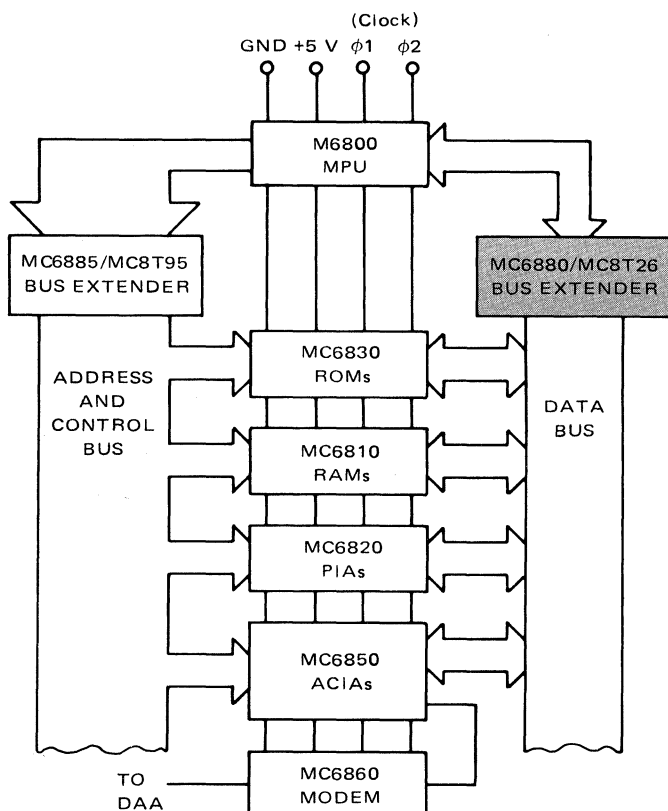


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CASE 620

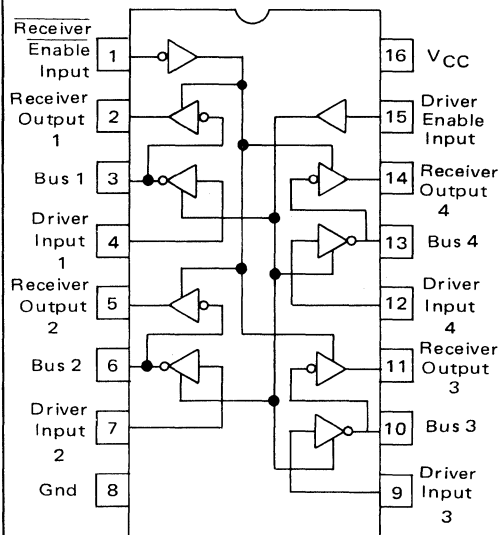


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MICROPROCESSOR BUS EXTENDER APPLICATION



PIN CONNECTIONS — MC6880 MC8T26



MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Rating	Symbol	Value	Unit
Power Supply Voltage	V_{CC}	8.0	Vdc
Input Voltage	V_I	5.5	Vdc
Power Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	1000 6.7	mW mW/ $^\circ\text{C}$
Operating Ambient Temperature Range	T_A	0 to +75	$^\circ\text{C}$
Storage Temperature Range	T_{stg}	-65 to +150	$^\circ\text{C}$

ELECTRICAL CHARACTERISTICS (Unless Otherwise Noted Specifications Apply $4.75\text{ V} \leq V_{CC} \leq 5.25\text{ V}$ and $0^\circ\text{C} \leq T_A \leq 75^\circ\text{C}$)

Characteristic	Symbol	Min	Typ	Max	Unit
Input Current — Low Logic State (Receiver Enable Input, $V_{IL}(\text{RE}) = 0.4\text{ V}$) (Driver Enable Input, $V_{IL}(\text{DE}) = 0.4\text{ V}$) (Driver Input, $V_{IL}(\text{D}) = 0.4\text{ V}$) (Bus (Receiver) Input, $V_{IL}(\text{B}) = 0.4\text{ V}$)	$I_{IL}(\overline{\text{RE}})$ $I_{IL}(\text{DE})$ $I_{IL}(\text{D})$ $I_{IL}(\text{B})$	— — — —	— — — —	-200 -200 -200 -200	μA
Input Current—High Logic State (Receiver Enable Input, $V_{IH}(\text{RE}) = 5.25\text{ V}$) (Driver Enable Input, $V_{IH}(\text{DE}) = 5.25\text{ V}$) (Driver Input, $V_{IH}(\text{D}) = 5.25\text{ V}$)	$I_{IH}(\overline{\text{RE}})$ $I_{IH}(\text{DE})$ $I_{IH}(\text{D})$	— — —	— — —	25 25 25	μA
Input Voltage — Low Logic State (Receiver Enable Input) (Driver Enable Input) (Driver Input)	$V_{IL}(\overline{\text{RE}})$ $V_{IL}(\text{DE})$ $V_{IL}(\text{D})$	0.85 0.85 0.85	— — —	— — —	V
Input Voltage — High Logic State (Receiver Enable Input) (Driver Enable Input) (Driver Input)	$V_{IH}(\overline{\text{RE}})$ $V_{IH}(\text{DE})$ $V_{IH}(\text{D})$	— — —	— — —	2.0 2.0 2.0	V
Output Voltage — Low Logic State (Bus (Driver) Output, $I_{OL}(\text{B}) = -10\text{ mA}$) (Receiver Output, $I_{OL}(\text{R}) = -2.0\text{ mA}$)	$V_{OL}(\text{B})$ $V_{OL}(\text{R})$	— —	— —	0.5 0.5	V
Output Voltage — High Logic State (Bus (Driver) Output, $I_{OH}(\text{B}) = 40\text{ mA}$) (Receiver Output, $I_{OH}(\text{R}) = 16\text{ mA}$)	$V_{OH}(\text{B})$ $V_{OH}(\text{R})$	2.6 2.6	3.1 3.1	— —	V
Output Disabled Leakage Current — High Logic State (Bus (Driver) Output, $V_{OH}(\text{B}) = 2.6\text{ V}$) (Receiver Output, $V_{OH}(\text{R}) = 2.6\text{ V}$)	$I_{OHL}(\text{B})$ $I_{OHL}(\text{R})$	— —	— —	100 100	μA
Input Clamp Voltage (Driver Enable Input $I_{IC}(\text{DE}) = -5.0\text{ mA}$) (Receiver Enable Input $I_{IC}(\text{RE}) = -5.0\text{ mA}$) (Driver Input $I_{IC}(\text{D}) = -5.0\text{ mA}$)	$V_{IC}(\text{DE})$ $V_{IC}(\text{RE})$ $V_{IC}(\text{D})$	— — —	— — —	-1.0 -1.0 -1.0	V
Output Short-Circuit Current, $V_{CC} = 5.25\text{ V}$ (1) (Bus (Driver) Output) (Receiver Output)	$I_{OS}(\text{B})$ $I_{OS}(\text{R})$	-50 -30	— —	-150 -75	mA
Power Supply Current ($V_{CC} = 5.25\text{ V}$)	I_{CC}	—	—	87	mA

(1) Only one output may be short-circuited at a time.



SWITCHING CHARACTERISTICS (Unless otherwise noted, specifications apply at $T_A = 25^\circ\text{C}$ and $V_{CC} = 5.0\text{ V}$)

Characteristic	Symbol	Figure	Min	Typ	Max	Unit
Propagation Delay Time from Receiver (Bus) Input to High Logic State Receiver Output	$t_{PLH(R)}$	1	—	13	18	ns
Propagation Delay Time from Receiver (Bus) Input to Low Logic State Receiver Output	$t_{PHL(R)}$	1	—	6.0	10	ns
Propagation Delay Time from Driver Input to High Logic State Driver (Bus) Output	$t_{PLH(D)}$	2	—	16	20	ns
Propagation Delay Time from Driver Input to Low Logic State Driver (Bus) Output	$t_{PHL(D)}$	2	—	16	20	ns
Propagation Delay Time from Receiver Enable Input to High Impedance (Open) Logic State Receiver Output	$t_{PLO(\overline{RE})}$	3	—	10	17	ns
Propagation Delay Time from Receiver Enable Input to Low Logic Level Receiver Output	$t_{POL(\overline{RE})}$	3	—	20	30	ns
Propagation Delay Time from Driver Enable Input to High Impedance Logic State Driver (Bus) Output	$t_{PLO(DE)}$	4	—	35	43	ns
Propagation Delay Time from Driver Enable Input to Low Logic State Driver (Bus) Output	$t_{POL(DE)}$	4	—	29	38	ns

FIGURE 1 — TEST CIRCUIT AND WAVEFORMS FOR PROPAGATION DELAY FROM BUS (RECEIVER) INPUT TO RECEIVER OUTPUT $t_{PLH(R)}$, $t_{PHL(R)}$

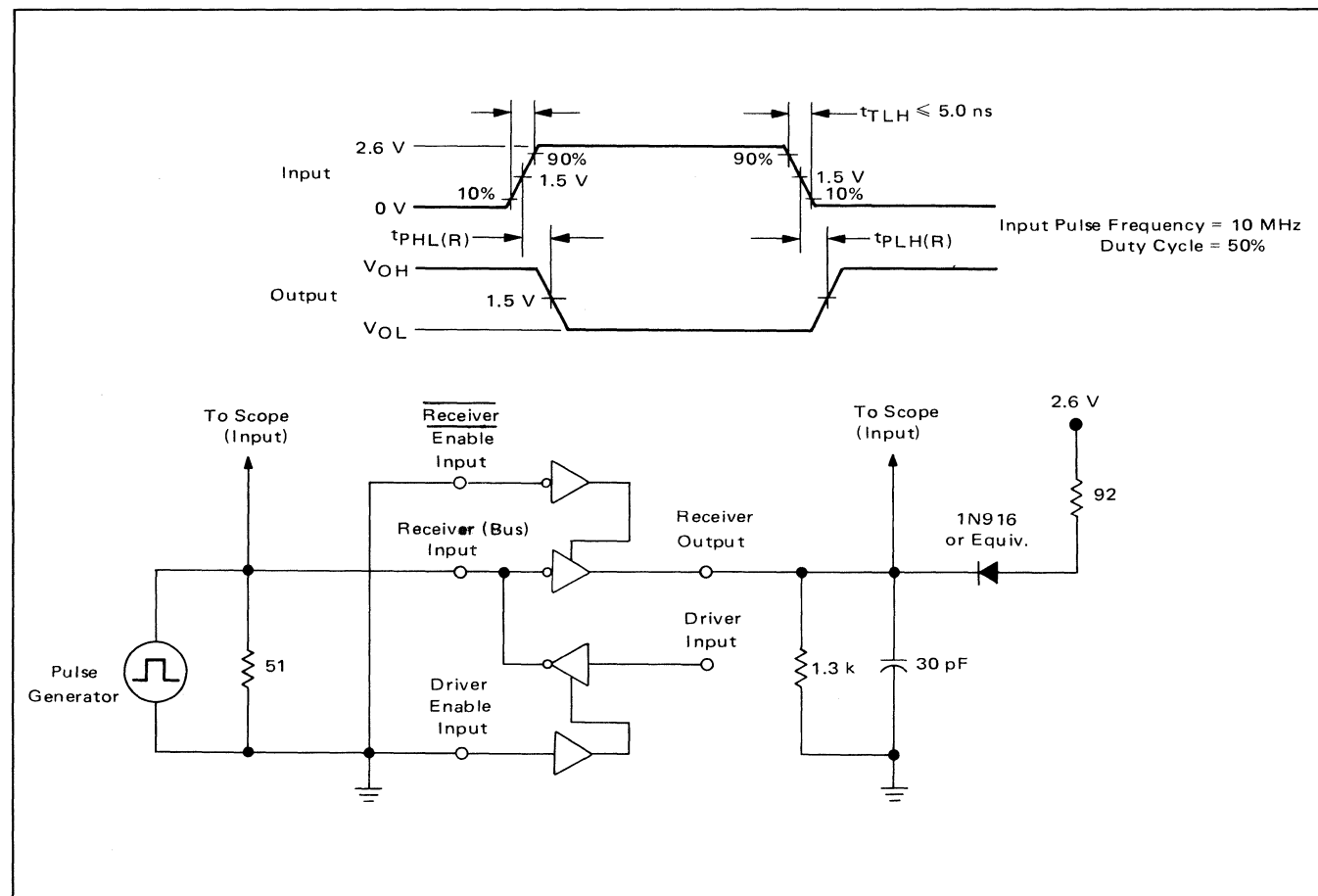


FIGURE 2 — TEST CIRCUIT AND WAVEFORMS FOR PROPAGATION DELAY TIME FROM DRIVER INPUT TO BUS (DRIVER) OUTPUT, $t_{PLH(D)}$ AND $t_{PLH(D)}$

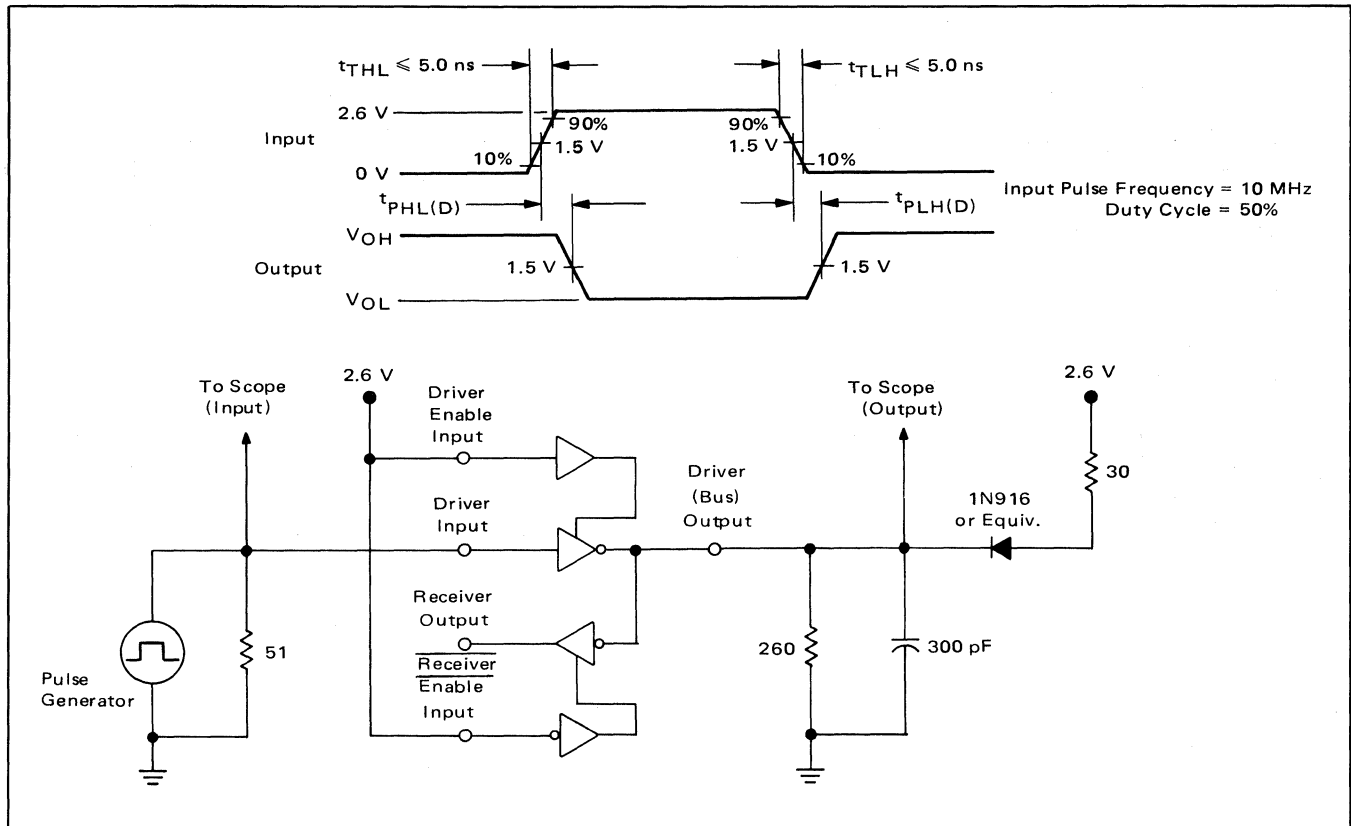


FIGURE 3 — TEST CIRCUIT AND WAVEFORMS FOR PROPAGATION DELAY TIME FROM RECEIVER ENABLE INPUT TO RECEIVER OUTPUT, $t_{PLO(RE)}$ AND $t_{POL(RE)}$

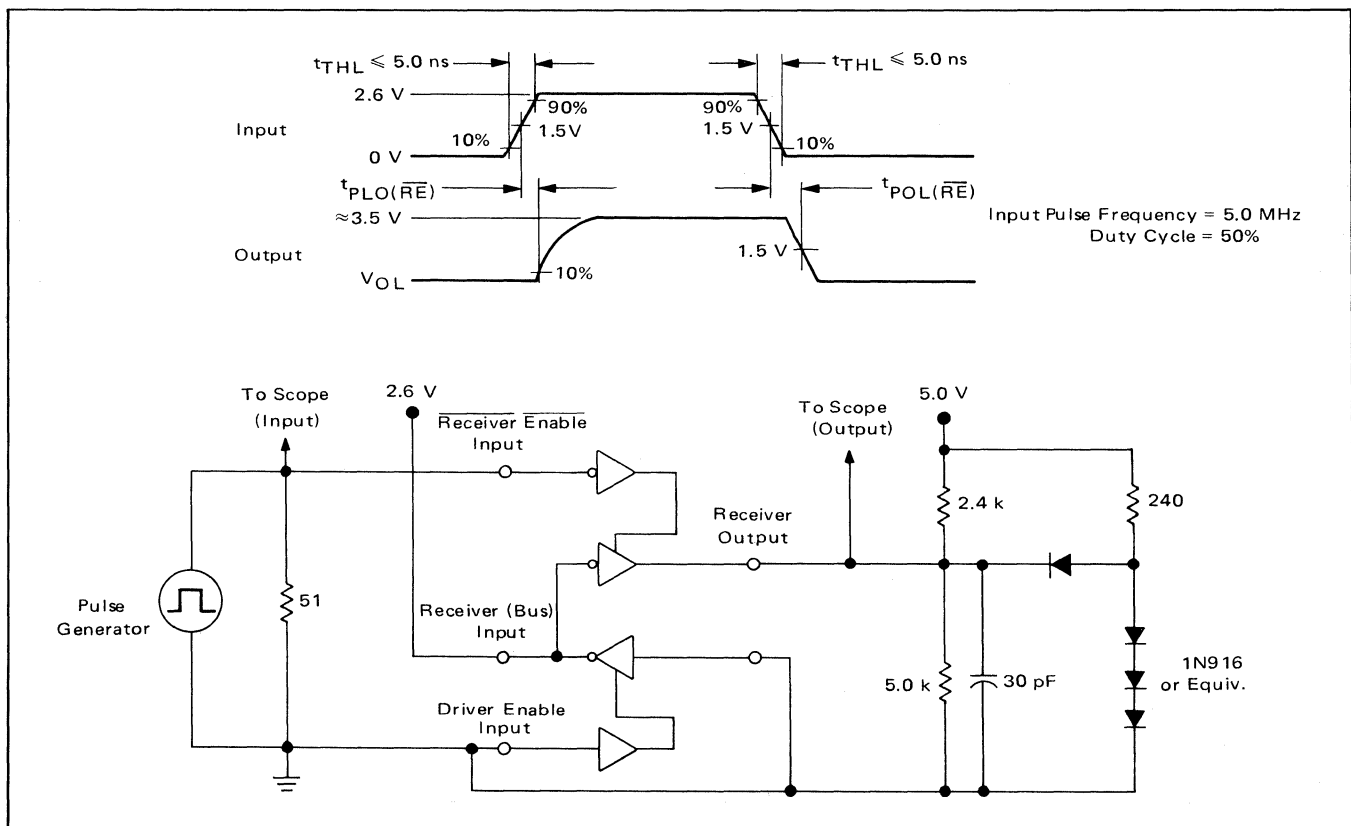


FIGURE 4 – TEST CIRCUIT AND WAVEFORMS FOR PROPAGATION DELAY TIMES FROM DRIVER ENABLE INPUT TO DRIVER (BUS) OUTPUT, $t_{PLO(DE)}$ AND $t_{POL(DE)}$

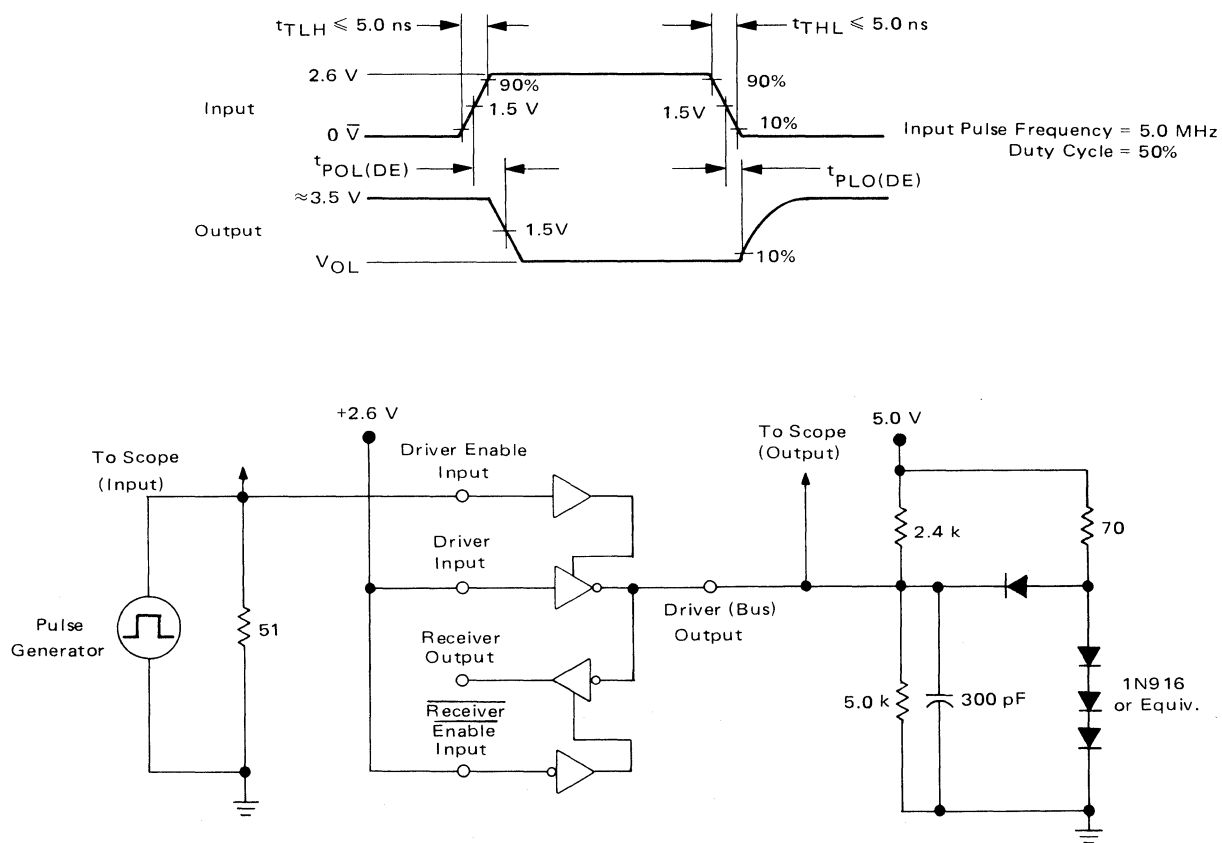
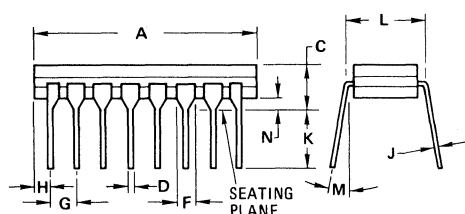
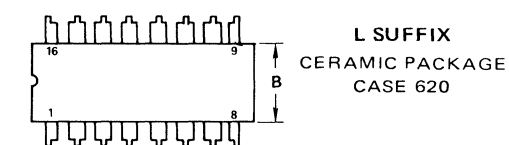
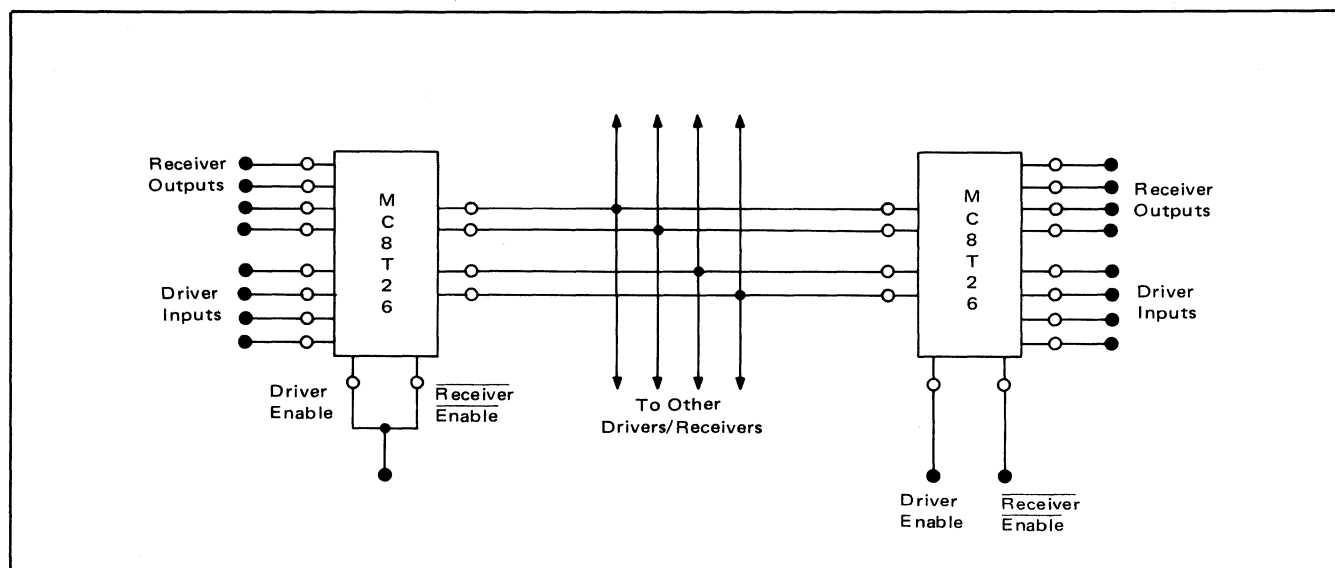


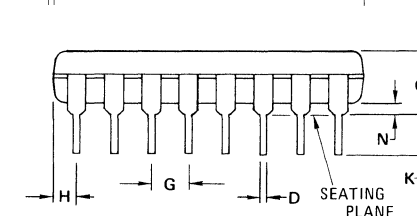
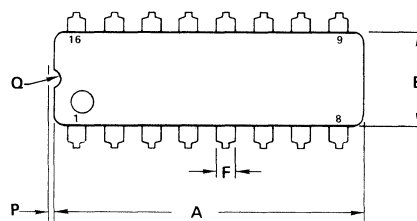
FIGURE 5 – BI-DIRECTIONAL BUS APPLICATIONS



NOTES:

- LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION
- PKG. INDEX: NOTCH IN LEAD NOTCH IN CERAMIC OR INK DOT
- DIM "L" TO CENTER OF LEADS WHEN FORMED PARALLEL

DIM	MIN	MAX	MIN	MAX
A	19.05	19.81	0.750	0.780
B	6.22	6.98	0.245	0.275
C	4.06	5.08	0.160	0.200
D	0.38	0.51	0.015	0.020
F	1.40	1.65	0.055	0.065
G	2.54 BSC		0.100 BSC	
H	0.51	1.14	0.020	0.045
J	0.20	0.30	0.008	0.012
K	3.18	4.06	0.125	0.160
L	7.37	7.87	0.290	0.310
M	—	15°	—	15°
N	0.51	1.02	0.020	0.040

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NOTES:

- LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION.
- DIMENSION "L" TO CENTER OF LEADS WHEN FORMED PARALLEL

DIM	MIN	MAX	MIN	MAX
A	20.70	21.34	0.815	0.840
B	6.10	6.60	0.240	0.260
C	4.06	4.57	0.160	0.180
D	0.38	0.51	0.015	0.020
F	1.02	1.52	0.040	0.060
G	2.54 BSC		0.100 BSC	
H	1.32	1.83	0.052	0.072
J	0.20	0.30	0.008	0.012
K	2.92	3.43	0.115	0.135
L	7.37	7.87	0.290	0.310
M	—	10°	—	10°
N	0.51	1.02	0.020	0.040
P	0.13	0.38	0.005	0.015
Q	0.51	0.76	0.020	0.030





MOTOROLA
Semiconductors

BOX 20912 • PHOENIX, ARIZONA 85036

Product Preview

TRIPLE BI-DIRECTIONAL BUS SWITCH

The XC6881/3449 is a three channel, non-inverting, bi-directional Bus Extender. It is designed to allow the bi-directional exchange of TTL level digital information between a selected pair of ports in a three port network. All three ports of each channel may be forced to a high impedance condition through that channel's Enable input.

Port pair selection and listener/talker status for the three channels is determined through the Control and Select inputs. All inputs are PNP buffered and M6800 Family compatible.

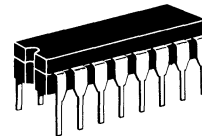
A summary of XC6881/3449 features include:

- Three Channels
- Noninverting Data Exchange
- Bi-Directional Operation
- Active Pull-Up with Three-State Capability
- High Impedance Inputs
- TTL Compatible (74LS)

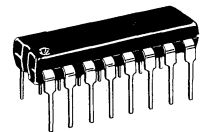
XC6881
XC3449

**BI-DIRECTIONAL
BUS EXTENDER/SWITCH**

**SILICON MONOLITHIC
INTEGRATED CIRCUITS**

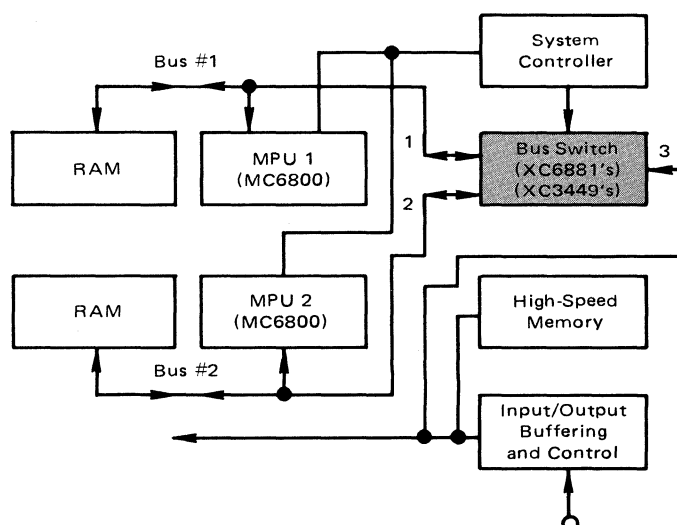


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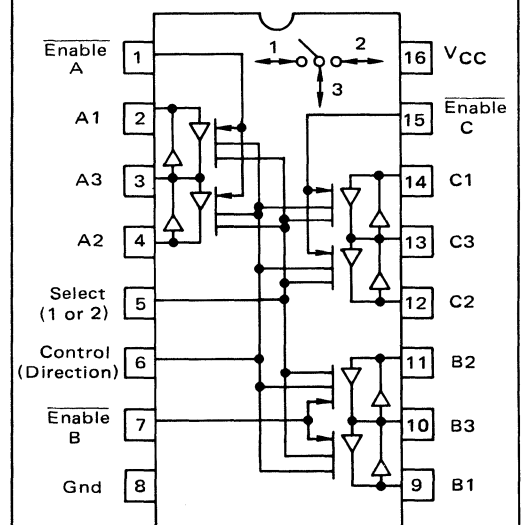


P SUFFIX
PLASTIC PACKAGE
CASE 648

MICROPROCESSOR BUS SWITCH APPLICATION



PIN CONNECTIONS

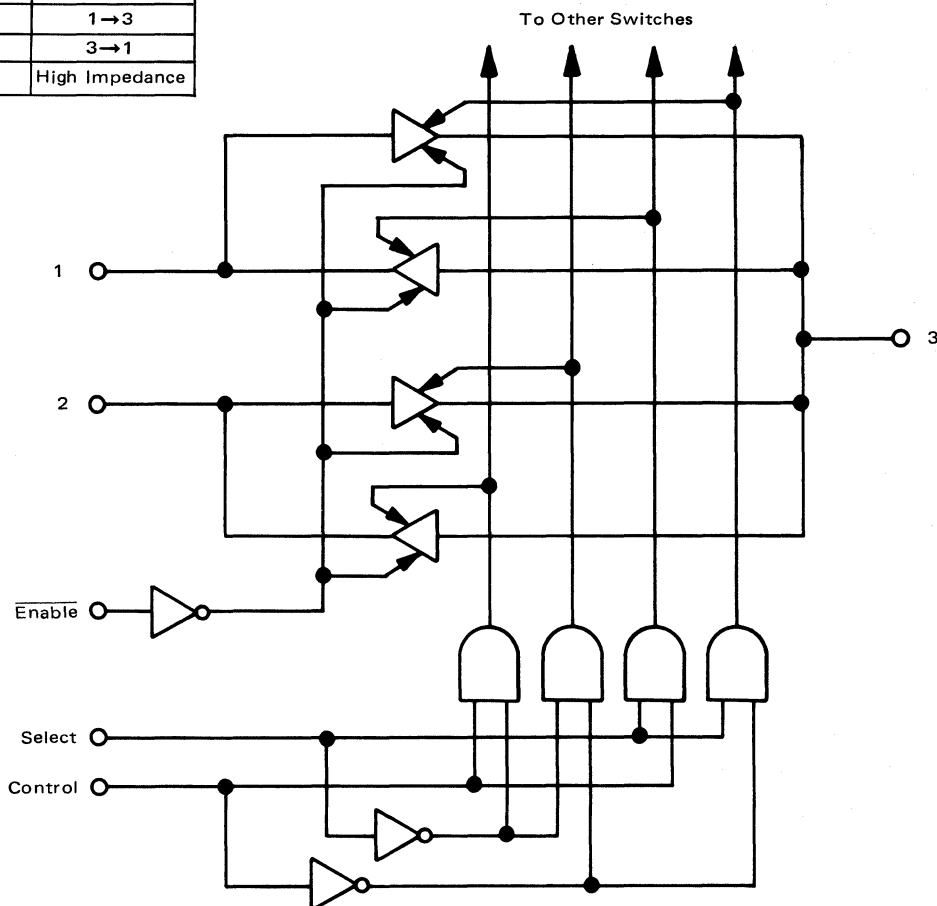


TRUTH TABLE

Enable	Select	Control	Data Flow
0	0	0	2→3
0	0	1	3→2
0	1	0	1→3
0	1	1	3→1
1	X	X	High Impedance

X - Don't Care

FUNCTIONAL DIAGRAM

TARGET SPECIFICATIONS ($4.75\text{ V} \leq V_{CC} \leq 5.25\text{ V}$, $0^\circ\text{C} \leq T_A \leq 70^\circ\text{C}$)

Characteristic	Symbol	Min	Max	Unit
Input Current — Low Logic State ($V_{IL} = 0.4\text{ V}$)	I_{IL}	—	—200	μA
Input Current — High Logic State ($V_{IH} = 5.25\text{ V}$)	I_{IH}	—	25	μA
Input Voltage — Low Logic State	V_{IL}	0.8	—	V
Input Voltage — High Logic State	V_{IH}	—	2.0	V
Output Voltage — Low Logic State ($I_{OL} = 8.0\text{ mA}$) ($I_{OL} = 16\text{ mA}$)	V_{OL}	— —	0.5 0.6	V
Output Voltage — High Logic State ($I_{OH} = -1.0\text{ mA}$)	V_{OH}	2.4	—	V
Output Disabled Current ($V_{OH} = 2.4\text{ V}$) ($V_{OL} = 0.5\text{ V}$)	I_{OD}	— —	25 -25	μA





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Semiconductors

BOX 20912 • PHOENIX, ARIZONA 85036

Product Preview

HEX THREE-STATE BUFFER/INVERTER

This series of devices combines four features usually found desirable in bus oriented systems. These features are: 1) — high impedance logic inputs insure that these devices do not seriously load the bus, 2) — three-state logic configuration allows buffers not being utilized to be effectively removed from the bus, 3) Schottky technology allows high-speed operation and 4) High-impedance output state maintained during power up/down. Particularly in unidirectional bus extenders and for both Address and Control functions.

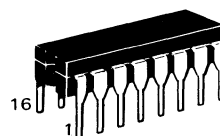
The devices differ in that the non-inverting XC8T95/XC6885 and inverting XC8T96/XC6886 provide a two-input Enable which controls all six buffers, while the non-inverting XC8T97/XC6887 and inverting XC8T98/XC6888 provide two Enable inputs — one controlling four buffers and the other controlling the remaining two buffers.

The units are well-suited for Address buffers on the M6800 or similar microprocessor application.

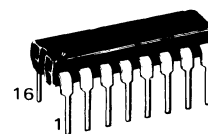
- High Speed — 8.0 ns (Typ)
- Three-State Logic Configuration
- Single +5 V Power Supply Requirement
- Compatible with 74LS Logic or M6800 Microprocessor Systems
- High Impedance PNP Inputs Assure Minimal Loading of the Bus

XC6885 XC8T95
XC6886 XC8T96
XC6887 XC8T97
XC6888 XC8T98

HEX THREE-STATE BUFFER/INVERTERS MONOLITHIC SCHOTTKY INTEGRATED CIRCUITS

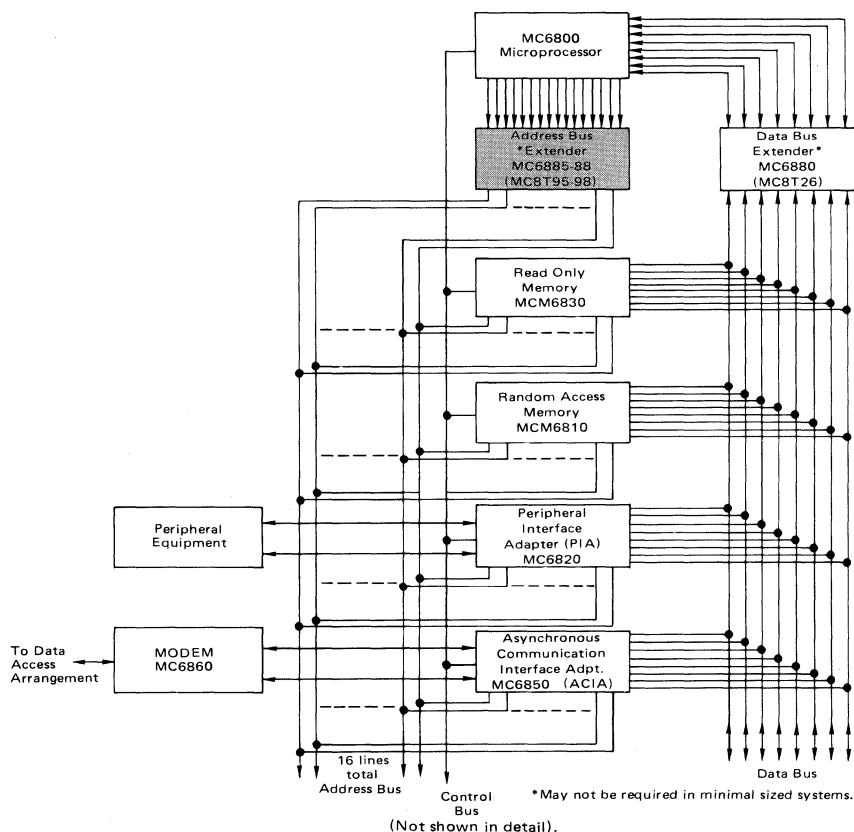


L SUFFIX
CERAMIC PACKAGE
CASE 620

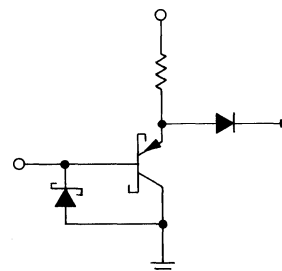


P SUFFIX
PLASTIC PACKAGE
CASE 648

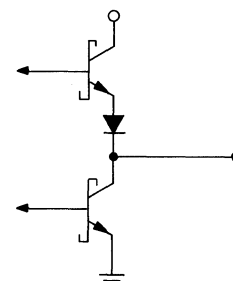
TYPICAL EXPANDED M6800 MPU SYSTEM



INPUT EQUIVALENT CIRCUIT

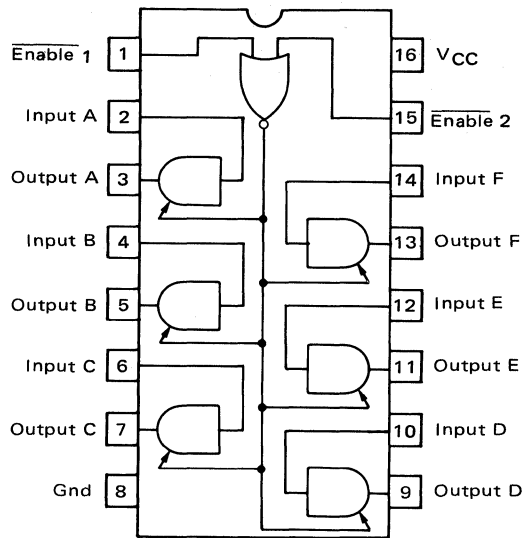


OUTPUT EQUIVALENT CIRCUIT



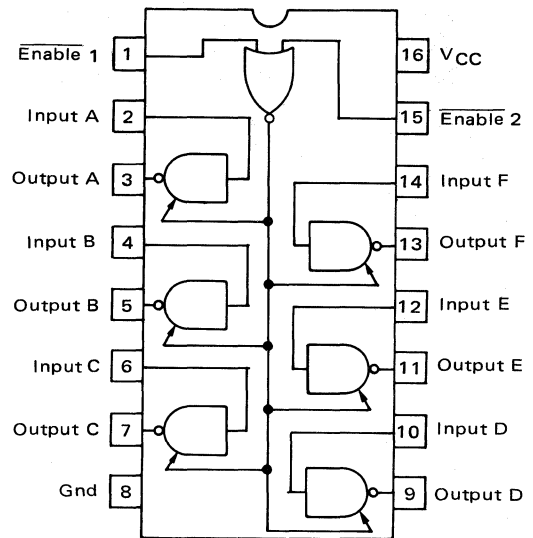
PIN CONNECTIONS AND TRUTH TABLES

XC6885/XC8T95



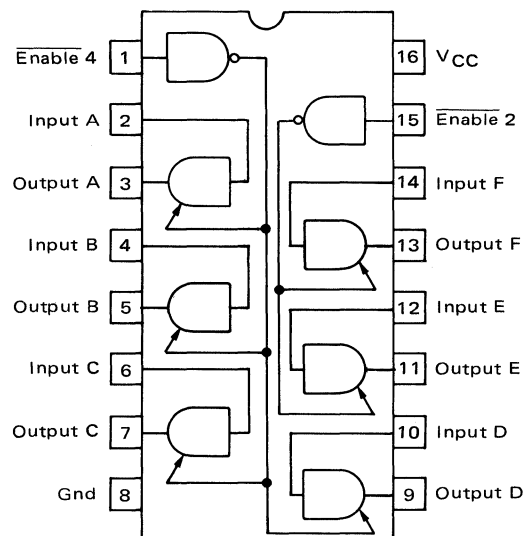
Enable 2	Enable 1	Input	Output
L	L	L	L
L	L	H	H
L	H	X	O
H	L	X	O
H	H	X	O

XC6886/XC8T96



Enable 2	Enable 1	Input	Output
L	L	L	H
L	L	H	O
L	H	X	O
H	L	X	O
H	H	X	O

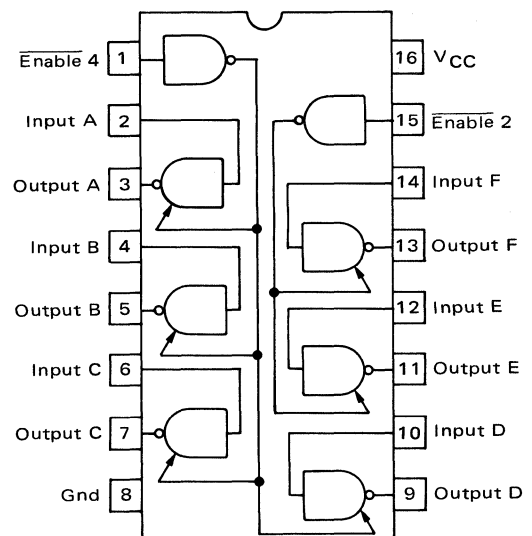
XC6887/XC8T97



Enable	Input	Output
L	L	L
L	H	H
H	X	O

L = Low Logic State
H = High Logic State
O = High Impedance State (open)
X = Irrelevant

XC6888/XC8T98



Enable	Input	Output
L	L	H
L	H	L
H	X	O

MAXIMUM RATINGS (T_A = 25°C unless otherwise noted.)

Rating	Symbol	Value	Unit
Power Supply Voltage	V _{CC}	8.0	Vdc
Input Voltage	V _I	5.5	Vdc
Operating Ambient Temperature Range	T _A	0 to +75	°C
Storage Temperature Range	T _{stg}	-65 to +150	°C
Operating Junction Temperature	T _J		°C
Plastic Package		150	
Ceramic Package		175	



ELECTRICAL CHARACTERISTICS (Unless otherwise noted, $0^{\circ}\text{C} \leq T_A \leq 75^{\circ}\text{C}$ and $4.75\text{ V} \leq V_{CC} \leq 5.25\text{ V}$)

Characteristic	Symbol	Min	Typ	Max	Unit
Input Voltage — High Logic State ($V_{CC} = 4.75\text{ V}$, $T_A = 25^{\circ}\text{C}$)	V_{IH}	2.0	—	—	V
Input Voltage — Low Logic State ($V_{CC} = 4.75\text{ V}$, $T_A = 25^{\circ}\text{C}$)	V_{IL}	—	—	0.8	V
Input Current — High Logic State ($V_{CC} = 5.25\text{ V}$, $V_{IH} = 2.4\text{ V}$)	I_{IH}	—	—	40	μA
Input Current — Low Logic State ($V_{CC} = 5.25\text{ V}$, $V_{IL} = 0.5\text{ V}$, $V_{IL}(\bar{E}) = 0.5\text{ V}$)	I_{IL}	—	—	-400	μA
Input Current — High Impedance State ($V_{CC} = 5.25\text{ V}$, $V_{IL}(I) = 0.5\text{ V}$, $V_{IH}(\bar{E}) = 2.0\text{ V}$)	$I_{IH}(\bar{E})$	—	—	-40	μA
Output Voltage — High Logic State ($V_{CC} = 4.75\text{ V}$, $I_{OH} = -5.2\text{ mA}$)	V_{OH}	2.4	—	—	V
Output Voltage — Low Logic State ($I_{OL} = 48\text{ mA}$)	V_{OL}	—	—	0.5	V
Output Current — High Impedance State ($V_{CC} = 5.25\text{ V}$, $V_{OH} = 2.4\text{ V}$) ($V_{CC} = 5.25\text{ V}$, $V_{OL} = 0.5\text{ V}$)	V_{OO}	— —	— —	40 -40	μA
Output Short-Circuit Current ($V_{CC} = 5.25\text{ V}$, $V_O = 0$) (only one output can be shorted at a time)	I_{OS}	-40	-80	-115	mA
Power Supply Current ($V_{CC} = 5.25\text{ V}$) XC8T95, XC8T97, XC6885, XC6887 XC8T96, XC8T98, XC6886, XC6888	I_{CC}	— —	65 59	98 89	mA
Input Clamp Voltage ($V_{CC} = 4.75\text{ V}$, $I_{IC} = -12\text{ mA}$)	V_{IC}	—	—	-1.5	V
Output V_{CC} Clamp Voltage ($V_{CC} = 0$, $I_{OC} = 12\text{ mA}$)	V_{OC}	—	—	1.5	V
Output Gnd Clamp Voltage ($V_{CC} = 0$, $I_{OC} = -12\text{ mA}$)	V_{OC}	—	—	-1.5	V
Input Voltage ($I_I = 1.0\text{ mA}$)	V_I	5.5	—	—	V

SWITCHING CHARACTERISTICS ($V_{CC} = 5.0\text{ V}$, $T_A = 25^{\circ}\text{C}$ unless otherwise noted).

Characteristic	Symbol	Min	Typ	Max	Unit
Propagation Delay Time — Input to Output ($R_L = 200\ \Omega$, $C_L = 50\text{ pF}$)	t_{PLH} t_{PHL}	— —	6.0 5.0	— —	ns ns
Propagation Delay Time — Enable to Output ($R_L = 200\ \Omega$, $C_L = 50\text{ pF}$) ($R_L = 200\ \Omega$, $C_L = 50\text{ pF}$) ($R_L = \infty$, $C_L = 50\text{ pF}$) ($R_L = 200\ \Omega$, $C_L = 50\text{ pF}$)	$t_{PHO}(\bar{E})$ $t_{PLO}(\bar{E})$ $t_{POH}(\bar{E})$ $t_{POL}(\bar{E})$	— — — —	10 12 10 12	— — — —	ns ns ns ns

FIGURE 1 — TEST CIRCUIT FOR SWITCHING CHARACTERISTICS

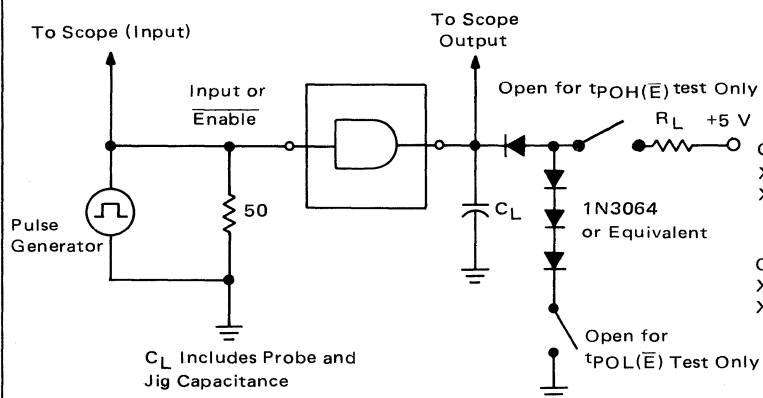


FIGURE 2 — WAVEFORMS FOR PROPAGATION DELAY TIMES INPUT TO OUTPUT

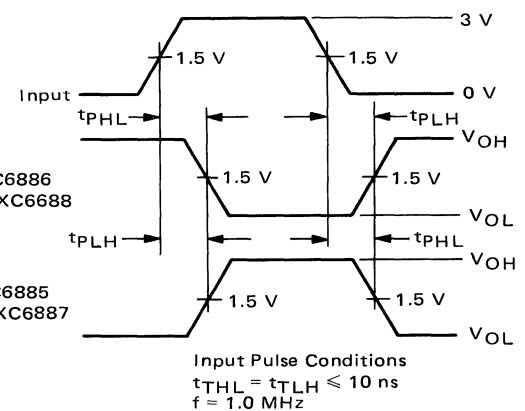


FIGURE 3 – WAVEFORMS FOR PROPAGATION DELAY TIMES – ENABLE TO OUTPUT

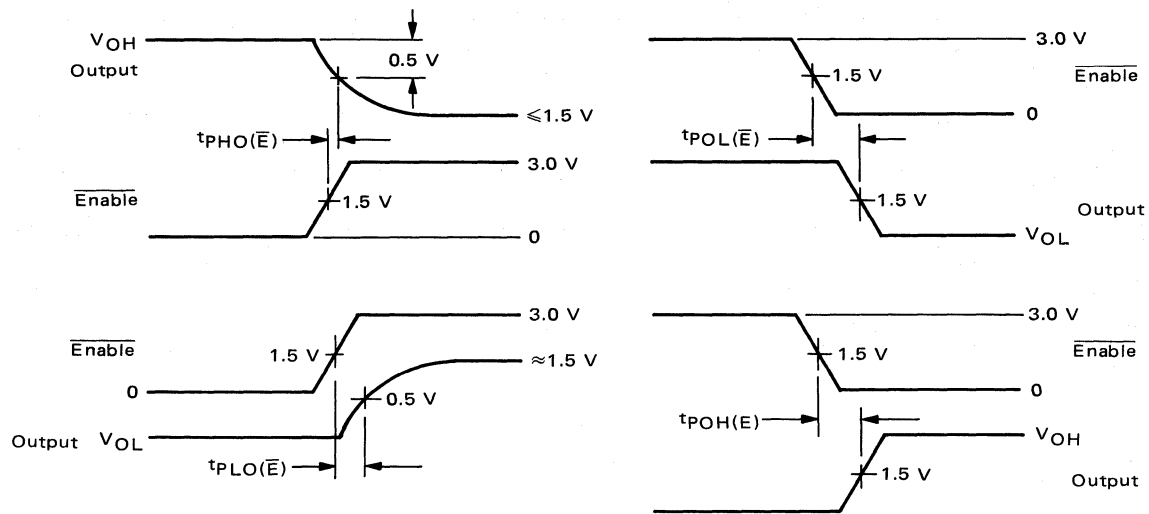
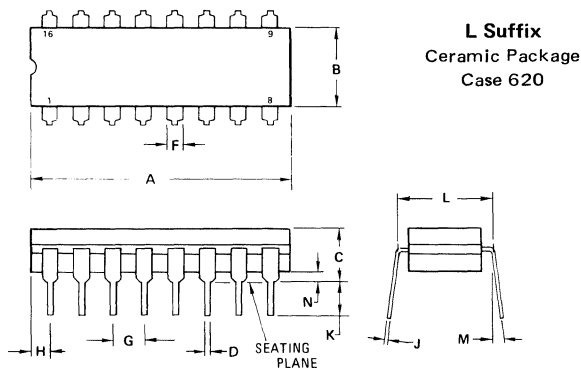
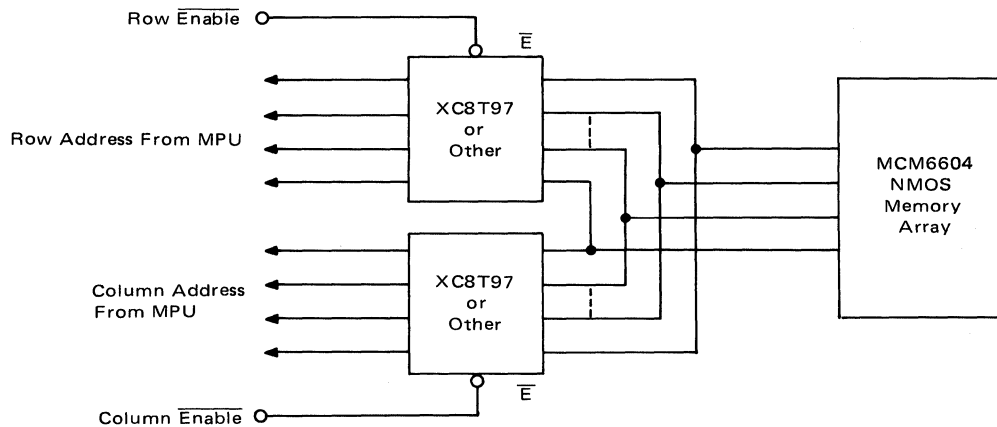


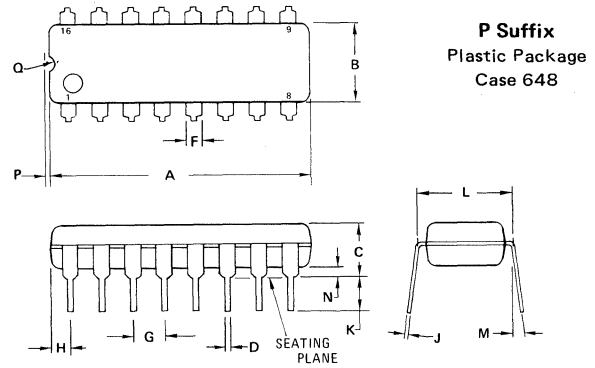
FIGURE 4 – ADDRESS MULTIPLEXER FOR 16-PIN 4K NMOS MEMORY



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	19.05	19.81	0.750	0.780
B	6.22	6.98	0.245	0.275
C	4.06	5.08	0.160	0.200
D	0.38	0.51	0.015	0.020
E	1.40	1.65	0.055	0.065
F	2.54	BSC	0.100	BSC
G	0.51	1.14	0.020	0.045
H	0.20	0.31	0.008	0.012
J	3.18	0.30	0.125	0.160
K	7.37	7.87	0.290	0.310
L	—	15°	—	15°
M	0.51	1.02	0.020	0.040
N	—	—	—	—

NOTES:

- LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION
- PKG. INDEX: NOTCH IN LEAD
- DIM "L" TO CENTER OF LEADS WHEN FORMED PARALLEL



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	20.70	21.34	0.815	0.840
B	6.10	6.60	0.240	0.260
C	4.06	4.57	0.160	0.180
D	0.38	0.51	0.015	0.020
E	1.02	1.52	0.040	0.060
F	2.54	BSC	0.100	BSC
G	1.32	1.83	0.052	0.072
H	0.20	0.30	0.008	0.012
J	2.92	3.43	0.115	0.135
K	7.37	7.87	0.290	0.310
L	—	10°	—	10°
M	0.51	1.02	0.020	0.040
N	0.13	0.38	0.005	0.015
P	0.51	0.76	0.020	0.030
Q	—	—	—	—

NOTES:

- LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION.
- DIMENSION "L" TO CENTER OF LEADS WHEN FORMED PARALLEL





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MCM6810A

(0 to 70°C; L or P Suffix)

MCM6810AC

(-40 to 85°C; L Suffix only)

128 X 8-BIT STATIC RANDOM ACCESS MEMORY

The MCM6810 is a byte-organized memory designed for use in bus-organized systems. It is fabricated with N-channel silicon-gate technology. For ease of use, the device operates from a single power supply, has compatibility with TTL and DTL, and needs no clocks or refreshing because of static operation.

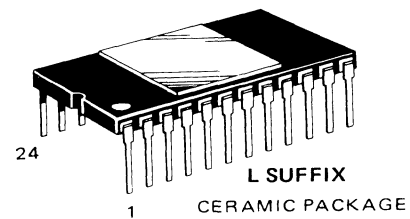
The memory is compatible with the M6800 Microcomputer Family, providing random storage in byte increments. Memory expansion is provided through multiple Chip Select inputs.

- Organized as 128 Bytes of 8 Bits
- Static Operation
- Bi-Directional Three-State Data Input/Output
- Six Chip Select Inputs (Four Active Low, Two Active High)
- Single 5-Volt Power Supply
- TTL Compatible
- Maximum Access Time = 350 ns – MCM6810AL1
450 ns – MCM6810AL

MOS

(N-CHANNEL, SILICON-GATE)

128 X 8-BIT STATIC RANDOM ACCESS MEMORY



NOT SHOWN: P SUFFIX
PLASTIC PACKAGE
CASE 709

PIN ASSIGNMENT

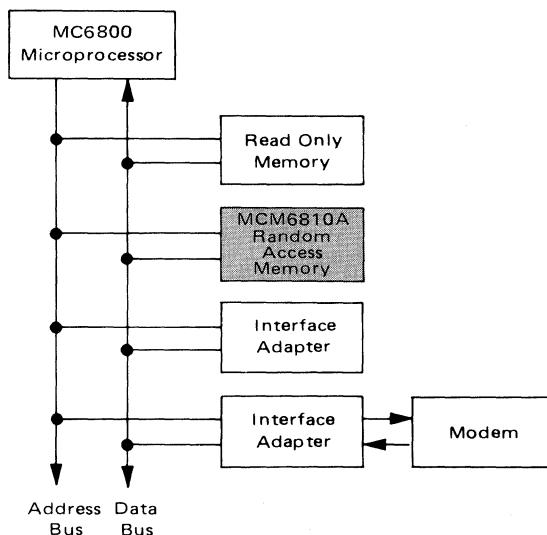
1	Gnd	0	V _{CC}	24
2	D0		A0	23
3	D1		A1	22
4	D2		A2	21
5	D3		A3	20
6	D4		A4	19
7	D5		A5	18
8	D6		A6	17
9	D7		R/W	16
10	CS0		CS5	15
11	CS1		CS4	14
12	CS2		CS3	13

ABSOLUTE MAXIMUM RATINGS (See Note 1)

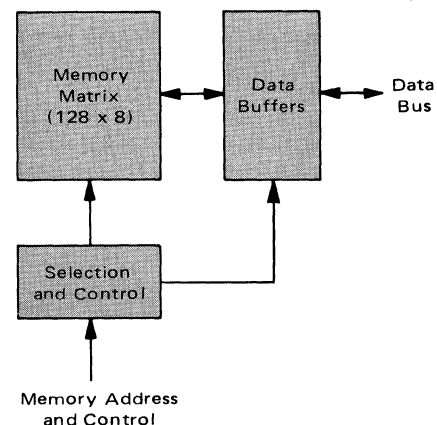
Rating	Symbol	Value	Unit
Supply Voltage	V _{CC}	-0.3 to +7.0	Vdc
Input Voltage	V _{in}	-0.3 to +7.0	Vdc
Operating Temperature Range	T _A	0 to +70	°C
Storage Temperature Range	T _{stg}	-65 to +150	°C

NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

M6800 MICROCOMPUTER FAMILY BLOCK DIAGRAM



MCM6810A RANDOM ACCESS MEMORY BLOCK DIAGRAM



DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{CC}	4.75	5.0	5.25	Vdc
Input High Voltage	V_{IH}	2.0	—	5.25	Vdc
Input Low Voltage	V_{IL}	-0.3	—	0.8	Vdc

DC CHARACTERISTICS

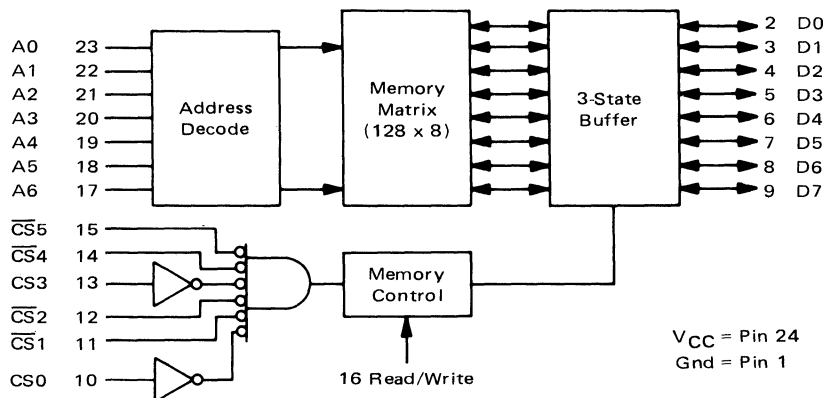
Characteristic	Symbol	Min	Typ	Max	Unit
Input Current (A_n , R/W, $\overline{CS}_n$, $\overline{CS}_n$) ($V_{in} = 0$ to 5.25 V)	I_{in}	—	—	2.5	μA_{dc}
Output High Voltage ($I_{OH} = -205 \mu A$)	V_{OH}	2.4	—	—	Vdc
Output Low Voltage ($I_{OL} = 1.6 mA$)	V_{OL}	—	—	0.4	Vdc
Output Leakage Current (Three-State) ($CS = 0.8 V$ or $\overline{CS} = 2.0 V$, $V_{out} = 0.4 V$ to 2.4 V)	I_{LO}	—	—	10	μA_{dc}
Supply Current ($V_{CC} = 5.25 V$, all other pins grounded, $T_A = 0^\circ C$) MCM6810AL MCM6810AL1	I_{CC}	—	—	70 80	mA_{dc}

CAPACITANCE ($f = 1.0 MHz$, $T_A = 25^\circ C$, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Max	Unit
Input Capacitance	C_{in}	7.5	pF
Output Capacitance	C_{out}	12.5	pF

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

BLOCK DIAGRAM



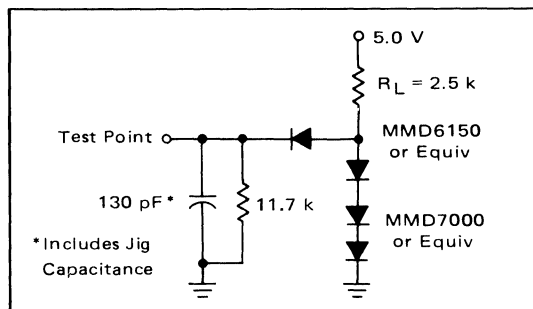
AC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature unless otherwise noted.)

AC TEST CONDITIONS

Condition	Value
Input Pulse Levels	0.8 V to 2.0 V
Input Rise and Fall Times	20 ns
Output Load	See Figure 1

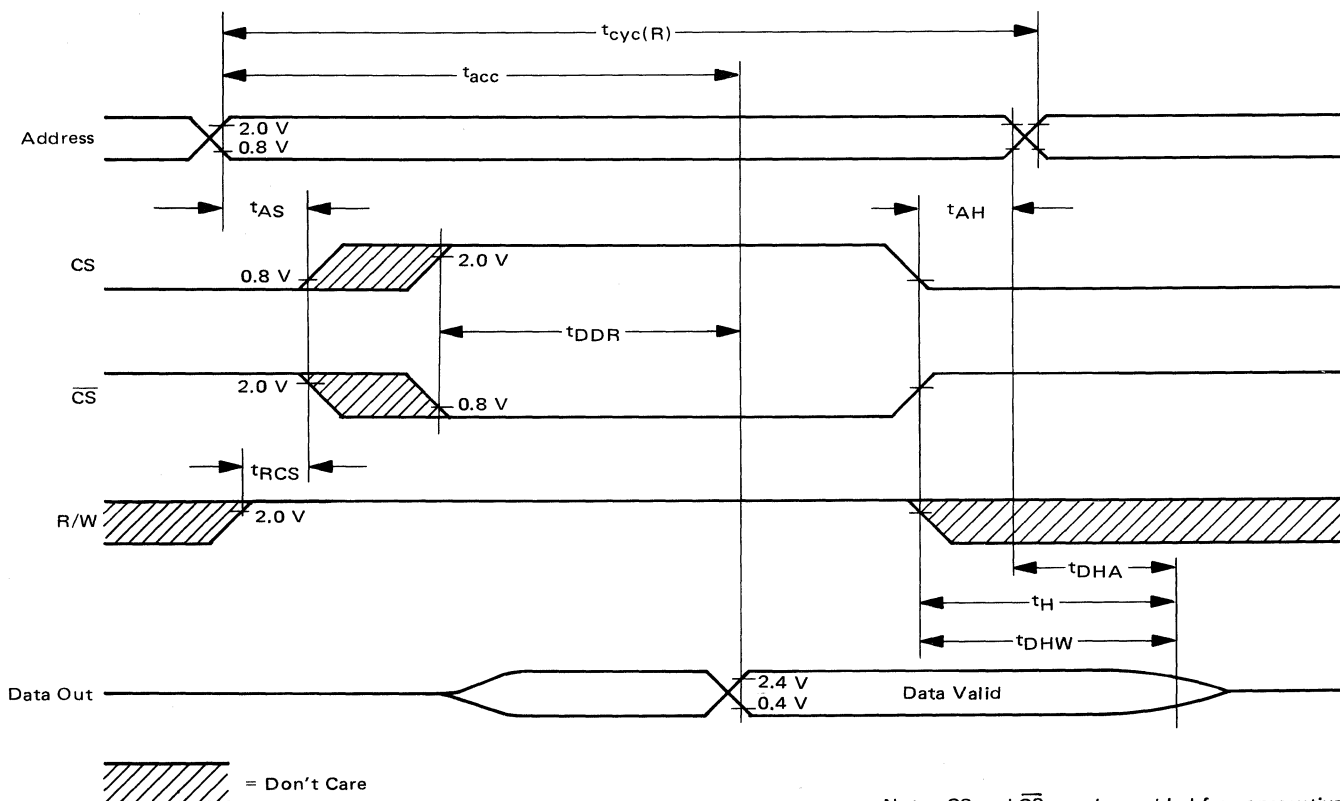
FIGURE 1 – AC TEST LOAD



READ CYCLE

Characteristic	Symbol	MCM6810AL		MCM6810AL1		Unit
		Min	Max	Min	Max	
Read Cycle Time	$t_{cyc(R)}$	450	—	350	—	ns
Access Time	t_{acc}	—	450	—	350	ns
Address Setup Time	t_{AS}	20	—	20	—	ns
Address Hold Time	t_{AH}	0	—	0	—	ns
Data Delay Time (Read)	t_{DDR}	—	230	—	180	ns
Read to Select Delay Time	t_{RCS}	0	—	0	—	ns
Data Hold from Address	t_{DHA}	10	—	10	—	ns
Output Hold Time	t_H	10	—	10	—	ns
Data Hold from Write	t_{DHW}	10	80	10	60	ns

READ CYCLE TIMING



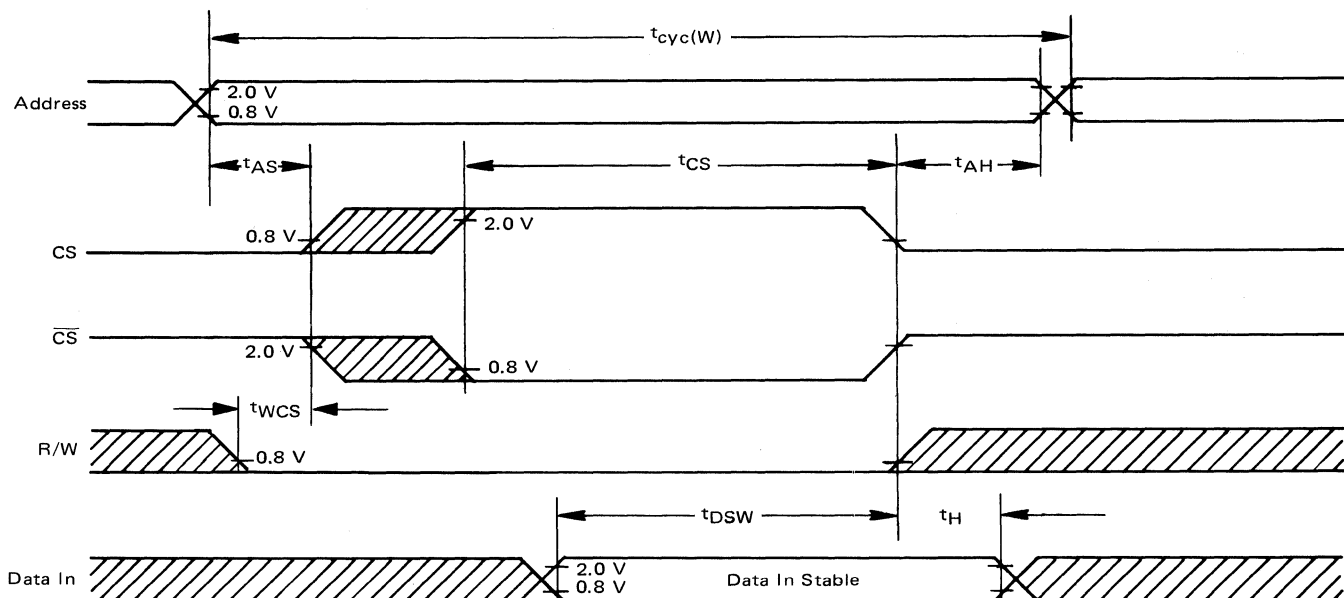
Note: CS and $\overline{CS}$ can be enabled for consecutive read cycles provided R/W remains at V_{IH} .



WRITE CYCLE

Characteristic	Symbol	MCM6810AL		MCM6810AL1		Unit
		Min	Max	Min	Max	
Write Cycle Time	$t_{cyc(W)}$	450	—	350	—	ns
Address Setup Time	t_{AS}	20	—	20	—	ns
Address Hold Time	t_{AH}	0	—	0	—	ns
Chip Select Pulse Width	t_{CS}	300	—	250	—	ns
Write to Chip Select Delay Time	t_{WCS}	0	—	0	—	ns
Data Setup Time (Write)	t_{DSW}	190	—	150	—	ns
Input Hold Time	t_H	10	—	10	—	ns

WRITE CYCLE TIMING



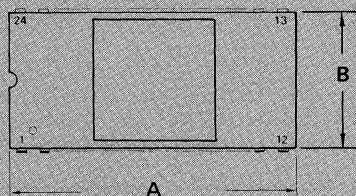
= Don't Care

Note: CS and $\overline{CS}$ can be enabled for consecutive write cycles provided R/W is strobed to V_{IH} before or coincident with the Address change, and remains high for time t_{AS} .

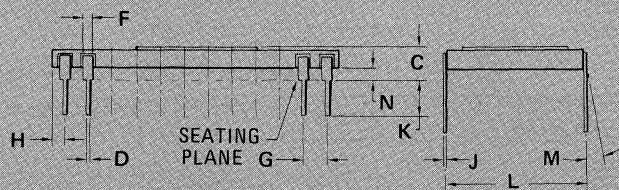
PACKAGE DIMENSIONS

CASE 716-02
(CERAMIC)

See Page 165 for
Plastic Package dimensions.



NOTE:
1. LEADS TRUE POSITIONED WITHIN
0.25mm (0.010) DIA (AT SEATING
PLANE) AT MAXIMUM MATERIAL
CONDITION.



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	29.97	30.99	1.180	1.220
B	14.88	15.62	0.585	0.615
C	3.05	4.19	0.120	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.30	0.008	0.012
K	2.54	4.19	0.100	0.165
L	14.88	15.37	0.585	0.605
M	—	10 ⁰	—	10 ⁰
N	0.51	1.52	0.020	0.060





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Advance Information

1024 X 8-BIT READ ONLY MEMORY

The MCM6830A is a mask-programmable byte-organized memory designed for use in bus-organized systems. It is fabricated with N-channel silicon-gate technology. For ease of use, the device operates from a single power supply, has compatibility with TTL and DTL, and needs no clocks or refreshing because of static operation.

The memory is compatible with the M6800 Microcomputer Family, providing read only storage in byte increments. Memory expansion is provided through multiple Chip Select inputs. The active level of the Chip Select inputs and the memory content are defined by the customer.

- Organized as 1024 Bytes of 8 Bits
- Static Operation
- Three-State Data Output
- Four Chip Select Inputs (Programmable)
- Single 5-Volt Power Supply
- TTL Compatible
- Maximum Access Time = 500 ns

ABSOLUTE MAXIMUM RATINGS (See Note 1)

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	Vdc
Input Voltage	V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	-65 to +150	°C

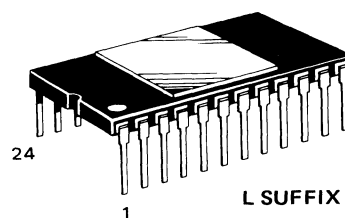
NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

MCM6830A

MOS

(N-CHANNEL, SILICON-GATE)

1024 X 8-BIT
READ ONLY MEMORY

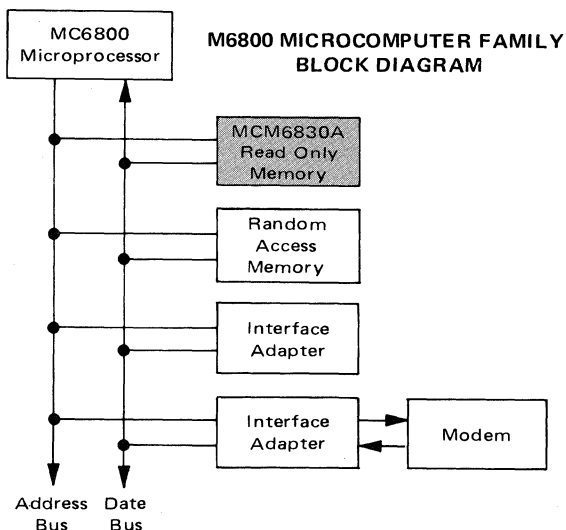


L SUFFIX
CERAMIC PACKAGE
CASE 716

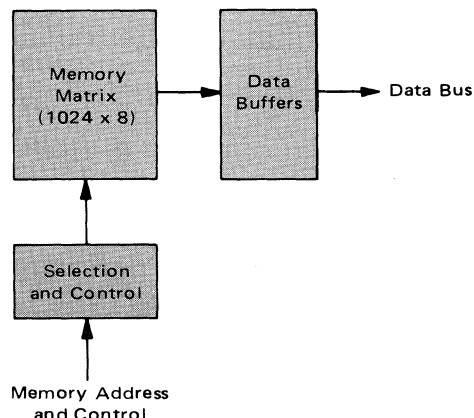
P SUFFIX
NOT SHOWN: PLASTIC PACKAGE
CASE 709

PIN ASSIGNMENT

1	Gnd	A0	24
2	D0	A1	23
3	D1	A2	22
4	D2	A3	21
5	D3	A4	20
6	D4	A5	19
7	D5	A6	18
8	D6	A7	17
9	D7	A8	16
10	CS0	A9	15
11	CS1	CS3	14
12	V_{CC}	CS2	13



MCM6830A READ ONLY MEMORY BLOCK DIAGRAM



DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{CC}	4.75	5.0	5.25	Vdc
Input High Voltage	V_{IH}	2.0	—	5.25	Vdc
Input Low Voltage	V_{IL}	-0.3	—	0.8	Vdc

DC CHARACTERISTICS

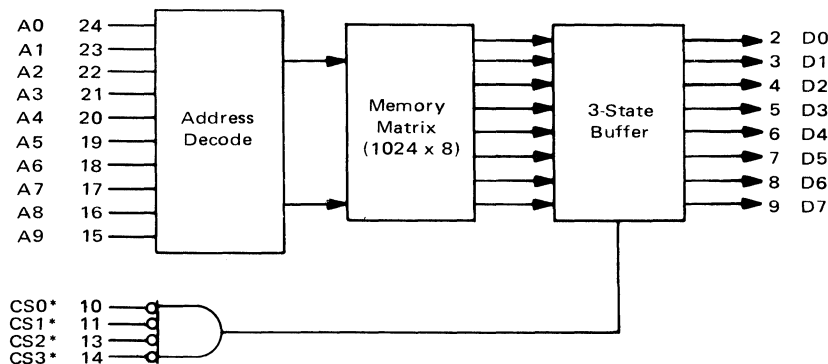
Characteristic	Symbol	Min	Typ	Max	Unit
Input Current ($V_{in} = 0$ to 5.25 V)	I_{in}	—	—	2.5	μA_{dc}
Output High Voltage ($I_{OH} = -205 \mu A$)	V_{OH}	2.4	—	—	Vdc
Output Low Voltage ($I_{OL} = 1.6$ mA)	V_{OL}	—	—	0.4	Vdc
Output Leakage Current (Three-State) ($CS = 0.8$ V or $\overline{CS} = 2.0$ V, $V_{out} = 0.4$ V to 2.4 V)	I_{LO}	—	—	10	μA_{dc}
Supply Current ($V_{CC} = 5.25$ V, $T_A = 0^\circ C$)	I_{CC}	—	—	130	mAdc

CAPACITANCE ($f = 1.0$ MHz, $T_A = 25^\circ C$, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Max	Unit
Input Capacitance	C_{in}	7.5	pF
Output Capacitance	C_{out}	12.5	pF

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

BLOCK DIAGRAM



* Active level defined by the customer.

V_{CC} = Pin 12
Gnd = Pin 1



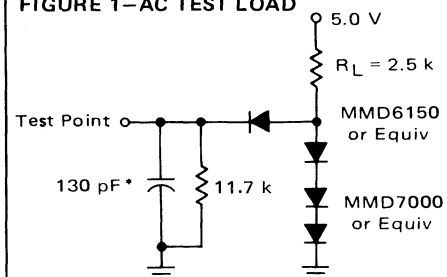
AC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature unless otherwise noted.)

(All timing with $t_r = t_f = 20$ ns, Load of Figure 1)

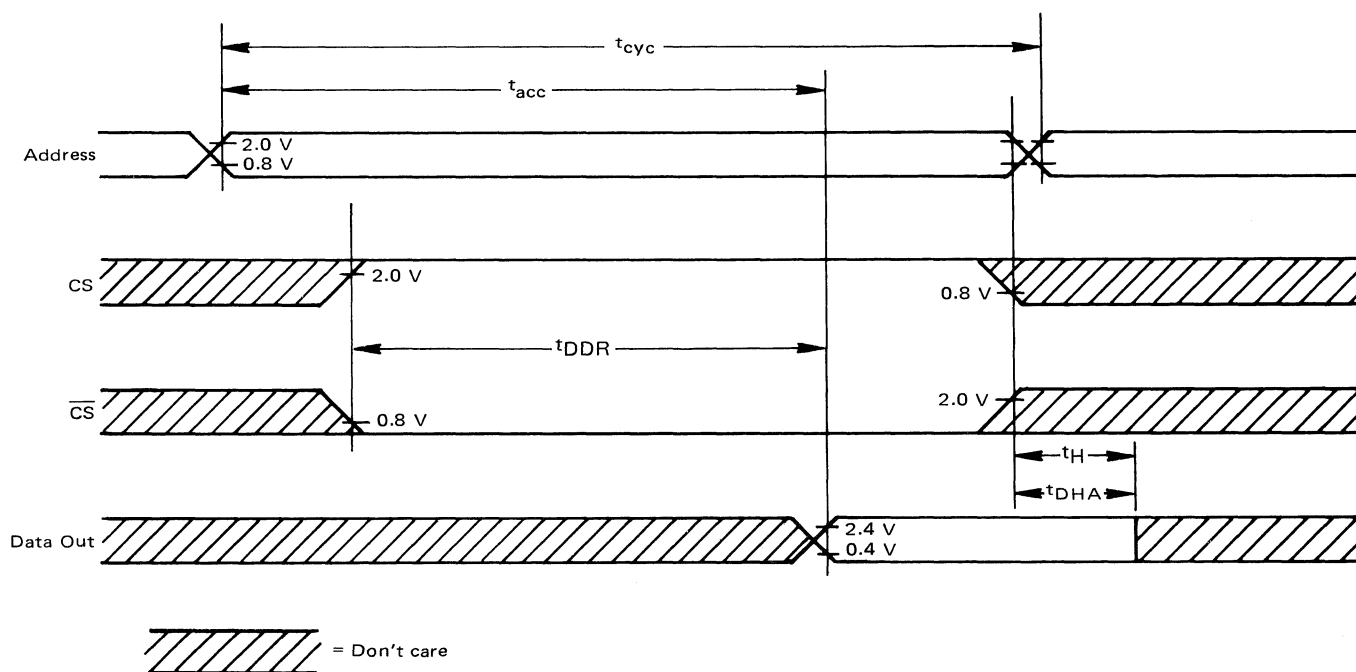
Characteristic	Symbol	Min	Max	Unit
Cycle Time	t_{cyc}	500	—	ns
Access Time	t_{acc}	—	500	ns
Data Delay Time (Read)	t_{DDR}	—	300	ns
Data Hold from Address	t_{DHA}	10	—	ns
Data Hold from Deselection	t_H	10	150	ns

FIGURE 1—AC TEST LOAD



*Includes Jig Capacitance

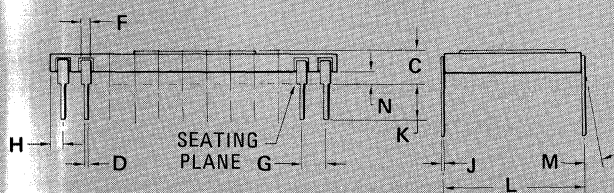
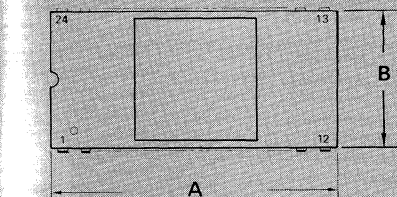
TIMING DIAGRAM



PACKAGE DIMENSIONS

CASE 716-02
(CERAMIC)

NOTE:

1. LEADS TRUE POSITIONED WITHIN
0.25mm (0.010) DIA (AT SEATING
PLANE) AT MAXIMUM MATERIAL
CONDITION.See Page 165 for
Plastic Package dimensions.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	29.97	30.99	1.180	1.220
B	14.88	15.62	0.585	0.615
C	3.05	4.19	0.120	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.30	0.008	0.012
K	2.54	4.19	0.100	0.165
L	14.88	15.37	0.585	0.605
M	—	10 ⁰	—	10 ⁰
N	0.51	1.52	0.020	0.060



CUSTOM PROGRAMMING

By the programming of a single photomask for the MCM6830A, the customer may specify the content of the memory and the method of enabling the outputs.

Information on the general options of the MCM6830A should be submitted on an Organizational Data form such as that shown in Figure 3.

Information for custom memory content may be sent to Motorola in one of two forms (shown in order of preference):

1. Paper tape output of the Motorola M6800 Software.
2. Hexadecimal coding using IBM Punch Cards.

PAPER TAPE

Included in the software packages developed for the M6800 Micromputer Family is the ability to produce a paper tape output for computerized mask generation. The assembler directives are used to control allocation of memory, to assign values for stored data, and for controlling the assembly process. The paper tape must specify the full 1024 bytes.

Note: Motorola can accept magnetic tape and truth table formats. For further information, contact your local Motorola sales representative.

FIGURE 2 – BINARY TO HEXADECIMAL CONVERSION

Binary Data				Hexadecimal Character
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A
1	0	1	1	B
1	1	0	0	C
1	1	0	1	D
1	1	1	0	E
1	1	1	1	F

IBM PUNCH CARDS

The hexadecimal equivalent (from Figure 2) may be placed on 80 column IBM punch cards as follows:

Step	Column	
1	12	Byte "0" Hexadecimal equivalent for outputs D7 thru D4 (D7 = M.S.B.)
2	13	Byte "0" Hexadecimal equivalent for outputs D3 thru D0 (D3 = M.S.B.)
3	14-75	Alternate steps 1 and 2 for consecutive bytes.
4	77-78	Card number (starting 01)
5	79-80	Total number of cards (32)

FIGURE 3 – FORMAT FOR PROGRAMMING GENERAL OPTIONS

ORGANIZATIONAL DATA MCM6830A MOS READ ONLY MEMORY			
Customer:			
Company			
Part No.			
Originator			
Phone No.			
Enable Options:			
	1	0	
CS0	☐	☐	1 is most positive 0 is most negative
CS1	☐	☐	
CS2	☐	☐	
CS3	☐	☐	

Motorola Use Only:

Quote:

Part No.:

Specif. No.:





MOTOROLA
Semiconductors

BOX 20912, PHOENIX, ARIZONA 85036

Advance Information

2048 x 8-BIT READ ONLY MEMORY

The MCM6832 is a mask-programmable byte-organized memory designed for use in bus-organized systems. It is fabricated with N-channel metal-gate technology. For ease of use, the device is compatible with TTL and DTL, and needs no clocks or refreshing because of static operation.

The memory is compatible with the M6800 Microcomputer Family, providing read only storage in byte increments. Memory expansion is provided through a Chip Select input. The active level of the Chip Select input and the memory content are defined by the customer.

- Organized as 2048 Bytes of 8 Bits
- Static Operation
- Three-State Data Output
- Programmable Chip Select
- TTL Compatible
- Maximum Access Time = 500 ns

ABSOLUTE MAXIMUM RATINGS¹ (Referenced to V_{SS})

Rating	Symbol	Value	Unit
Supply Voltages	V _{DD}	-0.3 to +15	Vdc
	V _{CC}	-0.3 to +6.0	
	V _{BB}	-10 to +0.3	
Address/Control Input Voltage	V _{in}	-0.3 to +15	Vdc
Operating Temperature Range	T _A	0 to +70	°C
Storage Temperature Range	T _{stg}	-55 to +125	°C

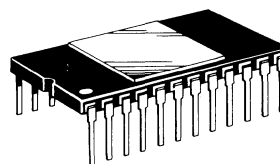
Note 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

MCM6832

MOS

(N-CHANNEL, LOW THRESHOLD)

2048 x 8-BIT
READ ONLY MEMORY

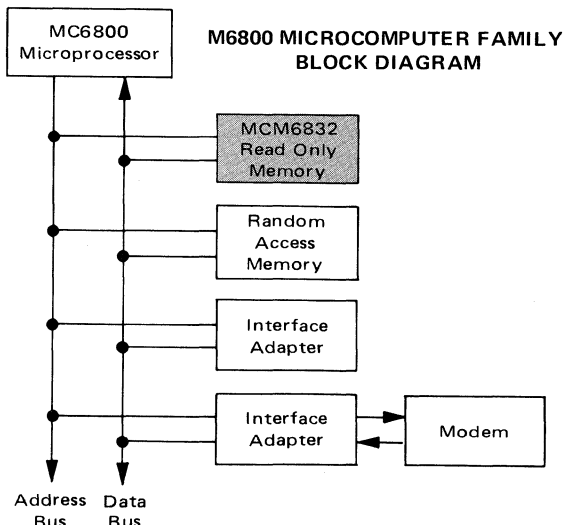


L SUFFIX
CERAMIC PACKAGE
CASE 716

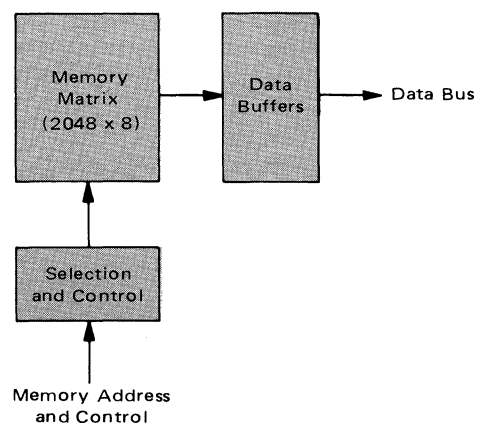
NOT SHOWN: **P SUFFIX**
PLASTIC PACKAGE
CASE 709

PIN ASSIGNMENT

1	V _{BB}	V _{CC}	24
2	A10	V _{DD}	23
3	CS	A9	22
4	D0	A8	21
5	D1	A7	20
6	D2	D4	19
7	D3	D5	18
8	A0	D6	17
9	A1	D7	16
10	A2	A6	15
11	A3	A5	14
12	V _{SS}	A4	13



MCM6832 READ ONLY MEMORY BLOCK DIAGRAM



DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS (Referenced to V_{SS} = Ground)

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage	V_{DD}	11.4	12	12.6	Vdc
	V_{CC}	4.75	5.0	5.25	Vdc
	V_{BB}	-5.25	-5.0	-4.75	Vdc
Input High Voltage (A_n , CS)	V_{IH}	3.0	—	V_{CC}	Vdc
Input Low Voltage (A_n , CS)	V_{IL}	-0.3	—	0.8	Vdc

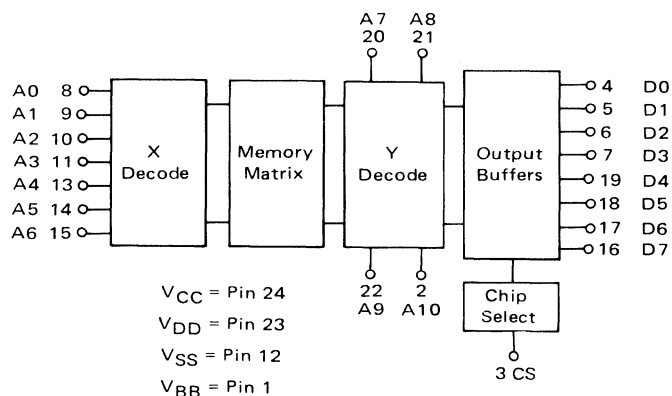
DC CHARACTERISTICS

Characteristic	Symbol	Min	Typ	Max	Unit
Input Leakage Current (A_n , CS) ($V_{in} = 0$ to 5.25 V)	I_{in}	—	—	10	μA_{dc}
Output Leakage Current (Three-State) ($V_O = 0.4$ V to -2.4 V, CS = 0.4 V or CS = 2.4 V)	I_{LO}	—	—	10	μA_{dc}
Output High Voltage ($I_{OH} = -100 \mu A$)	V_{OH}	3.7	—	V_{CC}	Vdc
Output Low Voltage ($I_{OL} = 1.6$ mA)	V_{OL}	0	—	0.4	Vdc
Supply Current (Chip Deselected or Selected)	I_{DD}	—	—	25	mA_{dc}
	I_{CC}	—	—	45	mA_{dc}
	I_{BB}	—	—	500	μA_{dc}

CAPACITANCE (Periodically Sampled Rather Than 100% Tested.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input Capacitance ($f = 1$ MHz)	C_{in}	—	5.0	7.5	pF
Output Capacitance ($f = 1$ MHz)	C_{out}	—	5.0	10	pF

BLOCK DIAGRAM



AC CHARACTERISTICS

(Full operating voltage and temperature unless otherwise noted. All timing with $t_r = t_f \leq 20$ ns;
Load = 1 TTL Gate (MC7400 Series) biased to draw 1.6 mA; $C_L = 130$ pF.)

Characteristic	Symbol	Min	Typ*	Max	Unit
Address Access Time	t_{acc}	—	320*	500	ns
Output Select Time	t_{OS}	—	175*	300	ns
Output Deselect Time	t_{OD}	30	100*	150	ns

*Typical values measured at 25°C and nominal supply voltages.

FIGURE 1 – AC TEST LOAD

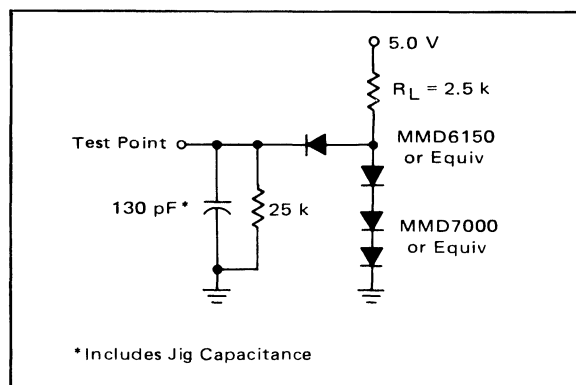
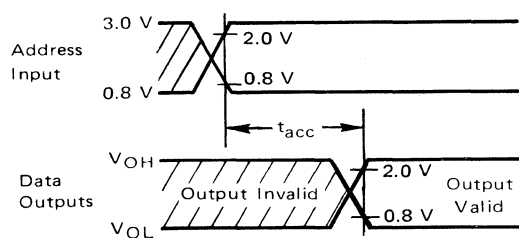
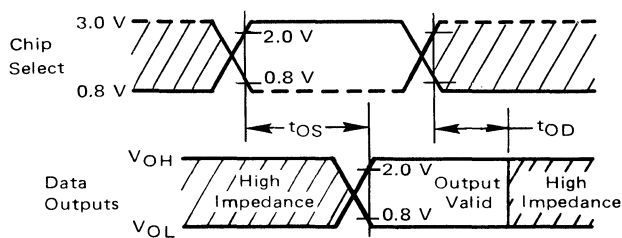


FIGURE 2 – TIMING DIAGRAM

A. ADDRESS ACCESS TIMING DIAGRAM
(Chip Selected)B. CHIP SELECT TIMING DIAGRAM
(Addresses Established)

PACKAGE DIMENSIONS

CASE 716-02
(CERAMIC)

NOTE:

- LEADS TRUE POSITIONED WITHIN 0.25mm (0.010) DIA (AT SEATING PLANE) AT MAXIMUM MATERIAL CONDITION.

SEE PAGE 165 FOR
PLASTIC PACKAGE
DIMENSIONS.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	29.97	30.99	1.180	1.220
B	14.88	15.62	0.585	0.615
C	3.05	4.19	0.120	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.30	0.008	0.012
K	2.54	4.19	0.100	0.165
L	14.88	15.37	0.585	0.605
M	—	10 ⁰	—	10 ⁰
N	0.51	1.52	0.020	0.060



CUSTOM PROGRAMMING

By the programming of a single photomask for the MCM6832, the customer may specify the content of the memory and the method of enabling the outputs.

Information on the general options of the MCM6832 should be submitted on an Organizational Data form such as that shown in Figure 4.

Information for custom memory content may be sent to Motorola in one of two forms (shown in order of preference):

1. Paper tape output of the Motorola M6800 Software.
2. Hexadecimal coding using IBM Punch Cards.

PAPER TAPE

Included in the software packages developed for the M6800 Microcomputer Family is the ability to produce a paper tape output for computerized mask generation. The assembler directives are used to control allocation of memory, to assign values for stored data, and for controlling the assembly process. The paper tape must specify the full 2048 bytes.

Note: Motorola can accept magnetic tape and truth table formats. For further information, contact your local Motorola sales representative.

FIGURE 3 — BINARY TO HEXADECIMAL CONVERSION

MSB D7 D3	D6 D2	D5 D1	LSB D4 D0	Hexadecimal Character
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A
1	0	1	1	B
1	1	0	0	C
1	1	0	1	D
1	1	1	0	E
1	1	1	1	F

0 = VOL
1 = VOH

IBM PUNCH CARDS

The hexadecimal equivalent (from Figure 3) may be placed on 80 column IBM punch cards as follows:

Step	Column	
1	12	Byte "0" Hexadecimal equivalent for outputs D7 thru D4 (D7 = M.S.B.)
2	13	Byte "0" Hexadecimal equivalent for outputs D3 thru D0 (D3 = M.S.B.)
3	14-75	Alternate steps 1 and 2 for consecutive bytes.
4	77-78	Card number (starting 01)
5	79-80	Total number of cards (64)

FIGURE 4 — FORMAT FOR PROGRAMMING GENERAL OPTIONS

ORGANIZATIONAL DATA MCM6832 MOS READ ONLY MEMORY	
Customer:	
Company _____	Motorola Use Only: Quote: _____ Part No.: _____ Specif. No.: _____
Part No. _____	
Originator _____	
Phone No. _____	
True Chip Select Options: I. 1 ☐ II. 0 ☐ 1 is most positive 0 is most negative	





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BOX 20912 • PHOENIX, ARIZONA 85036

MCM68308

Advance Information

1024 X 8-BIT READ ONLY MEMORY

The MCM68308 is a mask-programmable byte-organized memory designed for use in bus-organized systems. It is fabricated with N-channel silicon-gate technology. For ease of use, the device operates from a single power supply, has compatibility with TTL and DTL, and needs no clocks or refreshing because of static operation.

The memory is compatible with the M6800 Microcomputer Family, providing read only storage in byte increments. Memory expansion is provided through multiple Chip Select inputs. The active level of the Chip Select inputs and the memory content are defined by the customer.

- Static Operation
- Three-State Data Output
- Mask-Programmable Chip Selects for Simplified Memory Expansion
- Single 5-Volt Power Supply
- TTL Compatible
- Maximum Access Time = 500 ns
- Pin Compatible with MCM68708 Alterable ROM
- Pin Compatible with MCM68317 2048 x 8-Bit ROM

ABSOLUTE MAXIMUM RATINGS (See Note 1)

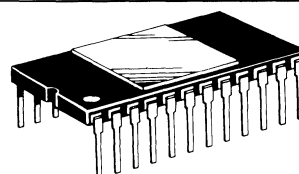
Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	Vdc
Input Voltage	V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	-65 to +150	°C

NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

MOS

(N-CHANNEL, SILICON-GATE)

1024 X 8-BIT READ ONLY MEMORY

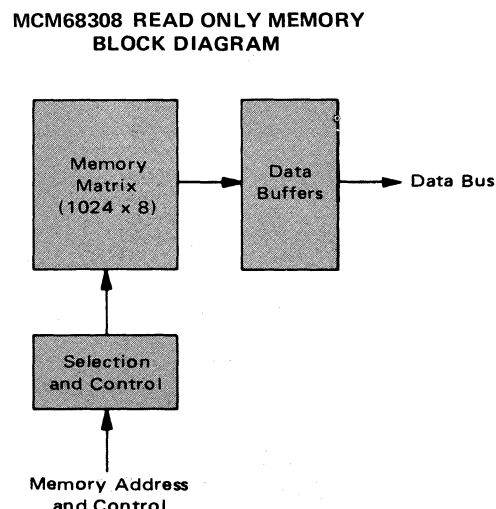
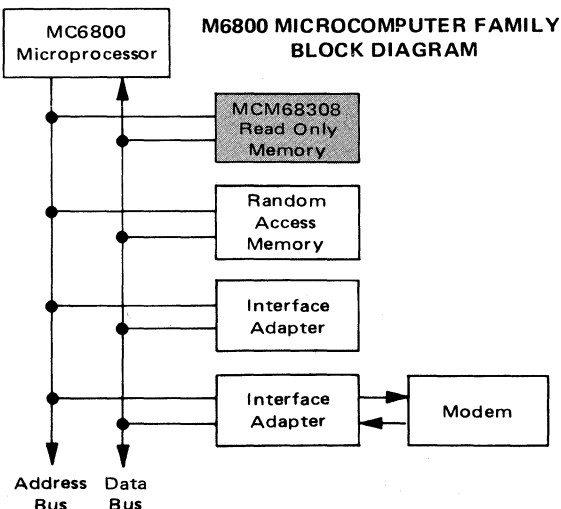


L SUFFIX
CERAMIC PACKAGE
CASE 716

NOT SHOWN: **P SUFFIX**
PLASTIC PACKAGE
CASE 709

PIN ASSIGNMENT

1	A7	V_{CC}	24
2	A6	A8	23
3	A5	A9	22
4	A4	NC	21
5	A3	CS1	20
6	A2	NC	19
7	A1	CS2	18
8	A0	D7	17
9	D0	D6	16
10	D1	D5	15
11	D2	D4	14
12	Gnd	D3	13



DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{CC}	4.75	5.0	5.25	Vdc
Input High Voltage	V_{IH}	2.0	—	5.25	Vdc
Input Low Voltage	V_{IL}	-0.3	—	0.8	Vdc

DC CHARACTERISTICS

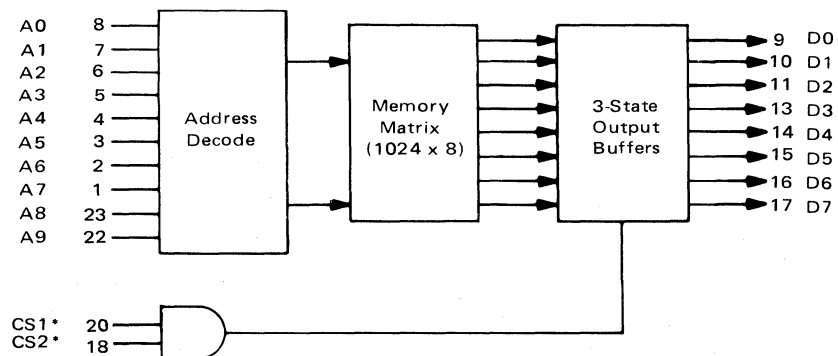
Characteristic	Symbol	Min	Typ	Max	Unit
Input Current ($V_{in} = 0$ to 5.25 V)	I_{in}	—	—	2.5	μA_{dc}
Output High Voltage ($I_{OH} = -205 \mu A$)	V_{OH}	2.4	—	—	Vdc
Output Low Voltage ($I_{OL} = 1.6$ mA)	V_{OL}	—	—	0.4	Vdc
Output Leakage Current (Three-State) ($CS = 0.8$ V or $\overline{CS} = 2.0$ V, $V_{out} = 0.4$ V to 2.4 V)	I_{LO}	—	—	10	μA_{dc}
Supply Current ($V_{CC} = 5.25$ V, $T_A = 0^\circ C$)	I_{CC}	—	—	130	mA_{dc}

CAPACITANCE ($f = 1.0$ MHz, $T_A = 25^\circ C$, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Max	Unit
Input Capacitance	C_{in}	7.5	pF
Output Capacitance	C_{out}	12.5	pF

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

BLOCK DIAGRAM



*Active level defined by the customer.

$V_{CC} = \text{Pin } 24$
Gnd = Pin 12

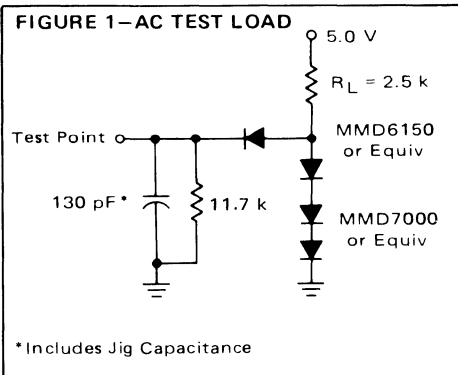


AC OPERATING CONDITIONS AND CHARACTERISTICS

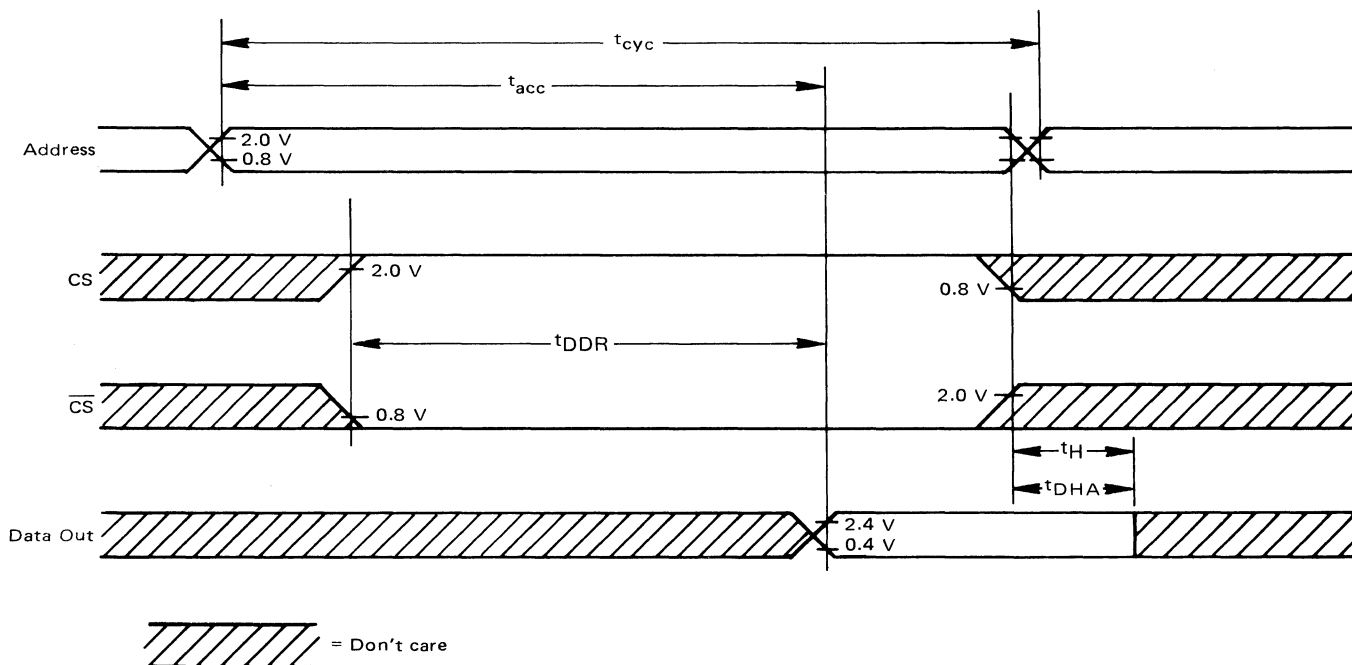
(Full operating voltage and temperature unless otherwise noted.)

(All timing with $t_r = t_f = 20$ ns, Load of Figure 1)

Characteristic	Symbol	Min	Max	Unit
Cycle Time	t_{cyc}	500	—	ns
Access Time	t_{acc}	—	500	ns
Data Delay Time (Read)	t_{DDR}	—	300	ns
Data Hold from Address	t_{DHA}	10	—	ns
Data Hold from Deselection	t_H	10	150	ns



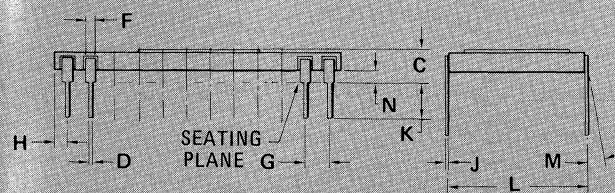
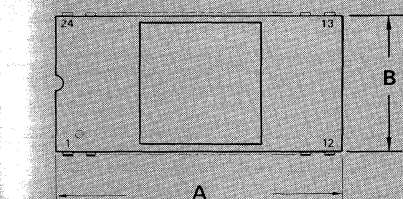
TIMING DIAGRAM



PACKAGE DIMENSIONS

CASE 716-02
(CERAMIC)

NOTE:

1. LEADS TRUE POSITIONED WITHIN
0.25mm (0.010) DIA (AT SEATING
PLANE) AT MAXIMUM MATERIAL
CONDITION.SEE PAGE 165 FOR
PLASTIC PACKAGE
DIMENSIONS.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	29.97	30.99	1.180	1.220
B	14.88	15.62	0.585	0.615
C	3.05	4.19	0.120	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.30	0.008	0.012
K	2.54	4.19	0.100	0.165
L	14.88	15.37	0.585	0.605
M	—	10 ⁰	—	10 ⁰
N	0.51	1.52	0.020	0.060



CUSTOM PROGRAMMING

By the programming of a single photomask for the MCM68308, the customer may specify the content of the memory and the method of enabling the outputs.

Information on the general options of the MCM68308 should be submitted on an Organizational Data form such as that shown in Figure 3.

Information for custom memory content may be sent to Motorola in one of two forms (shown in order of preference):

1. Paper tape output of the Motorola M6800 Software.
2. Hexadecimal coding using IBM Punch Cards.

PAPER TAPE

Included in the software packages developed for the M6800 Microcomputer Family is the ability to produce a paper tape output for computerized mask generation. The assembler directives are used to control allocation of memory, to assign values for stored data, and for controlling the assembly process. The paper tape must specify the full 1024 bytes.

Note: Motorola can accept magnetic tape and truth table formats. For further information, contact your local Motorola sales representative.

FIGURE 2 – BINARY TO HEXADECIMAL CONVERSION

Binary Data				Hexadecimal Character
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A
1	0	1	1	B
1	1	0	0	C
1	1	0	1	D
1	1	1	0	E
1	1	1	1	F

IBM PUNCH CARDS

The hexadecimal equivalent (from Figure 2) may be placed on 80 column IBM punch cards as follows:

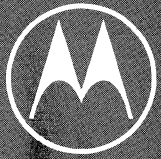
Step Column

- | | | |
|---|-------|--|
| 1 | 12 | Byte "0" Hexadecimal equivalent for outputs D7 thru D4 (D7 = M.S.B.) |
| 2 | 13 | Byte "0" Hexadecimal equivalent for outputs D3 thru D0 (D3 = M.S.B.) |
| 3 | 14-75 | Alternate steps 1 and 2 for consecutive bytes. |
| 4 | 77-78 | Card number (starting 01) |
| 5 | 79-80 | Total number of cards (32) |

FIGURE 3 – FORMAT FOR PROGRAMMING GENERAL OPTIONS

ORGANIZATIONAL DATA MCM68308 MOS READ ONLY MEMORY			
Customer: Company _____ Part No. _____ Originator _____ Phone No. _____			
Enable Options:		Motorola Use Only: Quote: _____ Part No.: _____ Specif. No.: _____	
CS1 CS2	1 0 ☐ ☐ ☐ ☐	1 is most positive 0 is most negative	





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Semiconductors

3501 ED BLUESTEIN BLVD., AUSTIN, TEXAS 78721

MCM68316E
MCM68316E1

Advance Information

2048 X 8-BIT READ ONLY MEMORY

The MCM68316E is a mask-programmable byte-organized memory designed for use in bus-organized systems. It is fabricated with N-channel silicon-gate technology. For ease of use, the device operates from a single power supply, has compatibility with TTL and DTL, and needs no clocks or refreshing because of static operation.

The memory is compatible with the M6800 Microcomputer Family, providing read only storage in byte increments. Memory expansion is provided through multiple Chip Select inputs. The active level of the Chip Select inputs and the memory content are defined by the user.

- Static Operation
- Two Temperature Range Options
- Three-State Data Output
- Mask-Programmable Chip Selects for Simplified Memory Expansion
- Single 5-Volt Power Supply
- TTL Compatible
- Maximum Access Time = 500 ns
- Directly Compatible with 2316E
- Pin Compatible with 2708 PROM

ABSOLUTE MAXIMUM RATINGS (See Note 1)

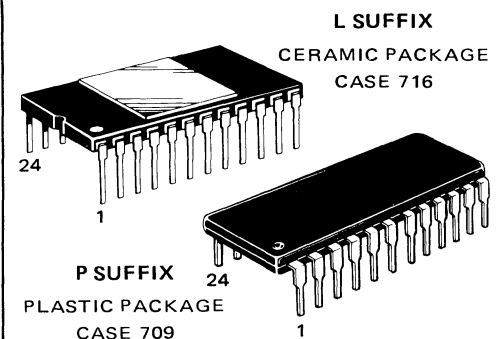
Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	Vdc
Input Voltage	V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range — MCM68316E1 MCM68316E	T_A	-40 to +85 0 to +70	°C
Storage Temperature Range	T_{stg}	-65 to +150	°C

NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

MOS

(N-CHANNEL, SILICON-GATE)

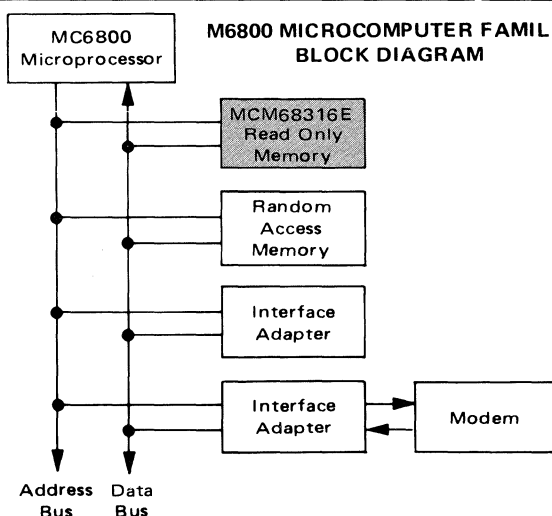
2048 X 8-BIT READ ONLY MEMORY



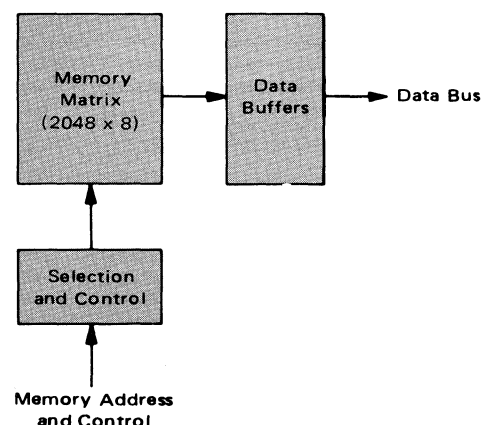
PIN ASSIGNMENT

1	A7	O	V_{CC}	24
2	A6		A8	23
3	A5		A9	22
4	A4	CS3/CS3		21
5	A3	CS1/CS1		20
6	A2		A10	19
7	A1	CS2/CS2		18
8	A0		D7	17
9	D0		D6	16
10	D1		D5	15
11	D2		D4	14
12	Gnd		D3	13

M6800 MICROCOMPUTER FAMILY BLOCK DIAGRAM



MCM68316E READ ONLY MEMORY BLOCK DIAGRAM



DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{CC}	4.75	5.0	5.25	Vdc
Input High Voltage	V_{IH}	2.0	—	5.25	Vdc
Input Low Voltage	V_{IL}	-0.3	—	0.8	Vdc

DC CHARACTERISTICS

Characteristic	Symbol	Min	Typ	Max	Unit
Input Current ($V_{in} = 0$ to 5.25 V)	I_{in}	—	—	2.5	μ Adc
Output High Voltage ($I_{OH} = -205 \mu$ A)	V_{OH}	2.4	—	—	Vdc
Output Low Voltage ($I_{OL} = 1.6$ mA)	V_{OL}	—	—	0.4	Vdc
Output Leakage Current (Three-State) ($CS = 0.8$ V or $\overline{CS} = 2.0$ V, $V_{out} = 0.4$ V to 2.4 V)	I_{LO}	—	—	10	μ Adc
Supply Current ($V_{CC} = 5.25$ V, $T_A = 0^\circ$ C)	I_{CC}	—	—	130	mAdc

CAPACITANCE ($f = 1.0$ MHz, $T_A = 25^\circ$ C, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Max	Unit
Input Capacitance	C_{in}	7.5	pF
Output Capacitance	C_{out}	12.5	pF

AC OPERATING CONDITIONS AND CHARACTERISTICS

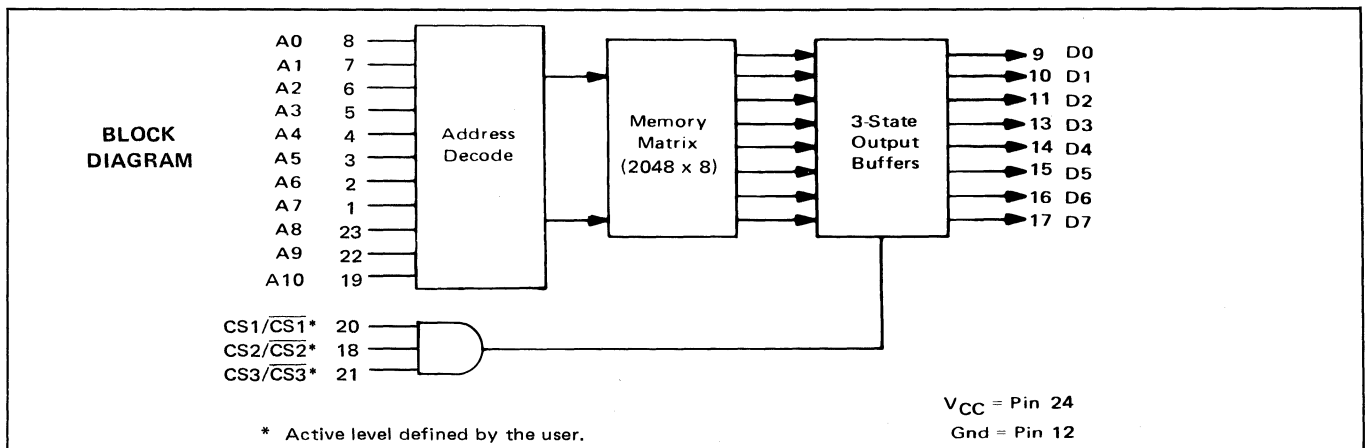
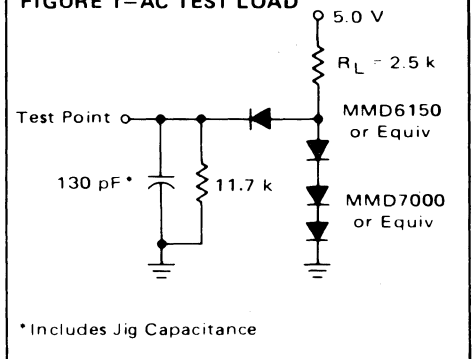
(Full operating voltage and temperature unless otherwise noted.)

(All timing with $t_r = t_f = 20$ ns, Load of Figure 1)

Characteristic	Symbol	Min	Max	Unit
Cycle Time	t_{cyc}	500	—	ns
Access Time	t_{acc}	—	500	ns
Chip Select to Output Delay	t_{CO}	—	300	ns
Data Hold from Address	t_{DHA}	10	—	ns
Data Hold from Deselection	t_H	10	150	ns

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

FIGURE 1—AC TEST LOAD





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MCM68708L

Product Preview

1024 X 8-BIT ALTERABLE READ ONLY MEMORY

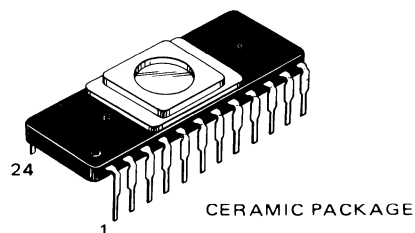
The MCM68708 is an 8192-bit Alterable Read Only Memory designed for system debug usage and similar applications requiring non-volatile memory that must be reprogrammed periodically. The transparent lid on the package allows the memory content to be erased with ultraviolet light. The memory can then be electrically reprogrammed.

- Organized as 1024 Bytes of 8-Bits
- Static Operation
- Standard Power Supplies of +12 V, +5 V, and -5 V.
- Access Time = 500 ns
- Low Power Dissipation
- Chip Select Input for Memory Expansion
- TTL Compatible
- Three-State Outputs
- Compatible with the 2708

MOS

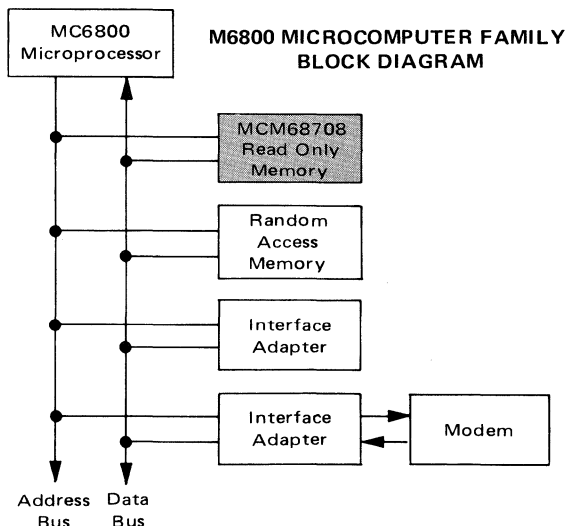
(N-CHANNEL, SILICON-GATE)

1024 X 8-BIT ALTERABLE READ ONLY MEMORY

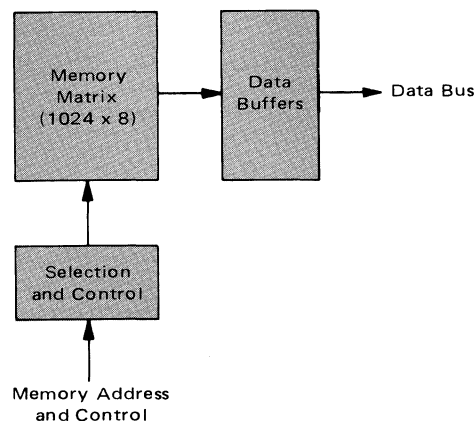


PIN ASSIGNMENT

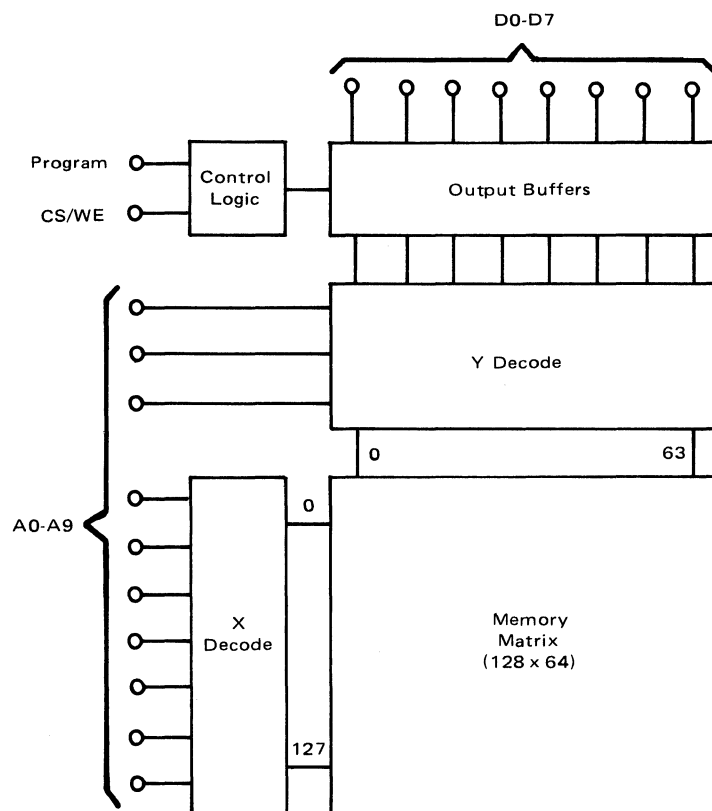
1	A7	O	V _{CC}	24
2	A6		A8	23
3	A5		A9	22
4	A4		V _{BB}	21
5	A3		CS/WE	20
6	A2		V _{DD}	19
7	A1		Progr.	18
8	A0		D7	17
9	D0		D6	16
10	D1		D5	15
11	D2		D4	14
12	V _{SS}		D3	13



MCM68708 READ ONLY MEMORY BLOCK DIAGRAM



BLOCK DIAGRAM





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Semiconductors

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Advance Information

4096-BIT DYNAMIC RANDOM ACCESS MEMORY

The MCM6604 is a 4096-bit high-speed dynamic Random Access Memory designed for high-performance, low-cost applications in mainframe and buffer memories and peripheral storage. Organized as 4096 one-bit words and fabricated using Motorola's highly reliable N-channel silicon gate technology, this device optimizes speed, power, and density tradeoffs.

By multiplexing row and column address inputs, the MCM6604 requires only six address lines and permits packaging in Motorola's standard 16-pin dual in-line packages. Complete address decoding is done on chip with address latches incorporated.

All inputs are TTL compatible, and the output is 3-state TTL compatible. The MCM6604 incorporates a one-transistor cell design and dynamic storage techniques, with each of the 64 row addresses requiring a refresh cycle every 2.0 milliseconds.

- Organized as 4096 Words of 1 Bit
- Maximum Access Time = 250 ns (L2,P2)
300 ns (L4,P4)
350 ns (L,P)
- Minimum Read and Write Cycle Time =
375 ns (L2,P2)
425 ns (L4,P4)
500 ns (L,P)
- Low Power Dissipation
630 mW Maximum (Active)
25 mW Maximum (Standby)
- TTL Compatible
- 3-State Output
- On-Chip Latches for Address, Chip Select, and Data In
- Power Supply Pins on Package Corners for Optimum Layout
- Standard 16-Pin Package
- Compatible with the Popular 2104/MK4096

ABSOLUTE MAXIMUM RATINGS (See Note 1)

Rating	Symbol	Value	Unit
Voltage on Any Pin Relative to V_{BB}	V_{in}, V_{out}	-0.3 to +20	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	-65 to +150	°C

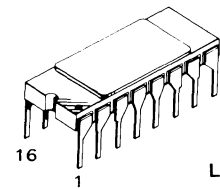
NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

MCM6604L/L2/L4
MCM6604P/P2/P4

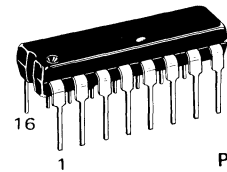
MOS

(N-CHANNEL, SILICON-GATE)

4096-BIT DYNAMIC RANDOM ACCESS MEMORY

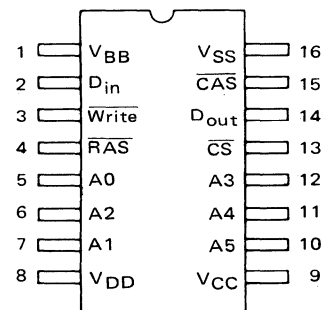


L SUFFIX
CERAMIC PACKAGE
CASE 690



P SUFFIX
PLASTIC PACKAGE
CASE 648

PIN ASSIGNMENT



This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

DC OPERATING CONDITIONS AND CHARACTERISTICS (Full operating voltage and temperature range unless otherwise noted.)

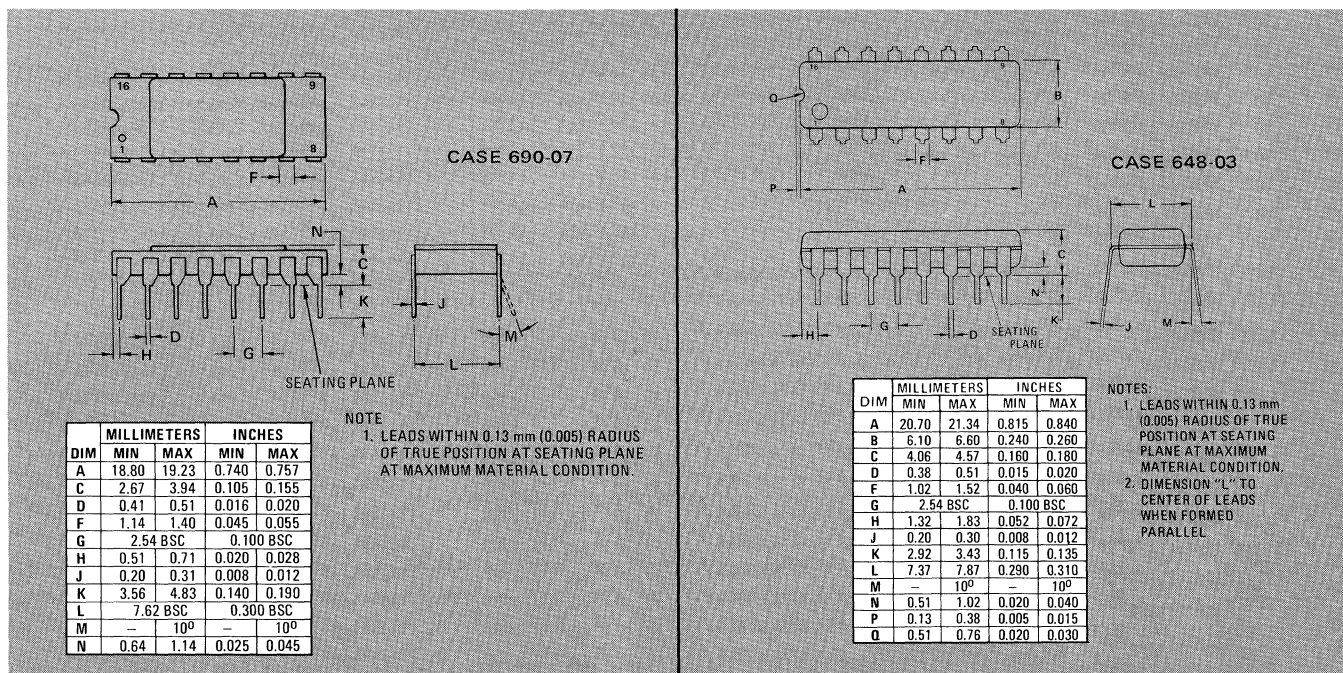
RECOMMENDED OPERATING CONDITIONS (Referenced to V_{SS} = Ground)

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{DD}	11.4	12.0	12.6	Vdc
	V_{CC}	4.5	5.0	5.5	Vdc
	V_{BB}	-4.5	-5.0	-5.5	Vdc
Input High Voltage An, $\overline{CS}$, D_{in} , RAS, CAS, Write	V_{IH}	2.4	—	V_{CC}	Vdc
		2.7	—	V_{CC}	
Input Low Voltage All Inputs	V_{IL}	-1.0	—	0.8	Vdc

DC CHARACTERISTICS ($V_{DD} = 12\text{ V} \pm 5\%$, $V_{CC} = 5.0\text{ V} \pm 10\%$, $V_{BB} = -5.0\text{ V} \pm 10\%$)

Characteristic	Symbol	Min	Typ	Max	Unit
Input Current, Any Input ($V_{in} = 0$ to V_{CC})	I_{in}	—	—	10	μA
Output High Voltage ($I_O = -3.0\text{ mA}$)	V_{OH}	2.4	—	—	Vdc
Output Low Voltage ($I_O = 2.0\text{ mA}$)	V_{OL}	—	—	0.4	Vdc
Output Leakage Current (Output Disabled by $\overline{CS}$ Input)	I_{LO}	—	—	10	μA
Average Supply Current, Active Mode ($T_{cyc}(W) = \text{min}$, $T_A = 70^\circ\text{C}$)	I_{DDA}	—	—	50	mA
	I_{CCA}	—	—	100	μA
	I_{BBA}	—	—	100	μA
Supply Current, Standby Mode ($T_A = 70^\circ\text{C}$)	I_{DDS}	—	—	2.0	mA
	I_{CCS}	—	—	10	μA
	I_{BBS}	—	—	100	μA

PACKAGE DIMENSIONS



EFFECTIVE CAPACITANCE (Full operating voltage and temperature range, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Max	Unit
Input Capacitance A_n $\overline{RAS}$, $\overline{CAS}$, D_{in} , $\overline{Write}$, $\overline{CS}$	$C_{in(EFF)}$	10 7.0	pF
Output Capacitance	$C_{out(EFF)}$	8.0	pF

AC OPERATING CONDITIONS AND CHARACTERISTICS
(Read, Write, and Read-Modify-Write Cycles)

RECOMMENDED AC OPERATING CONDITIONS ($V_{DD} = 12V \pm 5\%$, $V_{CC} = 5.0V \pm 10\%$, $V_{BB} = -5.0V \pm 10\%$, $T_A = 0$ to 70°C)

Parameter	Symbol	MCM6604L,P		MCM6604L2,P2		MCM6604L4,P4		Unit
		Min	Max	Min	Max	Min	Max	
Random Read or Write Cycle Time	t_{cyc}	500	—	375	—	425	—	ns
Read-Modify-Write Cycle Time	$t_{cyc(RMW)}$	700	—	540	—	620	—	ns
Row Address Strobe Precharge Time	t_{RP}	150	—	125	—	125	—	ns
Row to Column Strobe Lead Time (Note 1)	t_{RCL}	110	150	70	110	90	130	ns
Column Address Strobe Pulse Width	t_{CPW}	200	—	140	—	170	—	ns
Address Setup Time	t_{AS}	0	—	0	—	0	—	ns
Address Hold Time	t_{AH}	100	—	60	—	80	—	ns
$\overline{RAS}$ Address Release Time	t_{AR}	250	—	170	—	210	—	ns
Read Command Setup Time	t_{RCS}	0	—	0	—	0	—	ns
Read Command Hold Time	t_{RCH}	100	—	0	—	80	—	ns
Read Command Pulse Width	t_{RPW}	300	—	200	—	250	—	ns
Write Command Hold Time (Note 2)	t_{WCH}	150	—	110	—	130	—	ns
Write Command Pulse Width	t_{WP}	200	—	140	—	170	—	ns
Column to Row Strobe Lead Time	t_{CRL}	-50	+50	-40	+40	-45	+45	ns
Write Command to Column Strobe Lead Time	t_{CWL}	200	—	140	—	170	—	ns
Data In Setup Time	t_{DS}	0	—	0	—	0	—	ns
Data In Hold Time	t_{DH}	150	—	110	—	130	—	ns
Refresh Period	t_{REF}	—	2.0	—	2.0	—	2.0	ms
Modify Time	t_{MOD}	0	10	0	10	0	10	μs

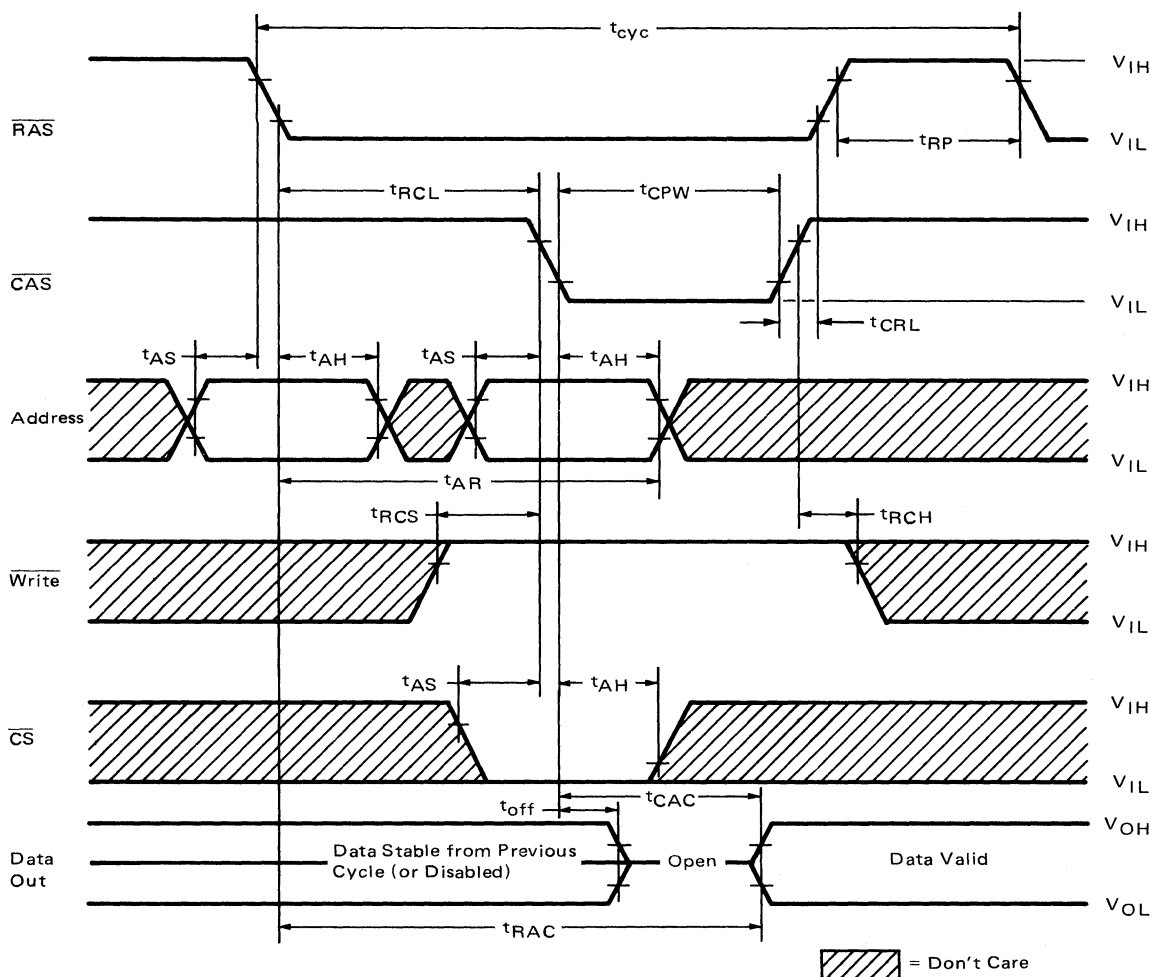
1. If t_{RCL} is greater than the maximum recommended value shown in this table, t_{cyc} and t_{RAC} will increase by the amount that t_{RCL} exceeds the value shown.
2. The Write Command Hold Time is important only when normal random write cycles are being performed. During a read-write or a read-modify-write cycle, the limiting parameter is the Write Command Pulse Width.

AC CHARACTERISTICS ($t_r = t_f = 20$ ns, Load = 1 MC74H00 Series TTL Gate, $C_L(EFF) = 50$ pF)

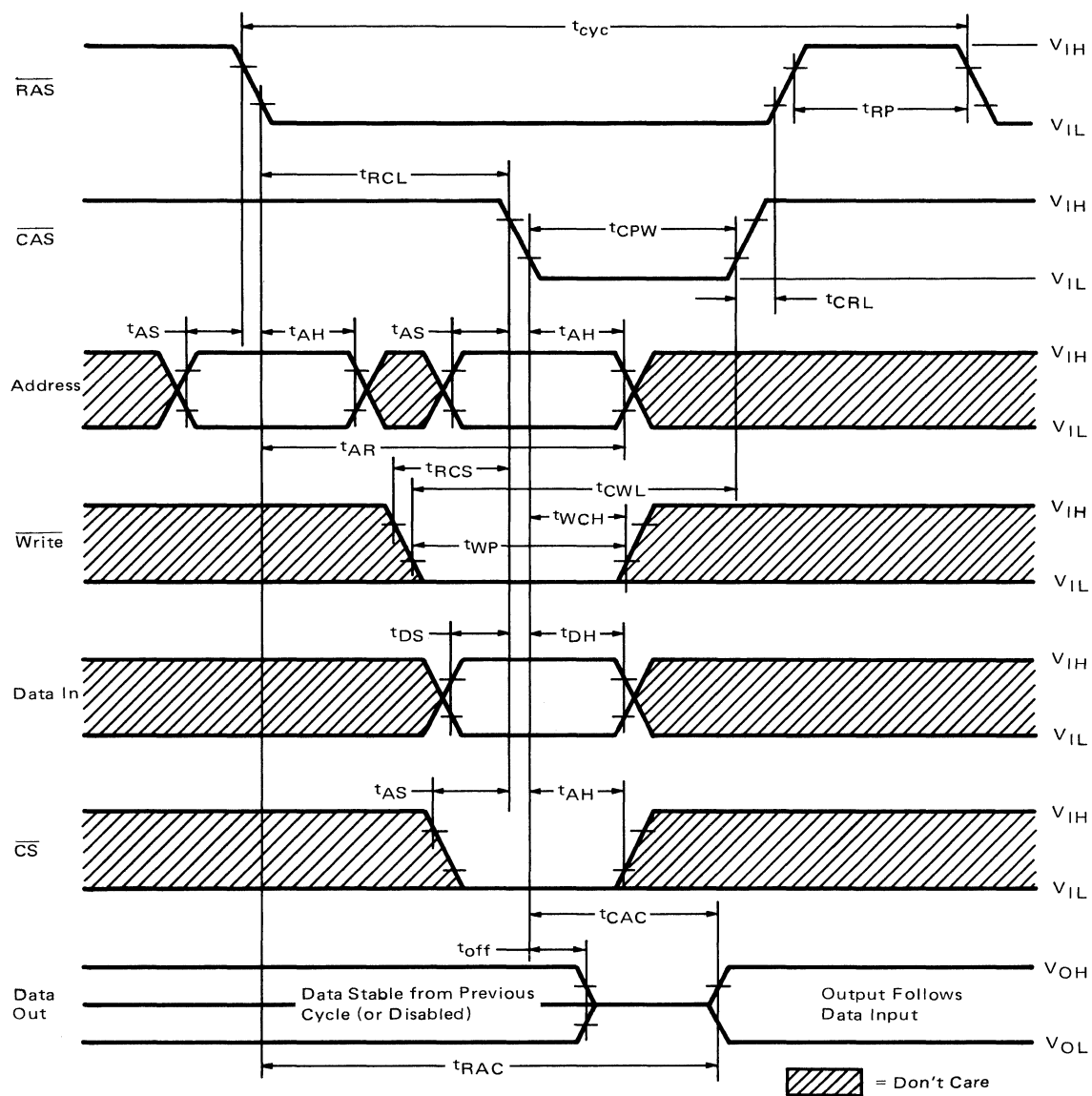
Characteristic	Symbol	MCM6604L,P	MCM6604L2,P2	MCM6604L4,P4	Unit
		Max	Max	Max	
Access Time from Row Address Strobe ($t_{RCL} \leq 150$ ns for MCM6604L,P) ($t_{RCL} \leq 110$ ns for MCM6604L2,P2) ($t_{RCL} \leq 130$ ns for MCM6604L4,P4)	t_{RAC}	350	250	300	ns
Access Time from Column Address Strobe	t_{CAC}	200	150	175	ns
Output Buffer Turn-Off Delay	t_{off}	100	65	85	ns



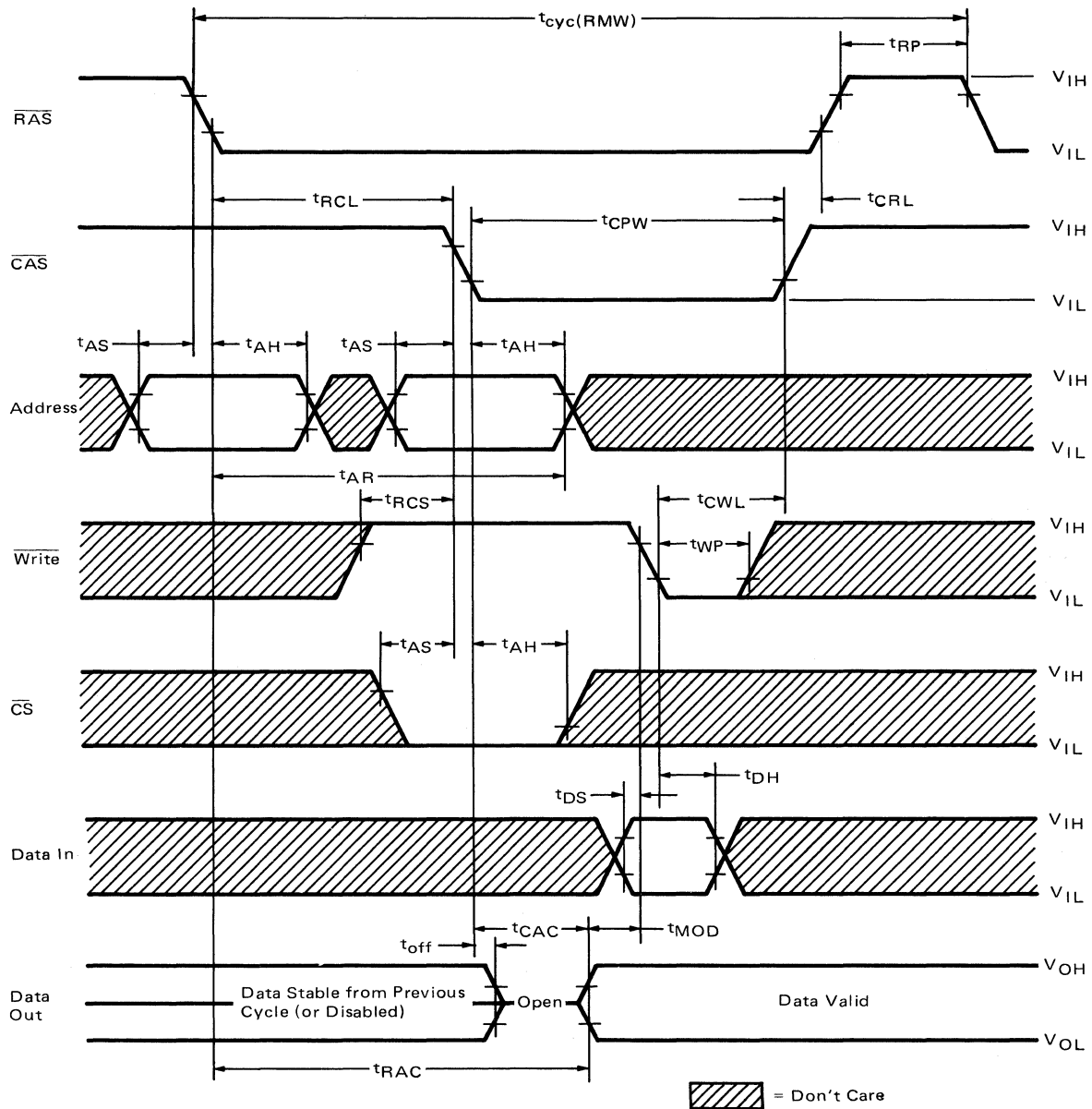
READ CYCLE TIMING



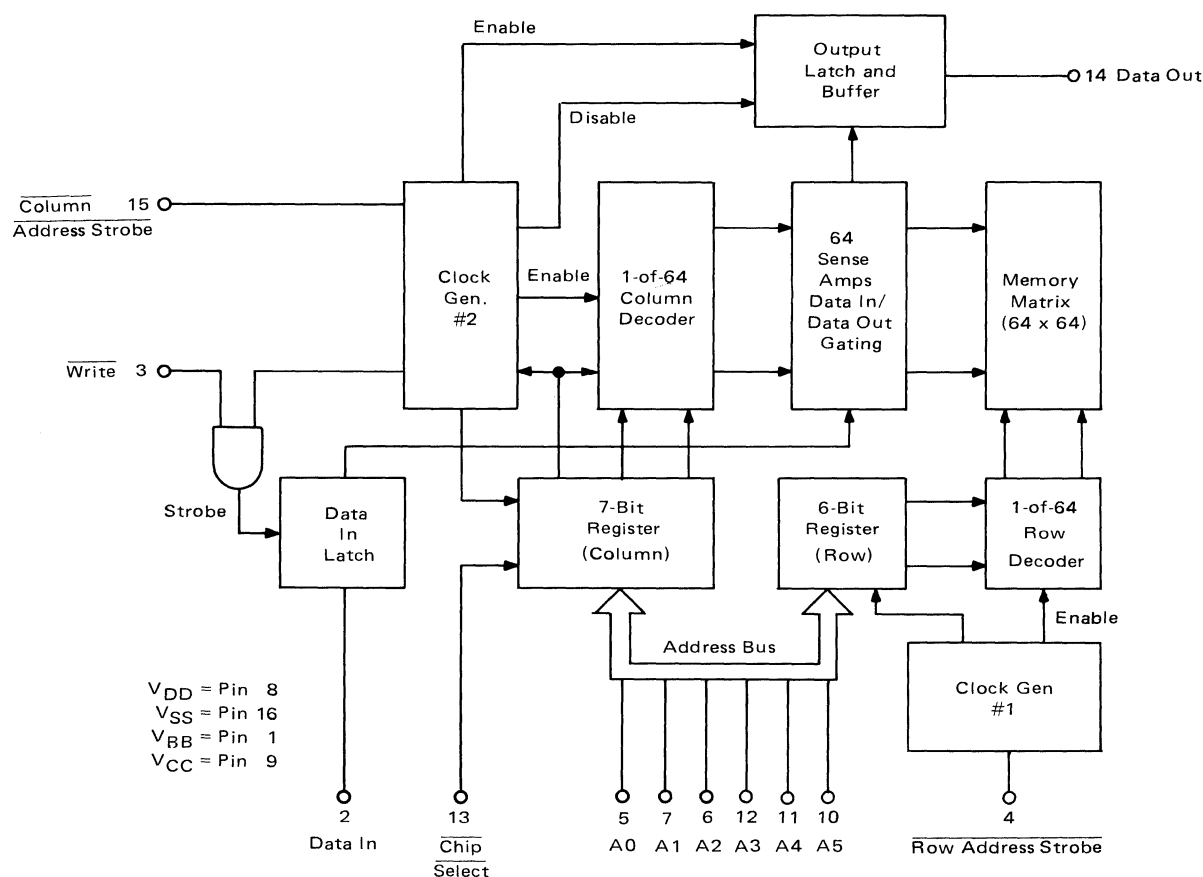
WRITE CYCLE TIMING



READ – MODIFY – WRITE TIMING



BLOCK DIAGRAM



ADDRESSING

The MCM6604 has six address inputs (A0 through A5) that are common to two address registers, one register for the row address and another for the column address. The column register has an additional latch that accommodates the $\overline{\text{Chip Select}}$ (CS) signal. At the start of a memory cycle, the row address is latched into the address register with the Row Address Strobe (RAS) signal. Next, the 6-bit column address is placed on the address bus along with the $\overline{\text{Chip Select}}$ signal, and they are latched into the column register with the Column Address Strobe (CAS). Since the $\overline{\text{Chip Select}}$ signal is latched well into the memory cycle, its decoding time will not increase the memory system access or cycle time.

DATA OUTPUT

In order to simplify the memory system design and reduce the total package count, the MCM6604 contains an input data latch and a buffered output data latch. The state of the output latch and buffer at the end of a memory cycle will depend on the type of memory cycle performed and whether the chip is selected or unselected for that memory cycle.

A chip will be unselected during a memory cycle if:

- (1) The chip receives both $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ signals, but no $\overline{\text{Chip Select}}$ signal.

- (2) The chip receives a $\overline{\text{CAS}}$ signal but no $\overline{\text{RAS}}$ signal. With this condition, the chip will be unselected regardless of the state of $\overline{\text{Chip Select}}$ input.

If, during a read, write, or read-modify-write cycle, the chip is unselected, the output buffer will be in the high impedance state at the end of the memory cycle. The output buffer will remain in the high impedance state until the chip is selected for a memory cycle.

For a chip to be selected during a memory cycle, it must receive the following signals: $\overline{\text{RAS}}$, $\overline{\text{CAS}}$, and $\overline{\text{Chip Select}}$. The state of the output latch and buffer of a selected chip during the following type of memory cycles would be:

- (1) **Read Cycle** — On the negative edge of $\overline{\text{CAS}}$, the output buffer will unconditionally go to a high impedance state. It will remain in this state until access time. At this time, the output latch and buffer will assume the logic state of the data read from the selected cell. This output state will be maintained until the chip receives the next $\overline{\text{CAS}}$ signal.
- (2) **Write Cycle** — If the $\overline{\text{Write}}$ input is switched to a logic 0 before the $\overline{\text{CAS}}$ transition, the output latch and buffer will be switched to



the state of the data input at the end of the access time. This logic state will be maintained until the chip receives the next $\overline{\text{CAS}}$ signal.

- (3) Read-Modify-Write — Same as a read cycle.

DATA INPUT

Data to be written into a selected storage cell of the memory chip is first stored in the on-chip data latch. The gating of this latch is performed with a combination of the $\overline{\text{Write}}$ and $\overline{\text{CAS}}$ signals. The last of these signals to make a negative transition will strobe the data into the latch. If the $\overline{\text{Write}}$ input is switched to a logic 0 at the beginning of a write cycle, the falling edge of $\overline{\text{CAS}}$ strobes the data into the latch. The data setup and hold times are then referenced to the negative edge of $\overline{\text{CAS}}$.

If a read-modify-write cycle is being performed, the $\overline{\text{Write}}$ input would not make its negative transition until after the $\overline{\text{CAS}}$ signal was enabled. Thus, the data would not be strobed into the latch until the negative transition of $\overline{\text{Write}}$. The data setup and hold times would now be referenced to the negative edge of the $\overline{\text{Write}}$ signal. The only other timing constraints for a write-type cycle is that both the $\overline{\text{CAS}}$ and $\overline{\text{Write}}$ signals remain in the logic 0 state for a sufficient time to accomplish the permanent storage of the data into the selected cell.

INPUT/OUTPUT LEVELS

All of the inputs to the MCM6604 are TTL compatible. The inputs feature high impedance and low capacitance (<10 pF) characteristics which will minimize the driver requirements in a memory system. The three-state data output buffer is TTL compatible and has sufficient current

sink capability (2 mA) to drive one high speed TTL load. The output buffer also has a separate V_{CC} pin so that it can be powered from the same supply as the logic being employed.

POWER DISSIPATION

Since the MCM6604 is a dynamic RAM, its power drain will be extremely small during the time the chip is unselected.

The power of the MCM6604 increases when selected and most of this increase is encountered on the address strobe edge. Hence, the power will be a function of the duty cycle.

In a memory system, the $\overline{\text{CAS}}$ signal must be supplied to all the memory chips to insure that the outputs of the unselected chips are switched to the high impedance state. Those chips that do not receive an $\overline{\text{RAS}}$ signal will not dissipate any power on the $\overline{\text{CAS}}$ edge except for that required to turn off the chip outputs. Thus, in order to insure minimum system power, the $\overline{\text{RAS}}$ signal should be decoded so that only the chips to be selected receive an $\overline{\text{RAS}}$ signal. If the $\overline{\text{RAS}}$ signal is decoded, then the chip select input of all the chips can be set to a logic 0 state.

REFRESH

The MCM6604 is refreshed by sequentially cycling through the 64 row addresses every 2 milliseconds or less. It is not necessary to supply the $\overline{\text{CAS}}$ to the chip while it is being refreshed. Any read, write, or read-modify-write cycle will refresh a selected row. However, if a write cycle is used to perform a refresh cycle the chip must be unselected.





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MCM6605A L/L1/L2
MCM6605A P/P1/P2

4096-BIT DYNAMIC RANDOM ACCESS MEMORY

The MCM6605A is a 4096-bit high-speed dynamic Random Access Memory designed for high-performance, low-cost applications in mainframe and buffer memories and peripheral storage. Organized as 4096 one-bit words, these memories are fabricated using selective oxidation N-channel silicon gate technology to optimize device speed, power and density tradeoffs.

All address and control inputs are TTL compatible except for a single high-level clock (Chip Enable). Complete address decoding is done on chip and address latches are incorporated for ease of use. Refresh of the entire memory can be accomplished by sequentially cycling through addresses A0-A4 (32 cycles) a maximum of every 2.0 milliseconds.

The MCM6605A uses a three-transistor memory cell to simplify internal sense amplifier requirements. Output data is inverted with respect to input data. The outputs are 3-state TTL configuration and require no external sense amplifier. Outputs are in the high impedance (floating) state when either the Chip Enable is in the low state or the Chip Select is in the high state.

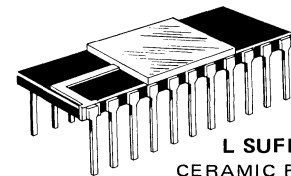
- Organized as 4096 Words of 1 Bit

	L1, P1	L2, P2	L, P
Maximum Access Time =	150 ns	200 ns	300 ns
Minimum Read Cycle Time =	290 ns	360 ns	470 ns
Minimum Write Cycle Time =	390 ns	490 ns	590 ns
Minimum Read Modify Write Cycle Time =	390 ns	490 ns	590 ns
Low Power Dissipation			
335 mW Typical (Active)			
2.6 mW Typical (Standby with Refresh)			
Easy Refresh – Only 32 Cycles Every 2.0 ms			
TTL Compatible			
3-State Output			
Address Latches On Chip			
Power Supply Pins on Package Corners for Layout Simplification			
Typical Applications:			
Main Memory			
Buffer Memory			
Peripheral Storage			

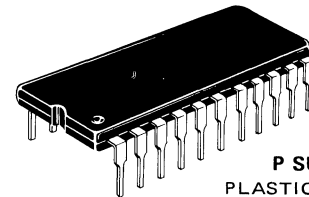
MOS

(N-CHANNEL, SILICON-GATE)

4096-BIT DYNAMIC RANDOM ACCESS MEMORY

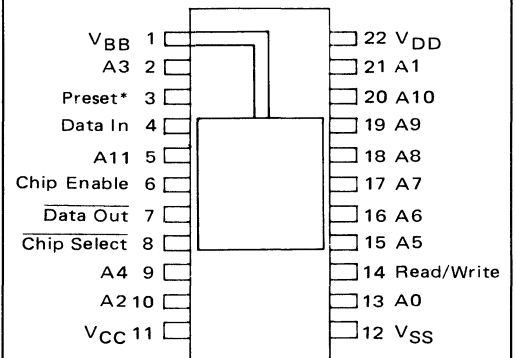


L SUFFIX
CERAMIC PACKAGE
CASE 677



P SUFFIX
PLASTIC PACKAGE
CASE 708

PIN ASSIGNMENT



*See Applications Information

ABSOLUTE MAXIMUM RATINGS (See Note 1)

Rating	Symbol	Value	Unit
Voltage on Any Pin Relative to V _{BB}	V _{in} , V _{out}	-0.3 to +20	V _{dc}
Operating Temperature Range	T _A	0 to +70	°C
Storage Temperature Range	T _{stg}	-65 to +150	°C

NOTE 1: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to RECOMMENDED OPERATING CONDITIONS. Exposure to higher than recommended voltages for extended periods of time could affect device reliability.

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

DC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature range unless otherwise noted.)

RECOMMENDED DC OPERATING CONDITIONS (Referenced to V_{SS}).

Parameter	Symbol	Min	Nom	Max	Unit
Supply Voltage	V_{DD}	11.4	12	12.6	Vdc
	V_{CC}	4.5	5.0	5.5	Vdc
	V_{SS}	0	0	0	Vdc
	V_{BB}	-5.25	-5.0	-4.75	Vdc
Logic Levels					
Input High Voltage ($A_n, D_{in}, R/W, \overline{CS}$)	V_{IH}	3.0	—	$V_{DD} + 0.6$	Vdc
Input Low Voltage ($A_n, D_{in}, R/W, \overline{CS}$)	V_{IL}	-1.0	—	0.8	Vdc
Chip Enable High Voltage	V_{CEH}	$V_{DD} - 0.6$	—	$V_{DD} + 0.6$	Vdc
Chip Enable Low Voltage	V_{CEL}	-1.0	—	0.8	Vdc

DC CHARACTERISTICS

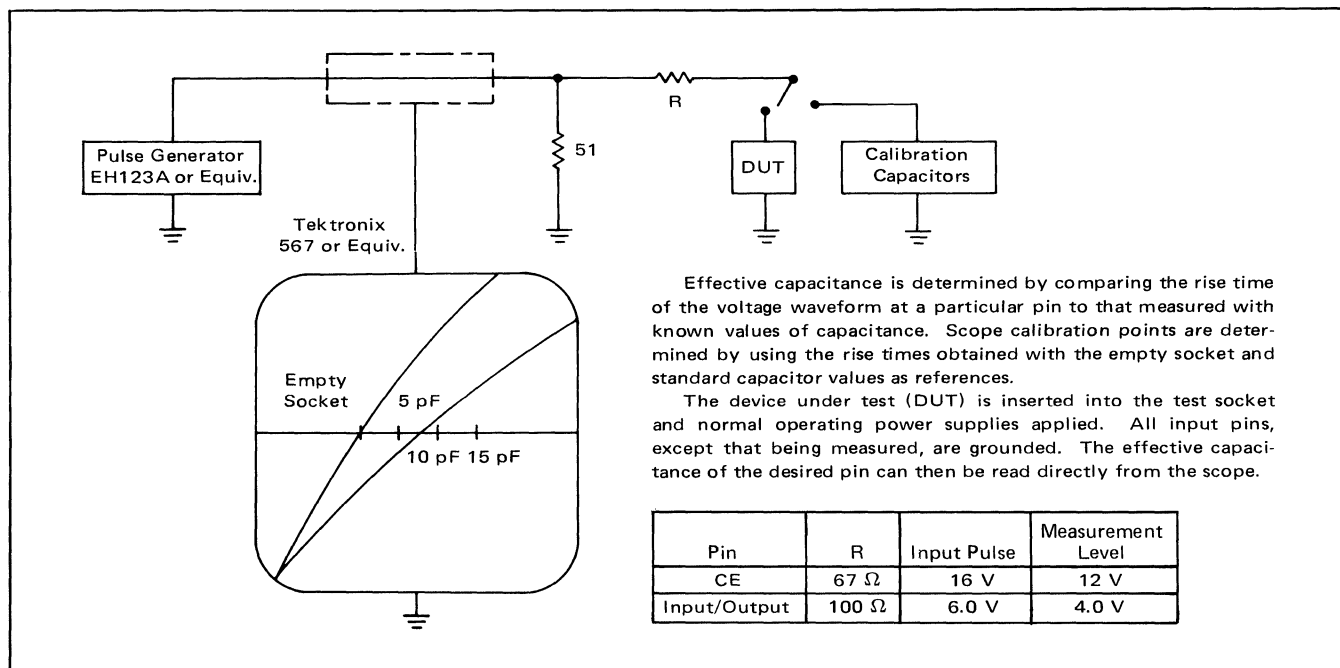
Characteristic	Symbol	Min	Typ	Max	Unit
Input Current ($A_n, D_{in}, R/W, \overline{CS}$, Preset) ($V_{in} = 0$ to $V_{DD} + 1.0$ V)	I_{in}	—	—	10	μA
Input Chip Enable Current ($V_{in} = 0$ to $V_{DD} + 1.0$ V)	I_{ICE}	—	—	10	μA
Output High Voltage ($I_O = -100 \mu A$)	V_{OH}	2.4	—	V_{CC}	Vdc
Output Low Voltage ($I_O = 2.0$ mA)	V_{OL}	V_{SS}	—	0.45	Vdc
Output Leakage Current ($V_O = 0.45$ V to V_{CC} , $CE = V_{CEL}$, or $\overline{CS} = V_{IH}$)	I_{LO}	—	—	10	μA
Average Supply Current, Active Mode ($T_{cyc}(W) = \min$)	I_{DDA}	—	28	36	mA
	I_{CCA}	—	0.05	1.0	mA
	I_{BBA}	—	—	100	μA
Supply Current, Standby Mode ($CE = 0.45$ V)	I_{DDS}	—	1.0	20	μA
	I_{CCS}	—	—	10	μA
	I_{BBS}	—	1.0	20	μA

EFFECTIVE CAPACITANCE (Test Circuit of Figure 1, full operating voltage and temperature range, periodically sampled rather than 100% tested.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input Capacitance ($A_n, D_{in}, R/W, \overline{CS}$, Preset)	$C_{in}(EFF)$	—	4.0	5.0	pF
Chip Enable Capacitance	$C_{CE}(EFF)$	—	25	30	pF
Output Capacitance	$C_{out}(EFF)$	—	4.0	5.0	pF



FIGURE 1 – MEASUREMENT OF EFFECTIVE CAPACITANCE



AC OPERATING CONDITIONS AND CHARACTERISTICS

(Full operating voltage and temperature unless otherwise noted.)

OPERATING MODES

Mode	Control States		Output
	R/W	CS	
Active (CE = High)			
Read Only	H	L	Valid
Read/Write	H $\rightarrow$ L	L	Valid
Write Only	L	L	Valid
Read Refresh	H $\rightarrow$ L	L $\rightarrow$ H	Valid $\rightarrow$ Floating
Refresh Only	L	H	Floating
Chip Disable (Unselected)	H	H	Floating
Standby (CE = Low)	X	X	Floating

X = Don't Care

RECOMMENDED AC OPERATING CONDITIONS (Read, Write, and Read Modify Write Cycles)

Parameter	Symbol	Min	Max	Unit
Address Setup Time	t_{AS}	0	—	ns
Address Hold Time	t_{AH}	60	—	ns
CE Pulse Transition Time	t_T	10	100	ns
CE Off Time	t_{SB}	120	—	ns
		90	—	
Chip Select Delay Time	t_{CSD}	—	70	ns
Chip Select Hold Time	t_{CSH}	0	—	ns
Read Write Delay Time	t_{RWD}	—	70	ns
Read Write Hold Time	t_{RWH}	0	—	ns
Time Between Refresh	t_{REF}	—	2.0	ms

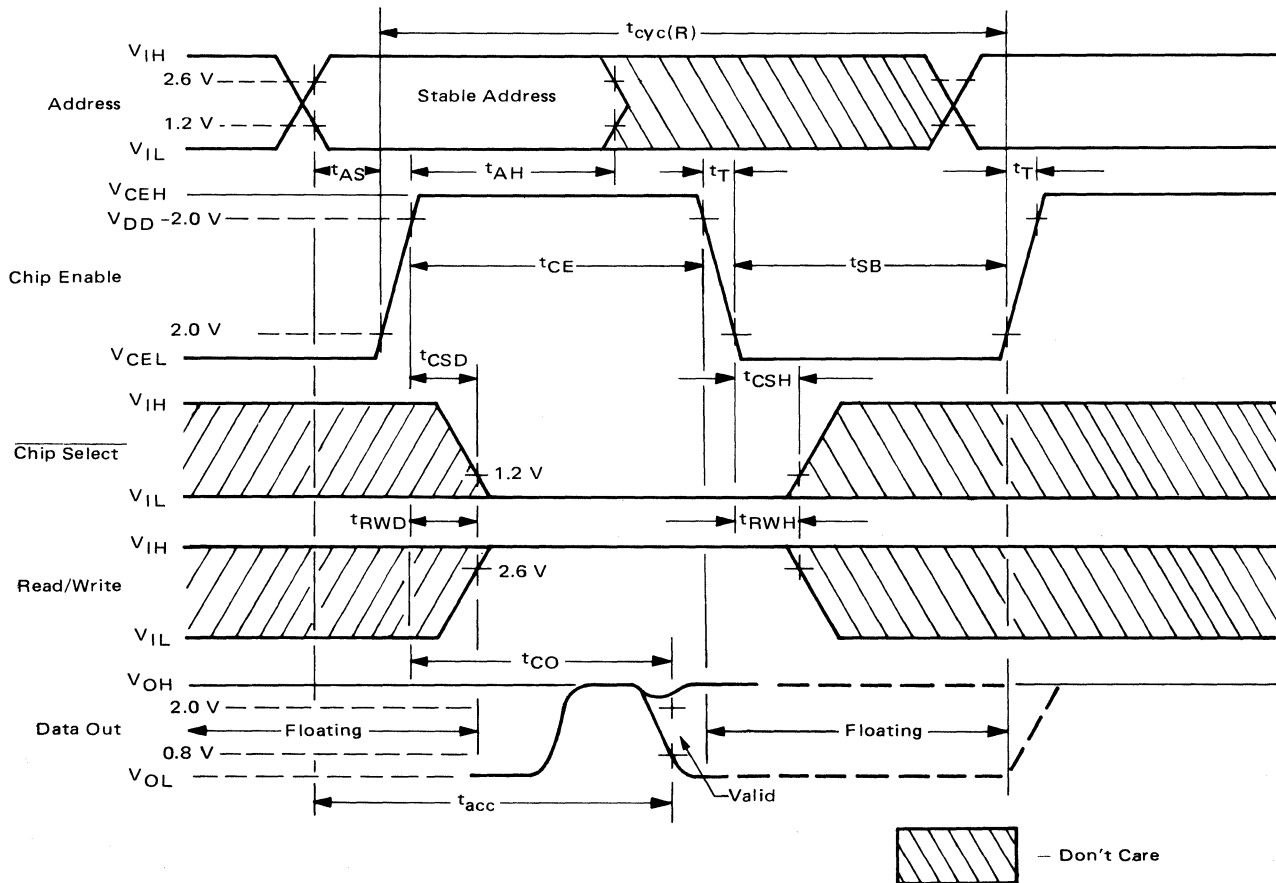


AC CHARACTERISTICS

[All timing with $t_T = 20$ ns; Load = 1 TTL Gate (MC74H00 Series), $C_L = 50$ pF (effective)]READ CYCLE (R/W = V_{IH} , $\overline{CS} = V_{IL}$)

Characteristic	Symbol	MCM6605AL,P		MCM6605AL1,P1		MCM6605AL2,P2		Unit
		Min	Max	Min	Max	Min	Max	
Read Cycle Time	$t_{cyc(R)}$	470	—	290	—	360	—	ns
Chip Enable On Time	t_{CE}	310	2000	160	2000	200	2000	ns
Chip Enable to Output Delay	t_{CO}	—	280	—	130	—	180	ns
Read Access Time	t_{acc}	—	300	—	150	—	200	ns

READ CYCLE TIMING

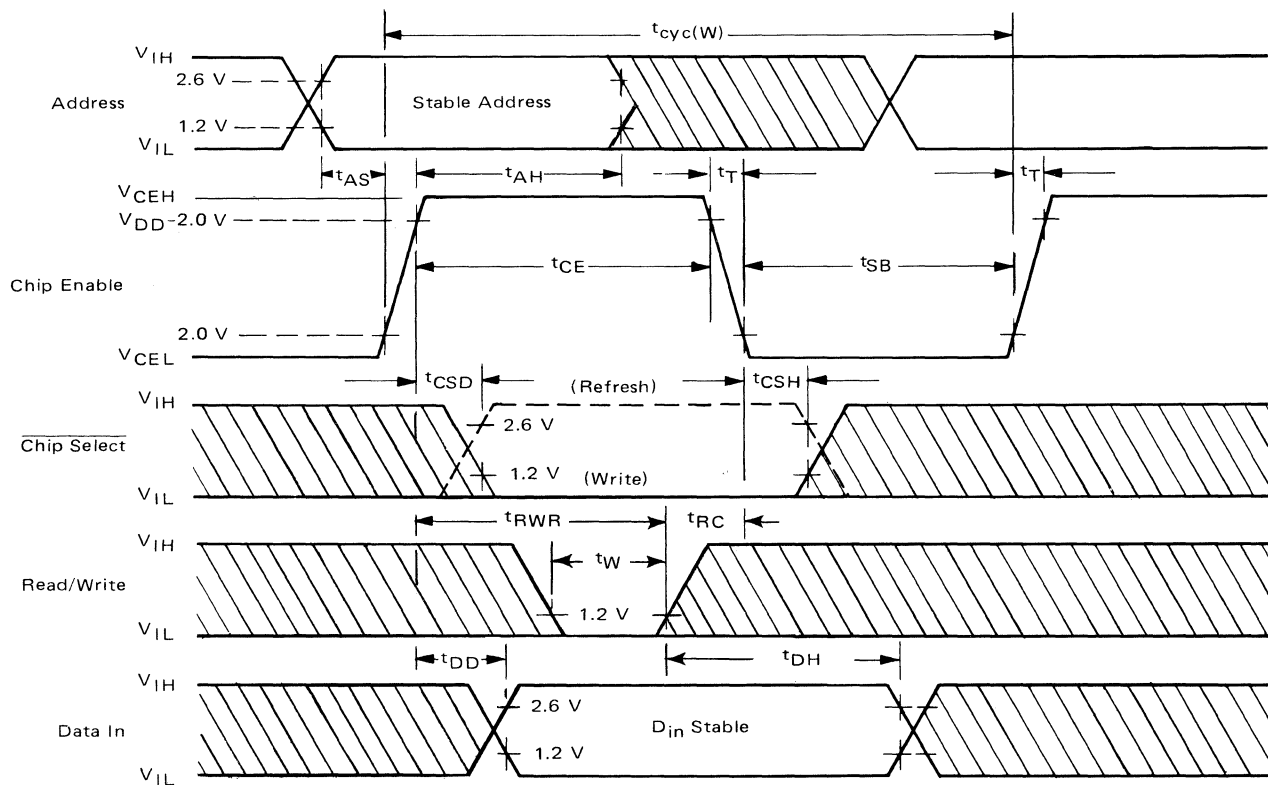


WRITE CYCLE (R/W = V_{IL} , $\overline{CS} = V_{IL}$)
 REFRESH CYCLE (R/W = V_{IL} , $\overline{CS} = V_{IH}$)

Characteristic	Symbol	MCM6605AL,P		MCM6605AL1,P1		MCM6605AL2,P2		Unit
		Min	Max	Min	Max	Min	Max	
Write Cycle Time	$t_{cyc(W)}$	590	—	390	—	490	—	ns
Chip Enable On Time	t_{CE}	430	2000	260	2000	330	2000	ns
Read-Write Release Time	t_{RWR}	410	2000	240	2000	310	2000	ns
Write Pulse Width	t_W	210	—	160	—	160	—	ns
Read-Write to Chip Enable Separation Time	t_{RC}	0	—	0	—	0	—	ns
Data Delay Time*	t_{DD}	—	70	—	70	—	70	ns
Data Hold Time	t_{DH}	50	—	20	—	50	—	ns

*If a write pulse (t_W) is employed on the R/W line during a write cycle, then the input data setup time is measured from the leading edge of the write pulse. The t_{DS} time is the same as that of the read-modify-write cycle.

WRITE AND REFRESH CYCLE TIMING



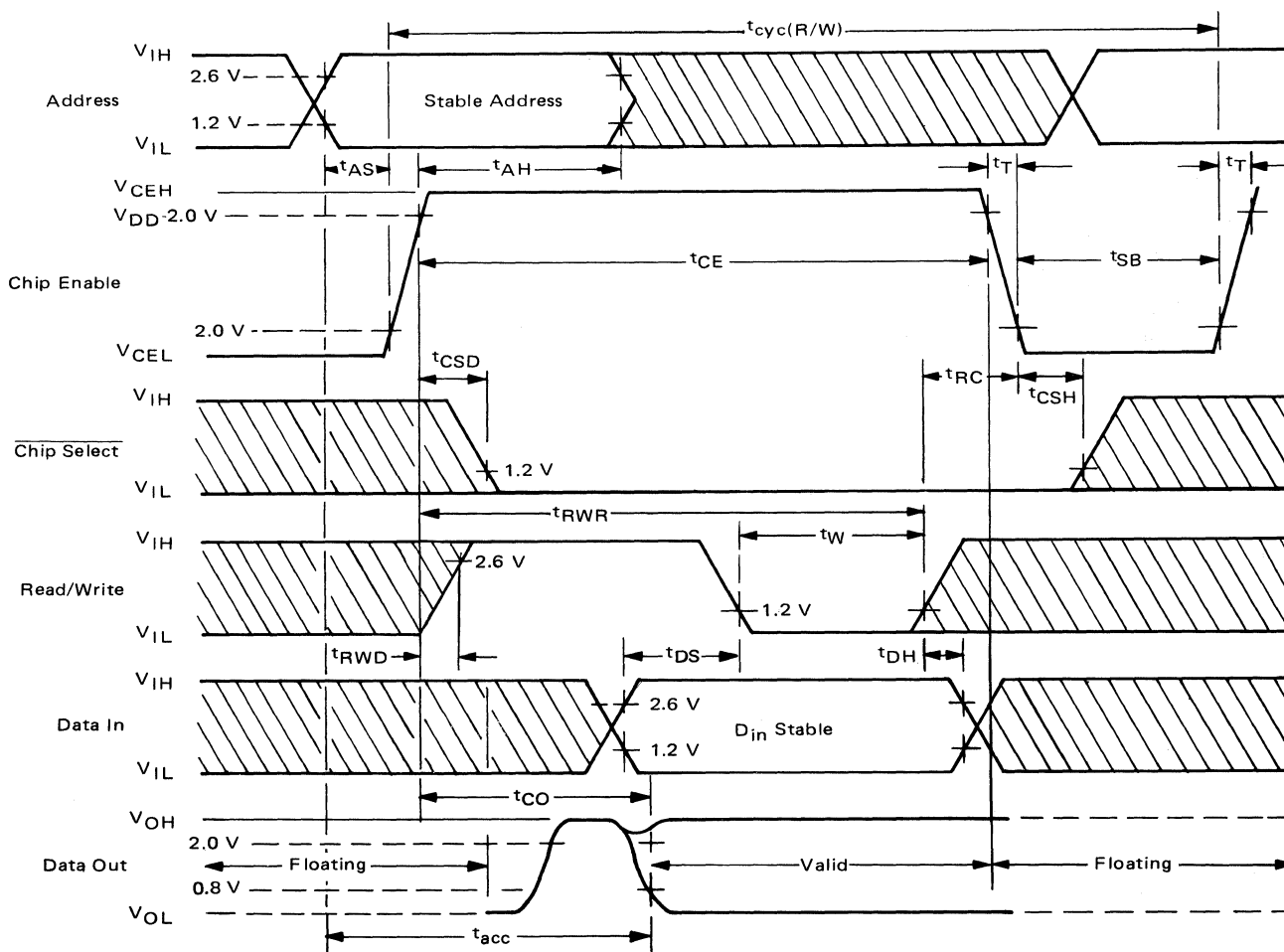
READ-MODIFY-WRITE (R/W = $V_{IH} \rightarrow V_{IL}$, $\overline{CS} = V_{IL}$)

READ REFRESH (See Note 1)

Characteristic	Symbol	MCM6605AL,P		MCM6605AL1,P1		MCM6605AL2,P2		Unit
		Min	Max	Min	Max	Min	Max	
Read-Modify-Write Cycle Time	$t_{cyc(R/W)}$	590	—	390	—	490	—	ns
Chip Enable On Time	t_{CE}	430	2000	260	2000	330	2000	ns
Read-Write Release Time	t_{RWR}	410	2000	240	2000	310	2000	ns
Write Pulse Width	t_W	210	—	160	—	160	—	ns
Data Setup Time	t_{DS}	0	—	0	—	0	—	ns
Data Hold Time	t_{DH}	50	—	20	—	50	—	ns
Read-Write to Chip Enable Separation Time	t_{RC}	0	—	0	—	0	—	ns
Chip Enable to Output Delay	t_{CO}	—	280	—	130	—	180	ns
Read Access Time	t_{acc}	—	300	—	150	—	200	ns

Note 1: A read refresh cycle is possible by bringing $\overline{CS}$ high after output data is valid and then bringing R/W low to the write position.

READ MODIFY WRITE TIMING



TYPICAL CHARACTERISTICS CURVES

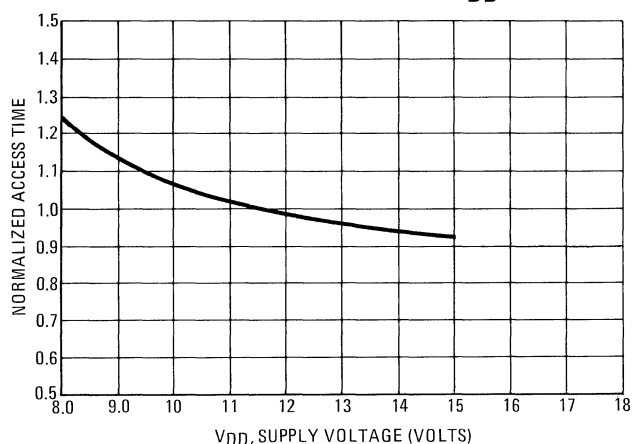
FIGURE 2 – ACCESS TIME versus V_{DD} 

FIGURE 3 – ACCESS TIME versus AMBIENT TEMPERATURE

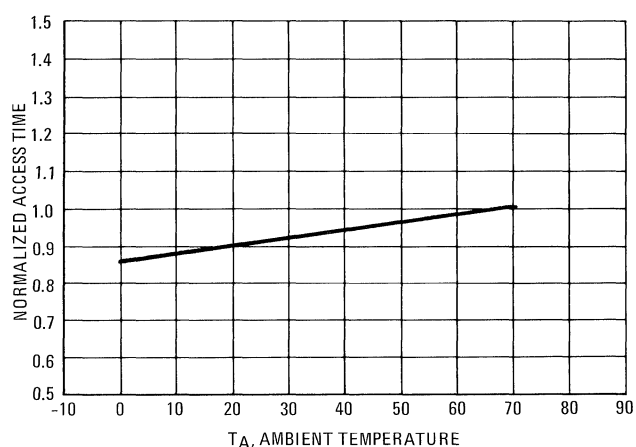
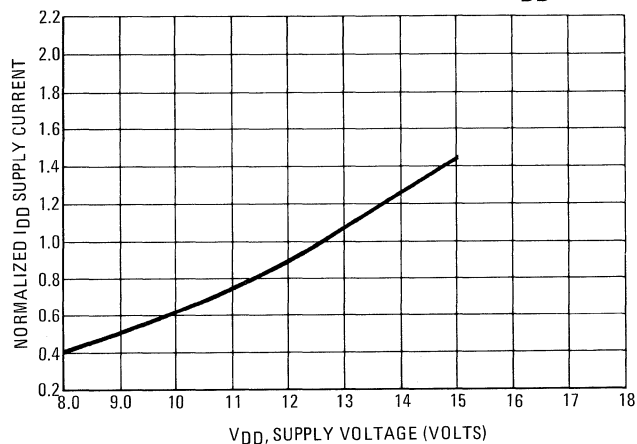
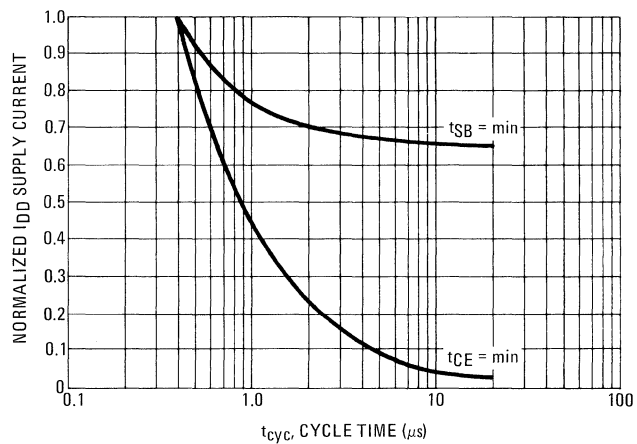
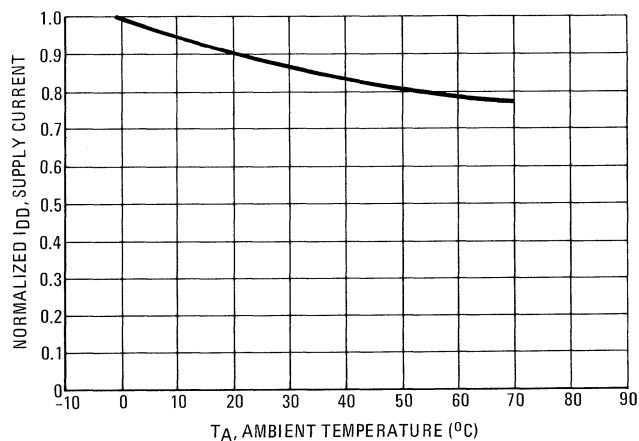
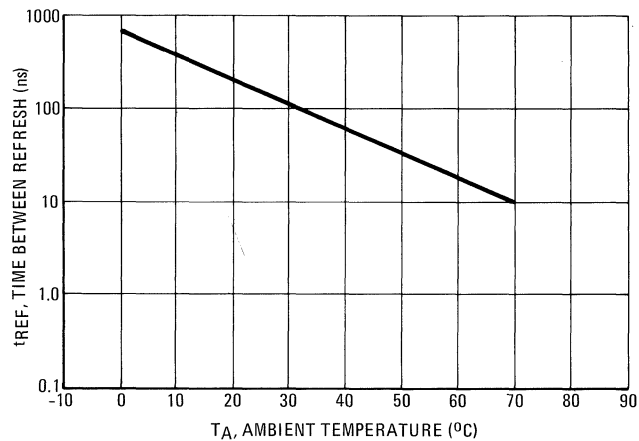
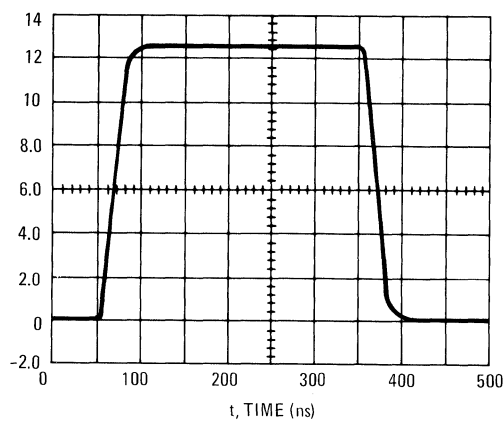
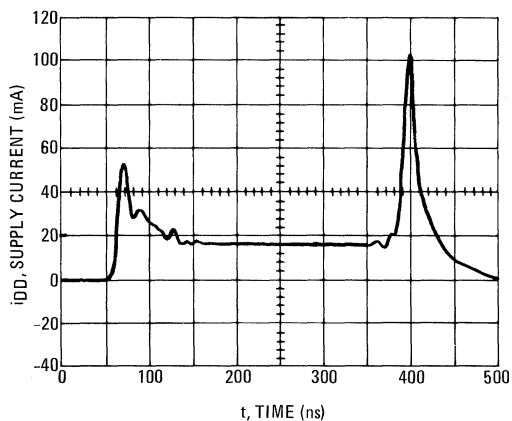
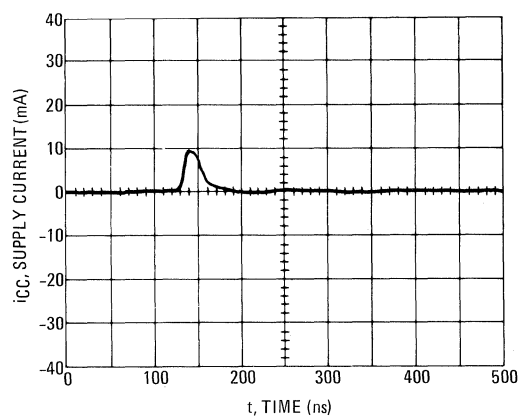
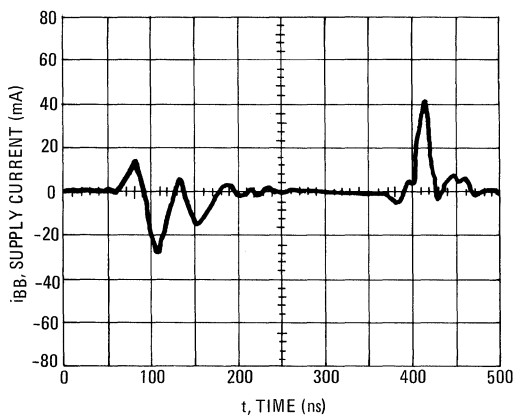
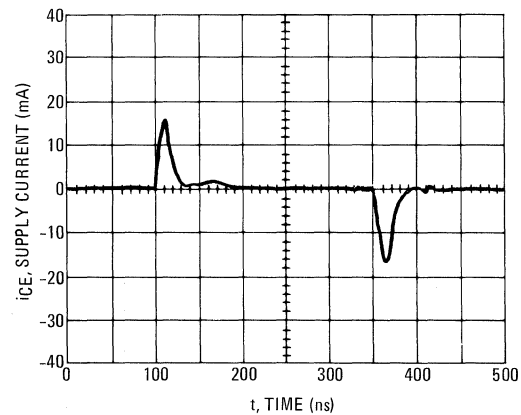
FIGURE 4 – I_{DD} SUPPLY CURRENT versus V_{DD} FIGURE 5 – I_{DD} SUPPLY CURRENT versus CYCLE TIMEFIGURE 6 – I_{DD} SUPPLY CURRENT
versus AMBIENT TEMPERATURE

FIGURE 7 – REFRESH TIME versus AMBIENT TEMPERATURE

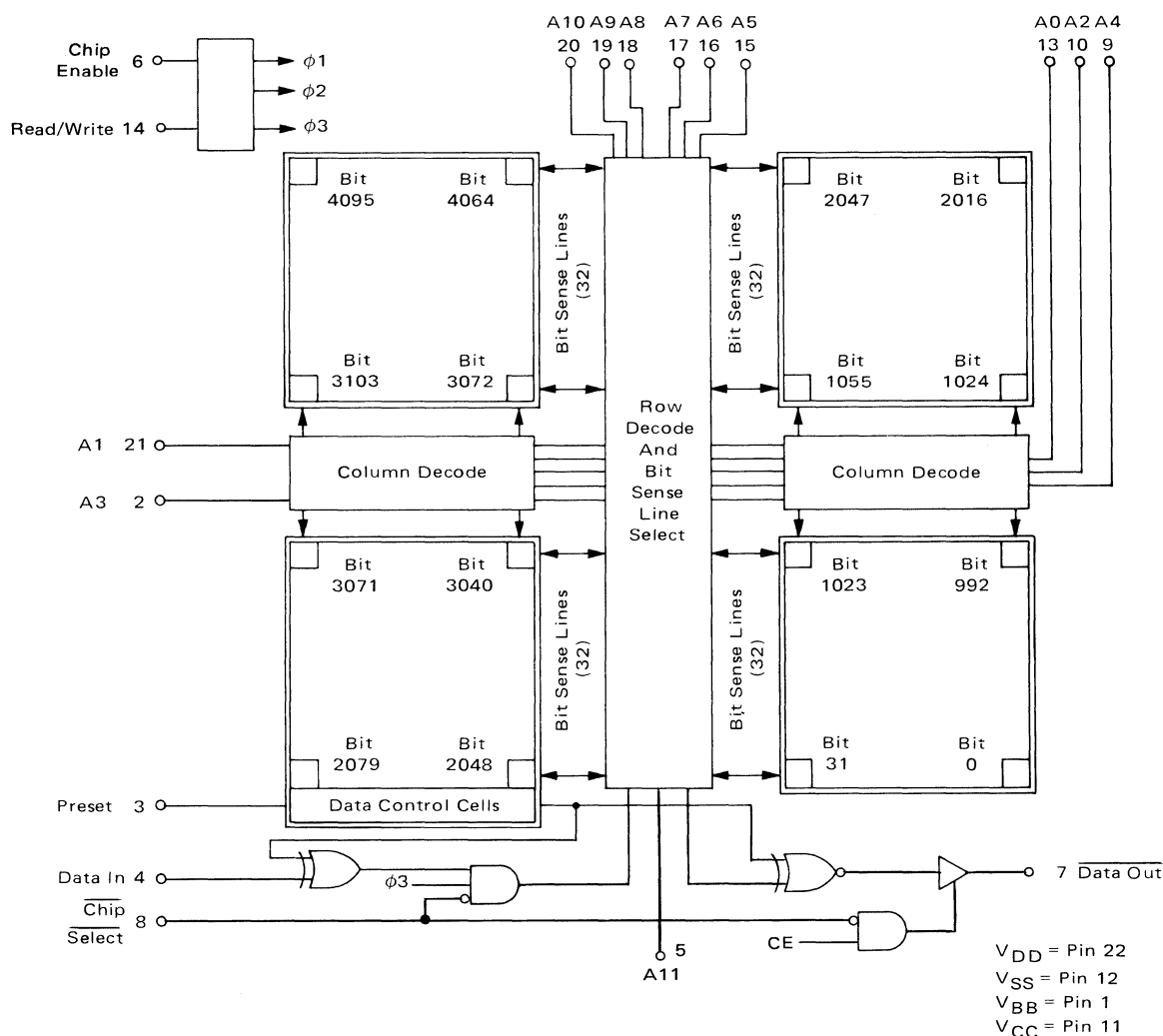


TYPICAL SUPPLY CURRENT TRANSIENT WAVEFORMS

FIGURE 8 – CHIP ENABLE VOLTAGE

FIGURE 9 – i_{DD} SUPPLY CURRENTFIGURE 10 – i_{CC} SUPPLY CURRENTFIGURE 11 – i_{BB} SUPPLY CURRENTFIGURE 12 – i_{CE} SUPPLY CURRENT

BLOCK DIAGRAM



FUNCTIONAL DESCRIPTION

The MCM6605A 4096-bit dynamic RAM uses a three transistor storage cell in an inverting cell configuration. The single high-level clock (Chip Enable) starts an internal three-phase clock generator which controls the read and write functions of the device. The $\phi 1$ signal, which is high when CE is low (standby mode), preconditions the nodes in the dynamic RAM in preparation for a memory cycle. The $\phi 2$ signal, which comes on as CE goes high, is the read control and transfers data from storage onto bit sense lines. The $\phi 3$ signal, which comes after $\phi 2$ only during a write or refresh cycle, transfers data from the bit sense lines back into storage. The $\phi 3$ signal occurs only if the R/W input is low.

To perform a read cycle, CE is brought high to initiate a $\phi 2$ signal and latch the input addresses. The column decoders select one column in each of the four storage quadrants (see the block diagram) and transfers data from storage onto the 128 bit sense lines. The row

decoder selects one of these 128 bit sense lines for read and write operations. During the $\phi 2$ signal, the data on this selected bit sense line is Exclusive ORed with the state of the appropriate data control cell to supply the correct output data. After this data is received by the external system, CE may be brought low to the standby position. This assumes that the R/W signal is held high to prevent an internal $\phi 3$ being generated.

To perform a write or refresh operation, CE is brought high and everything is identical to a read operation up until the 128 bit sense lines are charged with the selected columns of stored data. When R/W is brought low (if it is not already there), a $\phi 3$ signal is generated after $\phi 2$ is over. The $\phi 3$ signal takes the data from the 128 bit sense lines and returns it to the 128 storage locations it came from. Because of the design of the memory array, this $\phi 2$ - $\phi 3$, read-write operation inverts the data. Therefore, one extra row of memory cells, called data control cells, is used to



keep track of the polarity of stored data in order to be able to correctly recover it. During the write operation, the input data is Exclusive ORed with these control cells before being stored in the array. A refresh cycle does not modify any of the bit sense lines, but simply returns the data (now inverted) into storage.

All timing signals for the MCM6605A are specified around these operations. The following is a brief description of the input pins and relevant timing requirements.

Chip Enable — CE is a single high level clock which initiates all memory cycles. CE can remain low as long as desired for specific applications as long as the 2.0 ms refresh requirements are met.

Chip Select — This signal controls only the I/O buffers. When $\overline{CS}$ is high, the input is disconnected and the output is in the 3-state high-impedance state. A refresh cycle is, therefore, a write cycle with $\overline{CS}$ high. $\overline{CS}$ has no critical timing with respect to any other signal except that there is a finite delay between activation and data out.

Read/Write — When high, R/W inhibits the internal $\phi 3$ signal, thereby keeping the memory from writing. When R/W is low, a $\phi 3$ will occur soon after $\phi 2$ is finished. For a read cycle, R/W should be high within t_{RWD} of CE to insure that a $\phi 3$ does not start. The only timing requirement on the R/W input for writing is a minimum write pulse defined as the overlap of $\overline{CS}$, CE, and R/W. Refresh cycles require that $\overline{CS}$ be high to inhibit the input buffer before a $\phi 3$ occurs. Thus $\overline{CS}$ should be high within t_{CSD} for a refresh cycle, or before R/W goes low for a read-refresh cycle.

Data In — The input data must be valid for a sufficient time to override the data stored on the selected bit sense line. It must remain valid for the "write pulse" defined under Read/Write. Signals on the D_{in} pin are ignored when either $\overline{CS}$ or R/W is high, or CE is low.

Data Out — Output data is inverted from input data and is valid t_{acc} after CE goes high. The data will remain valid as long as CE is high and $\overline{CS}$ remains low. With either CE low or $\overline{CS}$ high, the output is in a high-impedance state. The data output is initially precharged high when CE goes high and is then either discharged to ground or left high depending on the stored data. This precharging followed by valid data occurs regardless of the state of the R/W input, making the write cycle actually a read-write cycle. The output will also try to precharge during a refresh cycle but will be kept at high impedance by the $\overline{CS}$ being high. If $\overline{CS}$ is originally low and is then brought high (within the t_{CSD} specification) the output may start to precharge before being cut off and returned to high impedance.

Addresses — The addresses are latched when CE goes high, and may be removed after an appropriate hold time.

VSS — Circuit ground.

VBB — The reverse bias substrate supply. Forward biasing this supply with respect to VSS will destroy the memory device.

VDD — Positive supply voltage.

VCC — Output buffer supply. This supply goes only to the data output buffer and draws current only when driving an output load high.

Preset — This pin should be tied to ground. During device testing Preset can be used to preset the data control cells to a logic zero. One 200 ns, 12 V pulse will set all 32 cells simultaneously. Preset has no system use; its only purpose is to ensure a good logic level in the control cells after first power up. In system use, this good logic level will come naturally after the first few refresh cycles.

APPLICATIONS INFORMATION

Power Supplies

The MCM6605A is a dynamic RAM which has essentially zero power drain when in the standby (CE low) mode. When operating, the VDD supply may experience transients in the order of 100 mA for a short time (Figure 9). The VBB supply, which has very low dc drain while operating, may see transients of about 40 mA during the edges of CE. Therefore, appropriate bypassing of both supplies is recommended. This bypassing has been simplified by the location of the power supply pins on the corners of the package.

The VCC line supplies only the input leakage of a TTL load on Data Out and should never exceed about 100 μA , presenting little bypassing requirement.

Power dissipation for a system of N chips is much lower than N times the 335 mW typical dissipation for a full speed operating chip. This is because the unselected rows in a memory array card are operating in the standby mode of near zero dissipation. This zero standby power is actually unachievable because of the requirements for refresh. Therefore, power dissipation for an array of N X M chips operating at t_1 cycle time, t_{REF} refresh increment, and maximum CE down time between cycles is:

$$P_D \approx M \left(\frac{490 \text{ ns}}{t_1 \text{ ns}} \right) 335 \text{ mW} + (N-1) (M) \left(\frac{15.7}{t_{REF} \mu s} \right) 335 \text{ mW}$$

For a 550-ns-cycle-time, 64 k by 16 system (16 by 16 chip array) with refresh at 2.0 ms, the approximate power dissipation is:

$$P_D \approx 16 \left(\frac{490}{550} \right) 335 + (15) (16) \left(\frac{15.7}{2000} \right) 335 \\ \approx 4775 \text{ mW} + 630 \text{ mW} = 5.4 \text{ W}$$

A similar one megabyte system, eight bytes wide, would have a dissipation of only 24 W. If the low standby power capability were not used, over 600 W would be dissipated.

Refresh

The MCM6605A is refreshed by performing a refresh (or write) cycle on each of the 32 combinations of the least significant address bits (A0-A4) within a 2.0 ms time period. (A5-A11 must remain constant at proper logic levels.) This refresh can be done in a burst mode (32 cycles starting every 2.0 ms) or in a distributed mode where one cycle is done every 62.5 μs .



A refresh abort can be accomplished by treating a refresh cycle as a read-modify-write cycle with $\overline{CS}$ high. This type of cycle can be aborted any time until the R/W signal has been brought low to allow a $\phi 3$ clock to begin.

Non-Volatile Storage

In many digital systems, it is extremely important to retain data during emergencies such as power failure. Unfortunately, however, most random access read/write semiconductor memories such as the MCM6605A are volatile. That is, if power is removed from the semiconductor memory, stored information is lost. Therefore, non-volatility for a specified period of time becomes highly desirable — as a necessity to maintain irreplaceable information or as a convenience to avoid the time consuming and troublesome task of having to reload the memory.

The extremely low standby power dissipation of the MCM6605A makes it ideal for main memory applications requiring battery backup for non-volatility. For example, the MCM6605A can be employed in an 8K byte non-volatile main memory system application for microprocessors. The memory system can be partitioned into three major sections as illustrated in Figure 13. The first section contains the address buffers and the Read/Write and Chip Select decoding logic. The second section consists of the

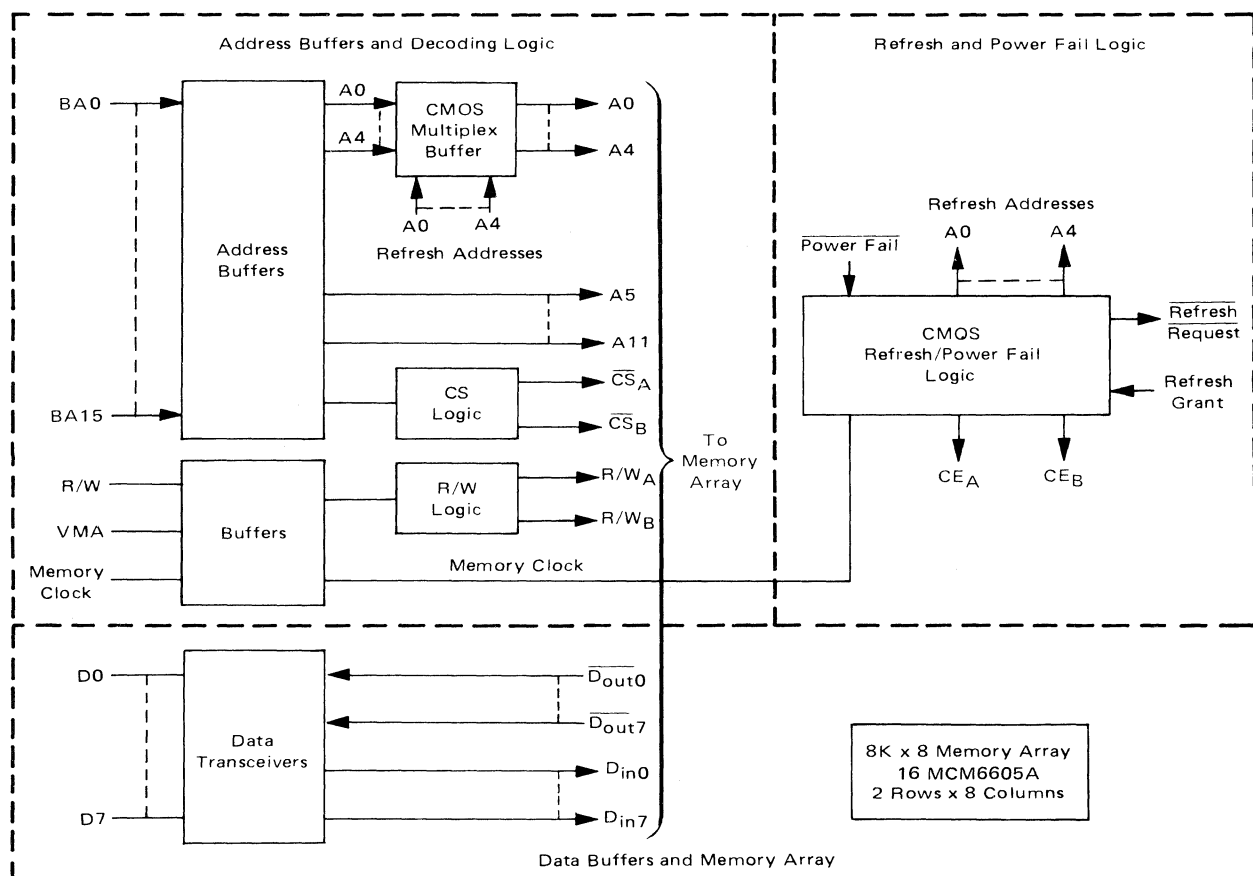
data bus buffering transceivers and the memory array (which consists of 16 MCM6605As) organized into two rows of 4K bytes each.

The third section of the block diagram comprises refresh and control logic for the memory system. This logic interfaces the timing of the refresh handshaking with the microprocessor (MPU) clock circuitry. It handles requests for refresh, the generation of refresh addresses, the synchronization of a Power Fail signal, the multiplexing of the external Memory Clock with the internal clock (used during standby), and the generation of a -5 V supply on the board using a charge-pump method.

The refresh control logic is illustrated in Figure 14. It handles the refreshing of the memory during both operating and standby modes. The timing for this logic is given in Figure 15. Figure 16 gives the memory timing for the standby mode only. Decoding of the memory clock (CE_A and CE_B) and the circuitry to synchronize the Power Fail signal are shown in Figure 17, with the timing given in Figure 18.

The memory device clock (CE_A and CE_B) during standby is created by a monostable multivibrator (MC14528) and buffered from the memory array by three MC14503 buffers in parallel. This clock is multiplexed with the Memory Clock by use of the three-state feature of the

FIGURE 13 — NON-VOLATILE MEMORY SYSTEM BLOCK DIAGRAM



MC14503. The Memory Clock (used during normal operation) is translated to 12 V levels by use of an MC3460 Clock Driver. Decoding of the $\overline{CE_A}$ and $\overline{CE_B}$ signals (i.e., clocking only the memory bank addressed) to conserve power is accomplished by the logic within the MC3460.

Since the $\overline{\text{Power Fail}}$ signal will occur asynchronously with both the Memory Clock and the refreshing operation (Refresh Clock), it is necessary to synchronize the $\overline{\text{Power Fail}}$ signal to the rest of the system in order to avoid aborting a memory access cycle or a refresh cycle. An MC14027 dual flip-flop is used as the basic synchronization device. The leading edge of the Refresh Clock triggers a 3 μs monostable multivibrator which is used as a refresh pretrigger. The trailing edge of this pretrigger triggers a 500 ns monostable which creates the CE pulse during standby operation. The 3 μs pretrigger signal is used to set half of the MC14027 flip-flop, the output of which, (B), then inhibits a changeover from the standby to the operating modes (or vice versa). This logic prevents the system from aborting a refresh cycle should the $\overline{\text{Power}}$

$\overline{\text{Fail}}$ signal change states just prior to or during a refresh cycle. The trailing edge of the 500 ns monostable clears the MC14027 flip-flop, enabling the second flip-flop in the package. The state of $\overline{\text{Power Fail}}$ and Power Fail is applied to the K and J inputs of this second flip-flop and is synchronized by clocking with Memory Clock. The outputs of this flip-flop, labeled Bat and $\overline{\text{Bat}}$, lock the system into the refresh mode and multiplex in the internal clock for standby operation when Bat = "1". The voltage to logic not required for the refresh only mode of operation is removed to conserve power.

By using CMOS for the refresh logic and capacitance drivers, and a low current refresh oscillator, the standby current required for the 8K byte system is extremely small, as noted in Table 1. This low standby current requirement can be easily supplied for several days with standard type +12 V batteries. For more detailed information on this system and a large mainframe memory system, see Application Notes AN-732 and AN-740.

FIGURE 14 – REFRESH CONTROL LOGIC

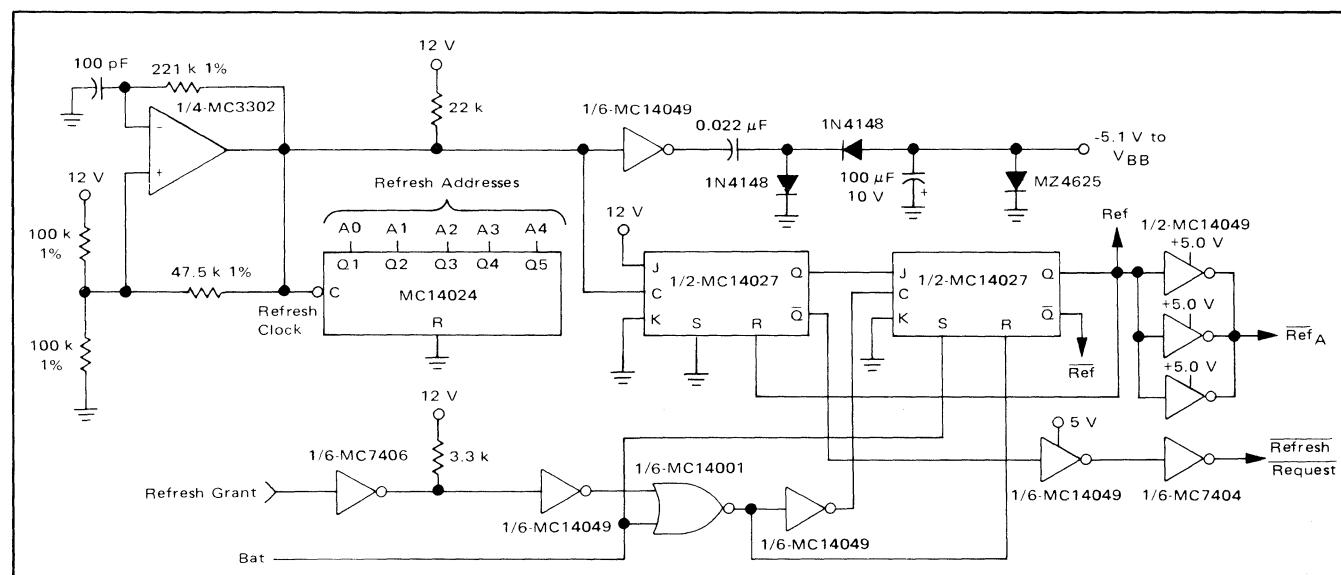


FIGURE 15 – REFRESH TIMING

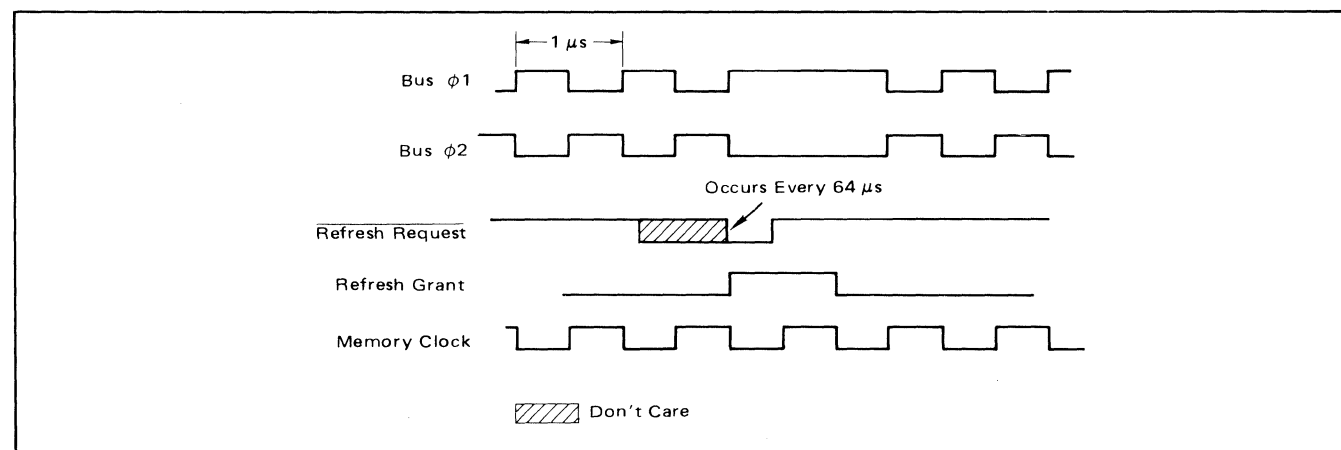


FIGURE 16 – MEMORY TIMING IN STANDBY MODE

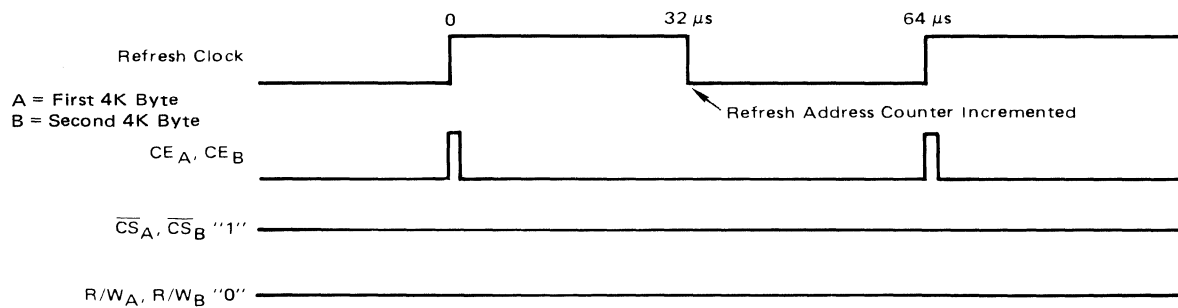


FIGURE 17 – POWER FAIL LOGIC AND CHIP ENABLE DRIVER

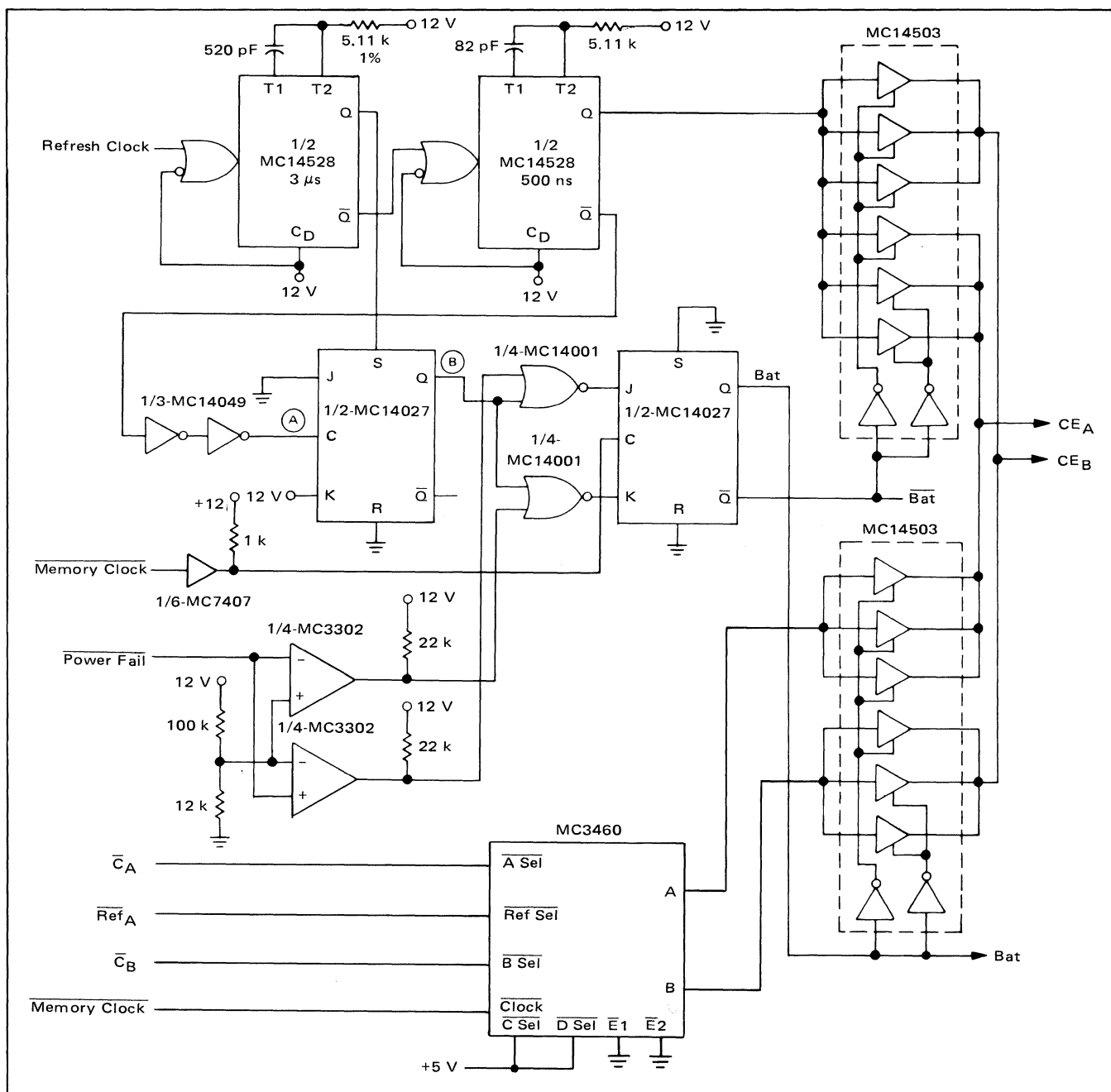


FIGURE 18 – POWER UP/DOWN SYNCHRONIZATION

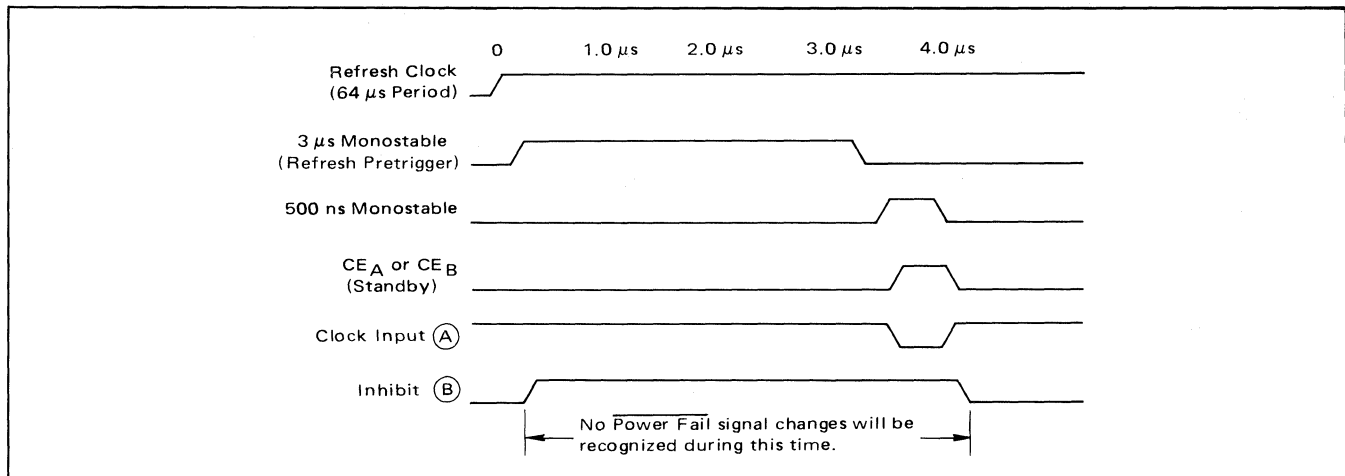
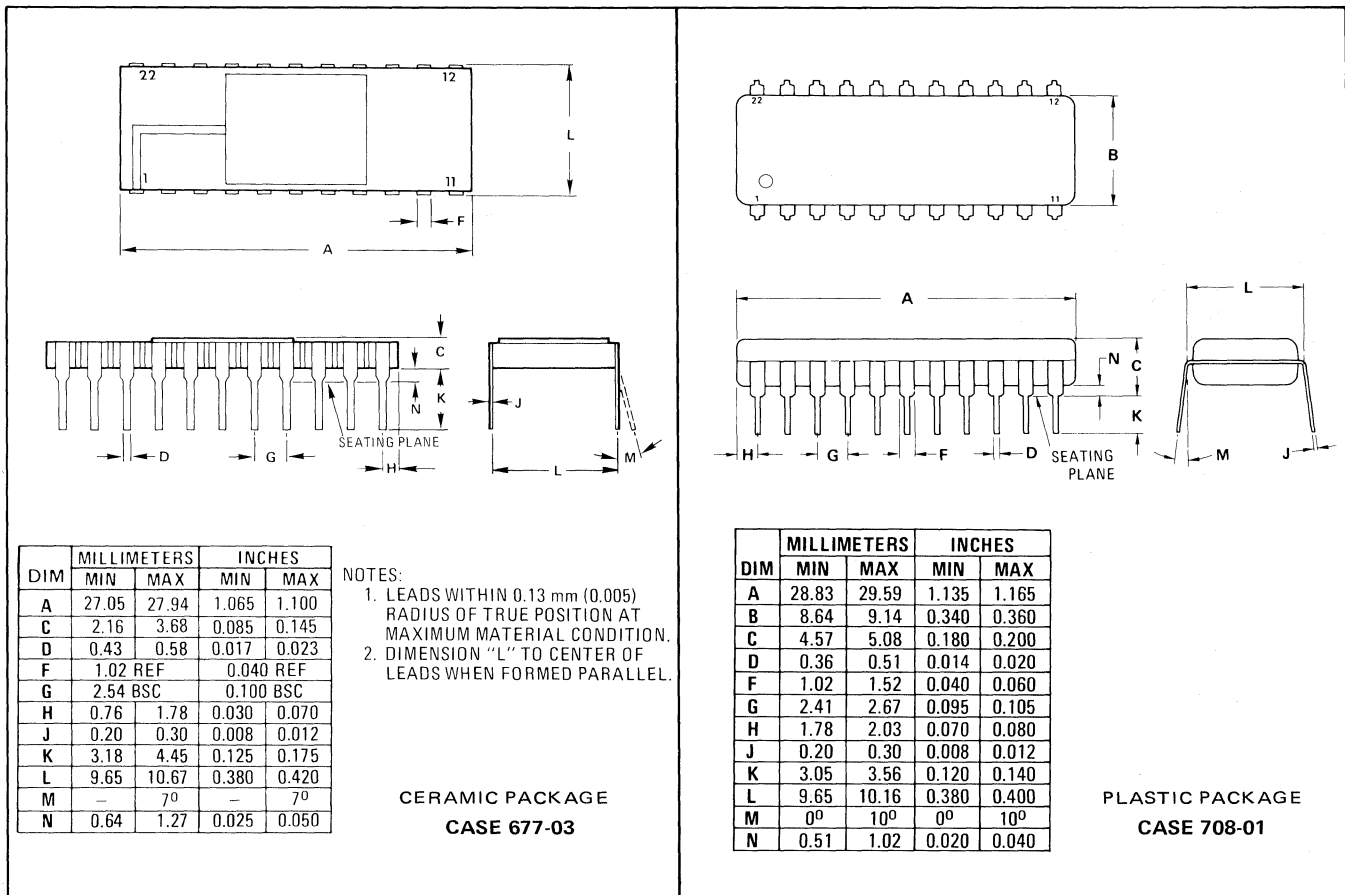


TABLE 1 – STANDBY MODE CURRENT ALLOCATION

Circuit Section	Typical Current
+12 V Current (V_{DD}) for 16 MCM6605A's	5 mA
Charge Pump	3 mA
Comparator	2 mA
Capacitance Drivers	4 mA
Total	14 mA

Circuit diagrams utilizing Motorola products are included as a means of illustrating typical semiconductor applications; consequently, complete information sufficient for construction purposes is not necessarily given. The information has been carefully checked and is believed to be entirely reliable. However, no responsibility is assumed for inaccuracies. Furthermore, such information does not convey to the purchaser of the semiconductor devices described any license under the patent rights of Motorola Inc. or others.

PACKAGE DIMENSIONS





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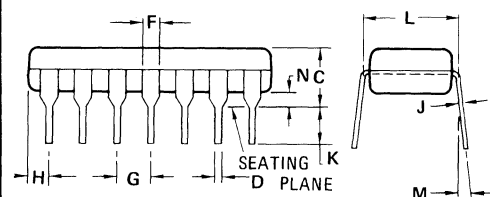
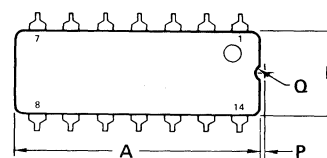
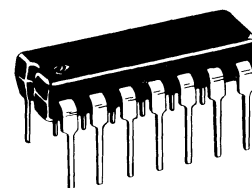
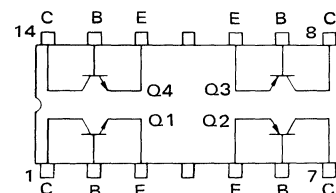
MPQ6842

MPU CLOCK BUFFER

The MPQ6842 is designed to provide the switching speed and saturation voltages necessary in the clock circuit for the $\phi 1$ and $\phi 2$ inputs of the MC6800 Microprocessor.

MPU CLOCK BUFFER

CONNECTION DIAGRAM



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	18.16	18.80	0.715	0.740
B	6.10	6.60	0.240	0.260
C	4.06	4.57	0.160	0.180
D	0.38	0.51	0.015	0.020
F	1.02	1.52	0.040	0.060
G	2.54 BSC		0.100 BSC	
H	1.32	1.83	0.052	0.072
J	0.20	0.30	0.008	0.012
K	2.92	3.43	0.115	0.135
L	7.37	7.87	0.290	0.310
M	10°		10°	
N	0.51	1.02	0.020	0.040
P	0.13	0.38	0.005	0.015
Q	0.51	0.76	0.020	0.030

NOTES:

- LEADS WITHIN 0.13 mm (0.005) RADIUS OF TRUE POSITION AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION.
- DIMENSION "L" TO CENTER OF LEADS WHEN FORMED PARALLEL

CASE 646
PLASTIC PACKAGE

MAXIMUM RATINGS

Rating	Symbol	Value		Unit
Collector-Emitter Voltage	V_{CEO}	30		Vdc
Collector-Base Voltage	V_{CB}	30		Vdc
Emitter-Base Voltage	V_{EB}	4.0		Vdc
Collector Current — Continuous	I_C	200		mA dc
		Each Transistor	Four Transistors Equal Power	
Total Power Dissipation @ $T_A = 25^\circ\text{C}$ (1) Derate above 25°C	P_D	500 4.0	900 7.2	mW mW/ $^\circ\text{C}$
Total Power Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	825 6.7	2400 19.2	mW mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150		$^\circ\text{C}$

(1) Second Breakdown occurs at power levels greater than 3 times the power dissipation rating.

THERMAL CHARACTERISTICS

Characteristic		Junction to Case	Junction to Ambient	Unit
Thermal Resistance	Each Die	151	250	$^\circ\text{C}/\text{W}$
	Effective, 4 Die	52	139	$^\circ\text{C}/\text{W}$
Coupling Factors	Q1-Q4 or Q2-Q3	34	70	%
	Q1-Q2 or Q3-Q4	2.0	26	%

THERMAL COUPLING AND EFFECTIVE THERMAL RESISTANCE

In multiple chip devices, coupling of heat between die occurs. The junction temperature can be calculated as follows:

$$(1) \Delta T_{J1} = R_{\theta 1} P_{D1} + R_{\theta 2} K_{\theta 2} P_{D2} + R_{\theta 3} K_{\theta 3} P_{D3} + R_{\theta 4} K_{\theta 4} P_{D4}$$

Where ΔT_{J1} is the change in junction temperature of die 1
 $R_{\theta 1}$ thru 4 is the thermal resistance of die 1 through 4
 P_{D1} thru 4 is the power dissipation in die 1 through 4
 $K_{\theta 2}$ thru 4 is the thermal coupling between die 1 and die 2 through 4.

An effective package thermal resistance can be defined as follows:

$$(2) R_{\theta (EFF)} = \Delta T_{J1} / P_{DT}$$

Where: P_{DT} is the total package power dissipation.

Assuming equal thermal resistance for each die, equation (1) simplifies to

$$(3) \Delta T_{J1} = R_{\theta 1} (P_{D1} + K_{\theta 2} P_{D2} + K_{\theta 3} P_{D3} + K_{\theta 4} P_{D4})$$

For the conditions where $P_{D1} = P_{D2} = P_{D3} = P_{D4}$, $P_{DT} = 4P_D$, equation (3) can be further simplified and by substituting into equation (2) results in

$$(4) R_{\theta (EFF)} = R_{\theta 1} (1 + K_{\theta 2} + K_{\theta 3} + K_{\theta 4}) / 4$$

Values for the coupling factors when either the case or the ambient is used as a reference are given in the table on page 1. If significant power is to be dissipated in two die, die at the opposite ends of the package should be used so that lowest possible junction temperatures will result.

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Collector-Emitter Breakdown Voltage ⁽¹⁾ ($I_C = 10 \text{ mAdc}$, $I_B = 0$)	BV_{CEO}	30	—	—	Vdc
Collector-Base Breakdown Voltage ($I_C = 10 \mu\text{Adc}$, $I_E = 0$)	BV_{CBO}	30	—	—	Vdc
Emitter-Base Breakdown Voltage ($I_E = 10 \mu\text{Adc}$, $I_C = 0$)	BV_{EBO}	4.0	—	—	Vdc
Collector Cutoff Current ($V_{CB} = 20 \text{ Vdc}$, $I_E = 0$)	I_{CBO}	—	—	50	nAdc
Emitter Cutoff Current ($V_{EB} = 3.0 \text{ Vdc}$, $I_C = 0$)	I_{EBO}	—	—	50	nAdc

ON CHARACTERISTICS (1)

DC Current Gain ($I_C = 0.5 \text{ mAdc}$, $V_{CE} = 1.0 \text{ Vdc}$) ($I_C = 1.0 \text{ mAdc}$, $V_{CE} = 1.0 \text{ Vdc}$) ($I_C = 10 \text{ mAdc}$, $V_{CE} = 1.0 \text{ Vdc}$)	h_{FE}	30 50 70	— — —	— — —	—
Collector-Emitter Saturation Voltage ($I_C = 0.5 \text{ mAdc}$, $I_B = 0.05 \text{ mAdc}$, $0^\circ\text{C} \leq T \leq 70^\circ\text{C}$)	$V_{CE(sat)}$	—	0.05	0.15	Vdc
Base-Emitter Saturation Voltage ($I_C = 0.5 \text{ mAdc}$, $I_B = 0.05 \text{ mAdc}$)	$V_{BE(sat)}$	—	0.65	0.9	Vdc

DYNAMIC CHARACTERISTICS

Current-Gain — Bandwidth Product ⁽¹⁾ ($I_C = 10 \text{ mAdc}$, $V_{CE} = 20 \text{ Vdc}$, $f = 100 \text{ MHz}$)	f_T	200	350	—	MHz
Output Capacitance ($V_{CB} = 5.0 \text{ Vdc}$, $I_E = 0$, $f = 100 \text{ kHz}$)	C_{ob}	—	3.0	4.5	pF
Input Capacitance ($V_{EB} = 0.5 \text{ Vdc}$, $I_C = 0$, $f = 100 \text{ kHz}$)	C_{ib}	—	5.0 4.0	10 8.0	pF

SWITCHING CHARACTERISTICS ($T_A = 25^\circ\text{C}$, $V_{CC} = 5.0 \text{ Vdc}$)

Propagation Delay Time (50% Points TP1 to TP3) (50% Points TP2 to TP4)	t_{PLH} t_{PHL}	— —	15 6.0	25 15	ns
Rise Time (0.3 V to 4.7 V, TP3 or TP4)	t_r	5.0	25	35	ns
Fall Time (4.7 V to 0.3 V, TP3 or TP4)	t_f	5.0	10	20	ns

(1) Pulse Test: Pulse Width $\leq 300 \mu\text{s}$, Duty Cycle $\leq 2.0\%$.



NPN

PNP

FIGURE 1 – DC CURRENT GAIN

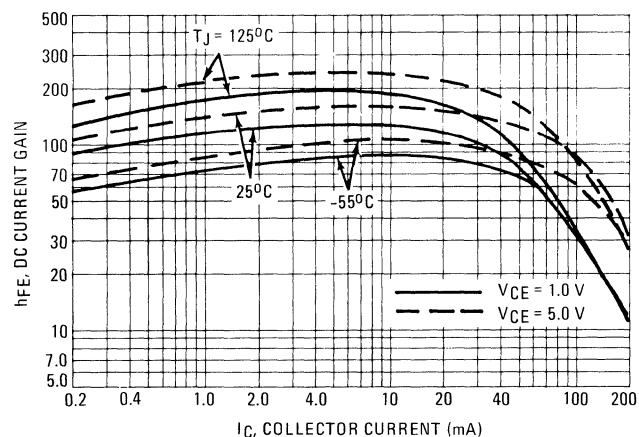
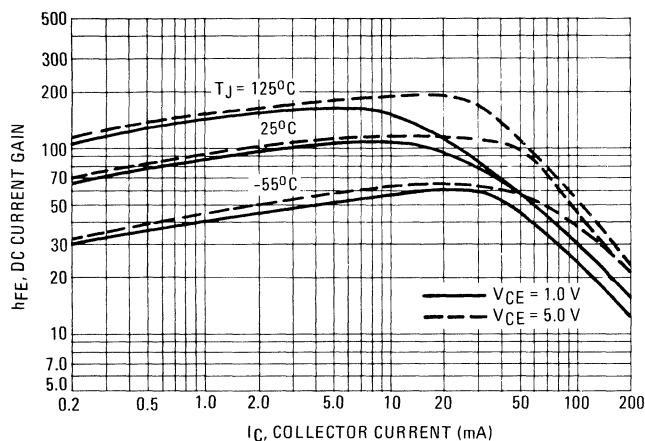


FIGURE 2 – "ON" VOLTAGE

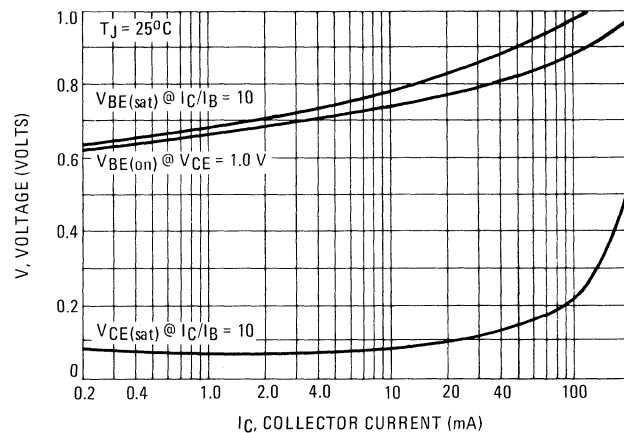
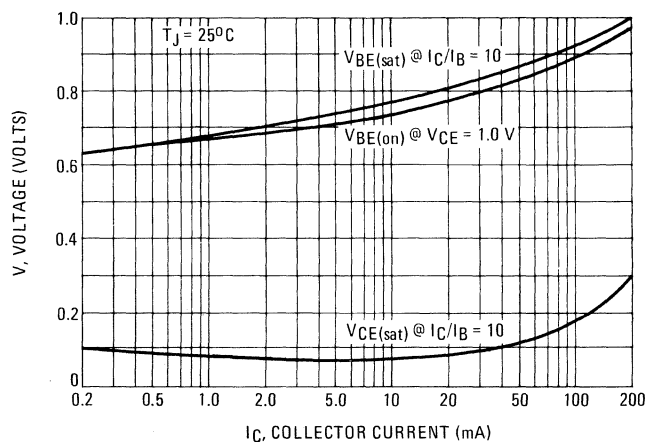
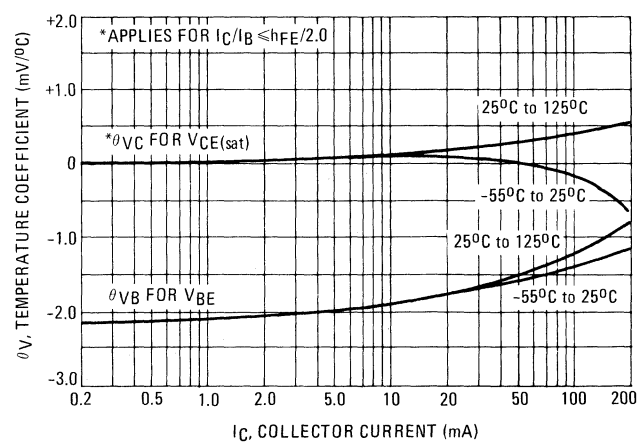
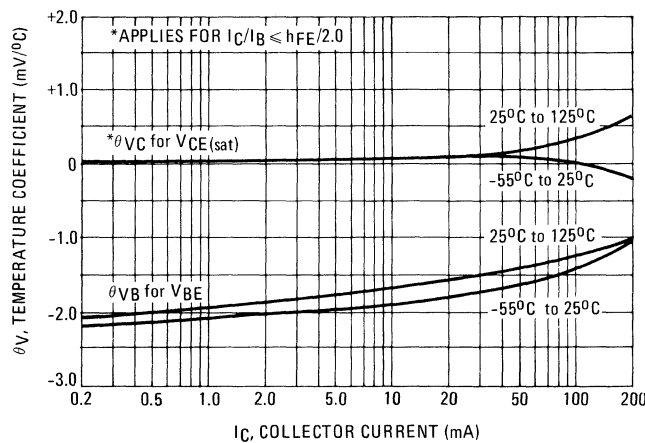
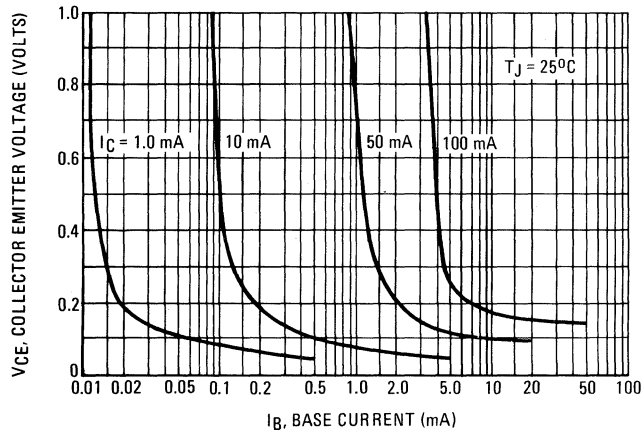


FIGURE 3 – TEMPERATURE COEFFICIENTS



NPN

FIGURE 4 – COLLECTOR SATURATION REGION



PNP

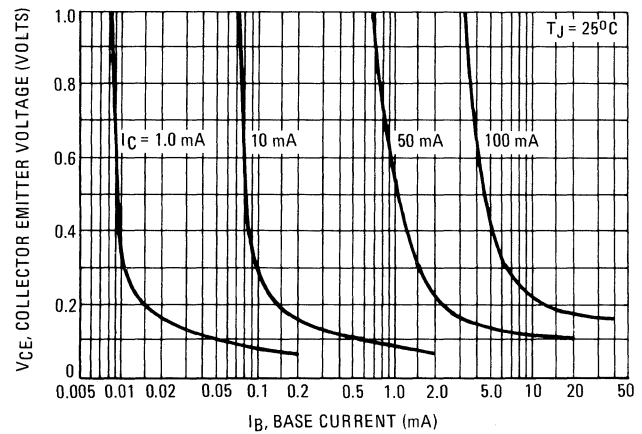
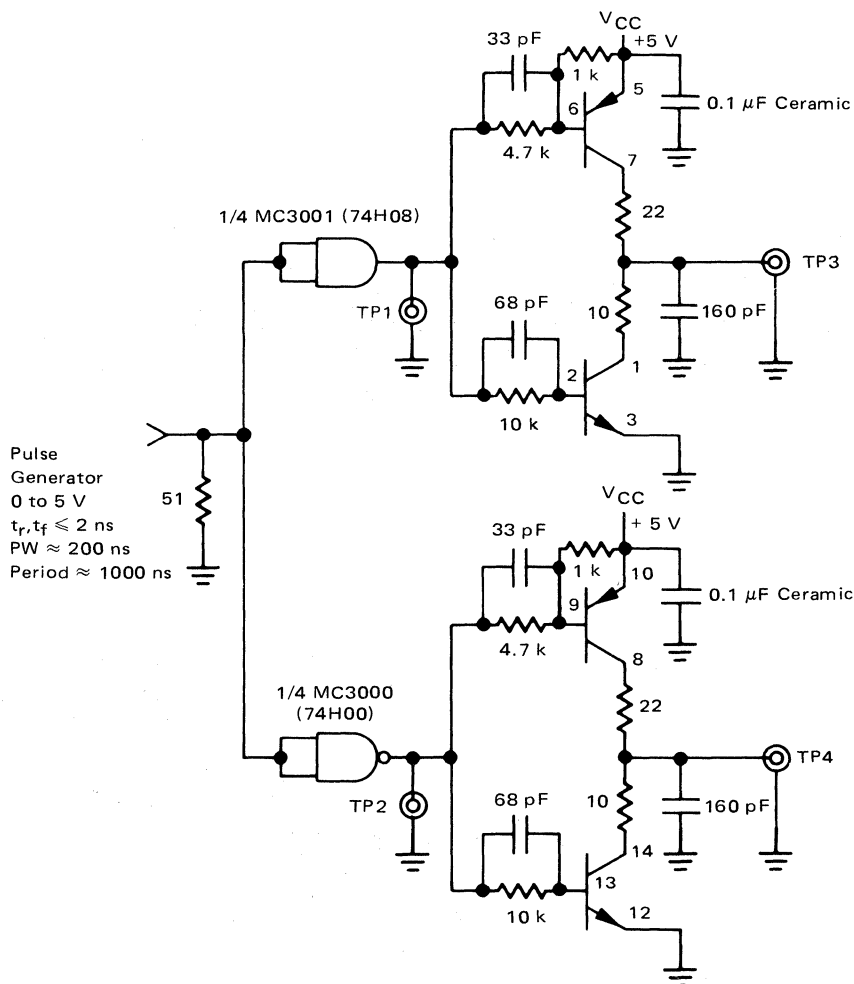
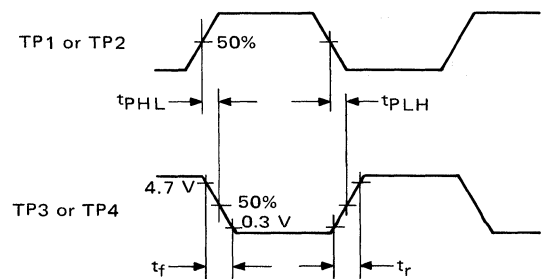


FIGURE 5 – SWITCHING TIMES TEST CIRCUIT AND WAVEFORMS



NOTES:

1. Unless otherwise noted, all resistors carbon composition $\frac{1}{4}\text{ W} \pm 5\%$, all capacitors dipped mica $\pm 2\%$.
2. Use short interconnect wiring with good power and ground busses.
3. TP1 thru TP4 are coaxial connectors to accept scope probe tip and provide a good ground.
4. Device under test is MPQ6842.
5. 160 pF load does not include stray or scope probe capacitance.
6. Scope probe resistance $> 5\text{ k}\Omega$.
Scope probe capacitance $< 10\text{ pF}$.



APPLICATIONS INFORMATION

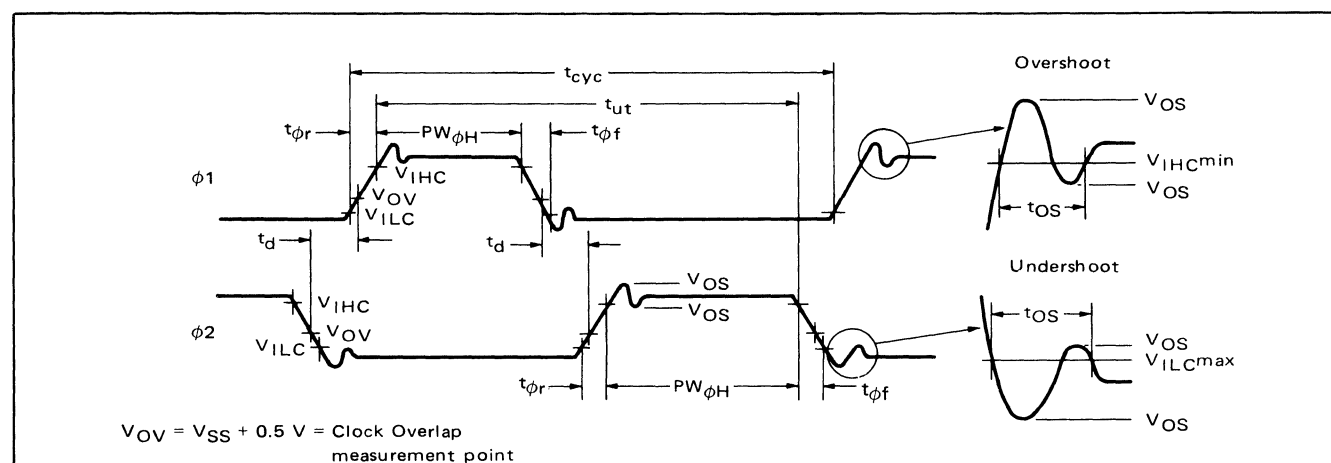
Figure 6 is a summary of the MC6800 Microprocessor clock waveform requirements. The $\phi 1$ and $\phi 2$ clock inputs require complementary 1 MHz, 5 volt non-overlapping clocks. The clock inputs of the MPU appear primarily capacitive, being 120 pF typical and 160 pF maximum with 100 μ Adc leakage. Provision is made in the specification for the undershoot and overshoot that will result from the generation of a high speed transition into a capacitive load.

The clock specifications which constrain the clock driver are the rise and fall times required to meet the pulse widths at the maximum operating frequency of 1 MHz, the non-overlapping requirement, the logic level requirements of $V_{SS} + 0.3$ volt and $V_{CC} - 0.3$ volt, the

overshoot specification, and the MPU input capacitance. The clock buffer circuit that drives the MPU clock inputs must be designed to meet the rise and fall time requirements as well as provide the proper logic levels into the load capacitance, within the overshoot constraints. The non-overlapping requirements of the clock signal can be met by the design of the control logic which drives the buffers. The MPQ6842 clock buffer can guarantee the clock designer the speed and saturation voltages necessary to design the clock circuit to meet the MPU clock requirements.

Figure 7 is a circuit designed with TTL logic devices and the MPQ6842 buffer to meet the MPU clock requirements while operating from a single +5 volt supply. The oscillator

FIGURE 6 — MPU CLOCK SPECIFICATION

ELECTRICAL CHARACTERISTICS ($V_{CC} = 5.0 \text{ V} \pm 5\%$, $V_{SS} = 0$, $T_A = 0$ to 70°C unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input High Voltage $\phi 1, \phi 2$	V_{IHC}	$V_{CC} - 0.3$	—	$V_{CC} + 0.1$	Vdc
Input Low Voltage $\phi 1, \phi 2$	V_{ILC}	$V_{SS} - 0.1$	—	$V_{SS} + 0.3$	Vdc
Clock Overshoot/Undershoot — Input High Level	V_{OS}	$V_{CC} - 0.5$	—	$V_{CC} + 0.5$	Vdc
— Input Low Level		$V_{SS} - 0.5$	—	$V_{SS} + 0.5$	Vdc
Input Leakage Current $\phi 1, \phi 2$ ($V_{in} = 0$ to 5.25 V , $V_{CC} = 0 \text{ V}$)	I_{in}	—	—	100	μAdc
Capacitance* ($V_{in} = 0$, $T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$)	C_{in}	80	120	160	pF
Frequency of Operation	f	0.1	—	1.0	MHz
Clock Timing					
Cycle Time	t_{cyc}	1.0	—	10	μs
Clock Pulse Width (Measured at $V_{CC} - 0.3 \text{ V}$)	$PW_{\phi H}$				ns
$\phi 1$		430	—	4500	
$\phi 2$		450	—	4500	
Total $\phi 1$ and $\phi 2$ Up Time	t_{ut}	940	—	—	ns
Rise and Fall Times $\phi 1, \phi 2$ (Measured between $V_{SS} + 0.3 \text{ V}$ and $V_{CC} - 0.3 \text{ V}$)	$t_{\phi R}, t_{\phi F}$	5.0	—	50	ns
Dealy Time or Clock Separation (Measured at $V_{OV} = V_{SS} + 0.5 \text{ V}$)	t_d	0	—	9100	ns
Overshoot Duration	t_{OS}	0	—	40	ns

*Capacitances are periodically sampled rather than 100% tested.

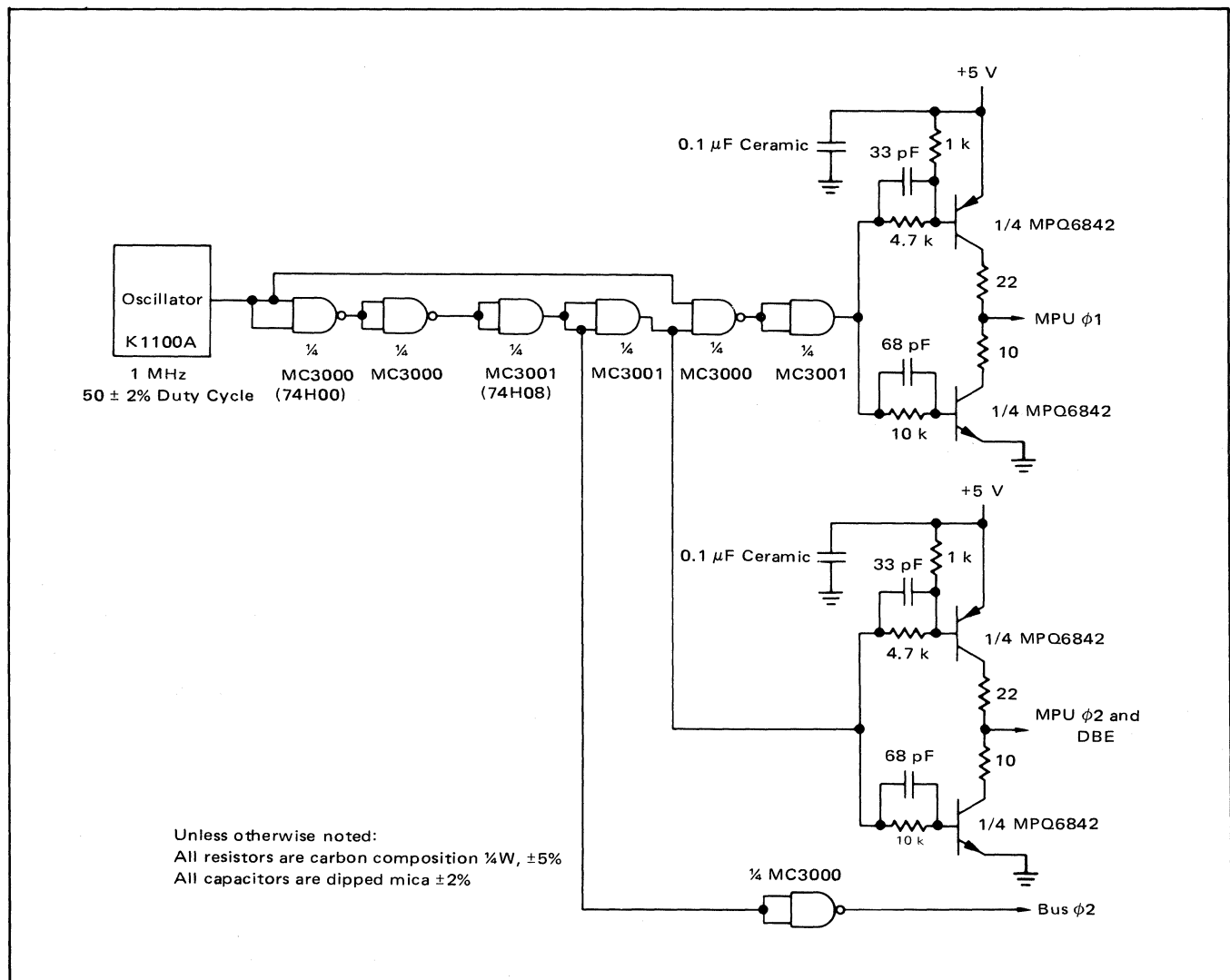
#For further information on M6800 system usage, refer to the M6800 Applications Manual.



can be any source with a maximum frequency of 1 MHz, TTL logic levels and a 50% duty cycle. This oscillator signal source could vary from a commercial oscillator, such as a K1100A available from Motorola's Component Products Department,¹ to a signal derived from a higher frequency signal already available in the system. The TTL gates shown are standard MC3000 and MC3001 (74H00 and 74H08) gates which were chosen for their speed and drive characteristics. The discrete buffers require good "1" level pullup and drive capability which is provided by the MC3001. The circuit was constructed on a wirewrap board and tested on an EXORciser.² Good power and ground distribution practice was followed but no special care was taken in parts layout.

Waveforms typical of the circuit in Figure 7 are shown in Figures 8a and 8b, with $T_A = 20^\circ\text{C}$ and $V_{CC} = 5.0$ volts. These figures depict the logic levels and pulse widths achieved by this circuitry with V_{CC} and Ground as reference levels. Figure 8c superimposes the two clock waveforms so that their phase relationship can be seen. Figure 8d shows the phase relationship of Bus $\phi 2$ and MPU $\phi 2$. Figures 8e and 8f examine the non-overlap regions as well as rise and fall times typical of this clock driver circuit. Table 1 presents test data taken over a voltage range of 4.75 volts to 5.25 volts and over a temperature range of 0°C to 70°C . Note the stability of these measured parameters, and that the logic levels achieved will provide noise margin on the system clocks.

FIGURE 7 — MPU CLOCK CIRCUIT

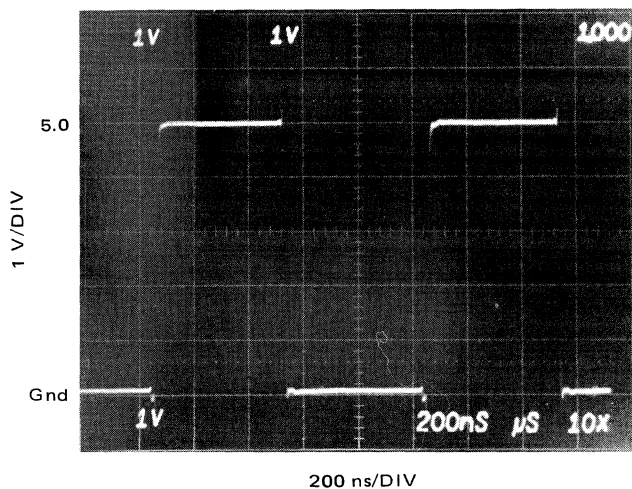
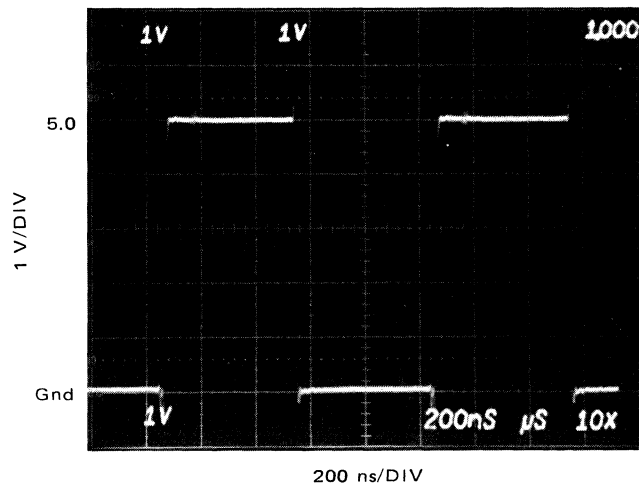
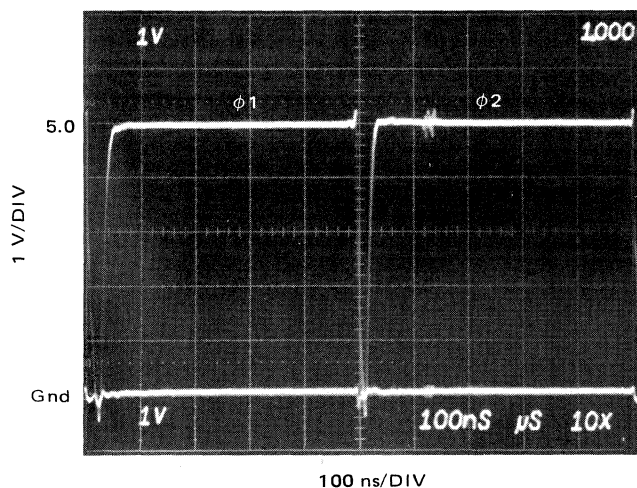
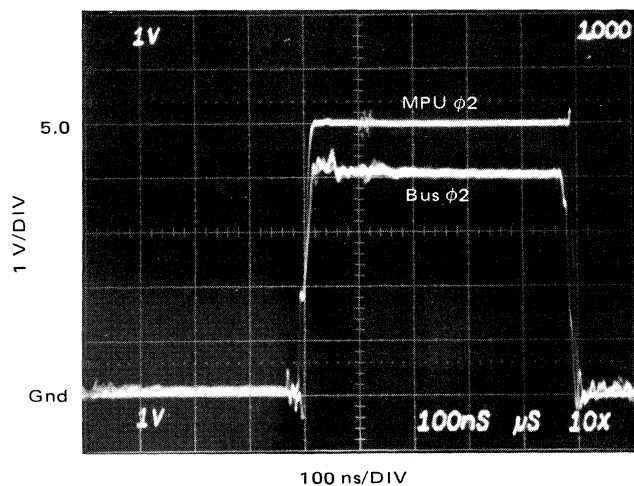
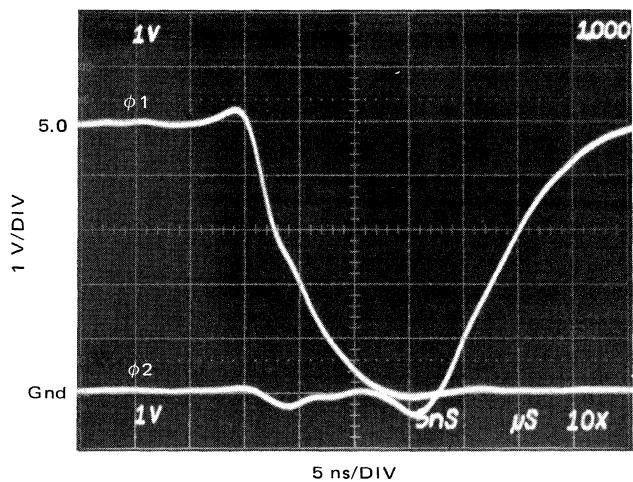
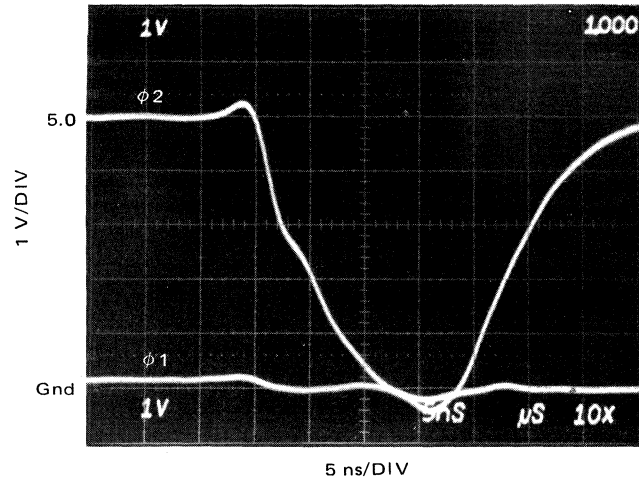


1. 2553 N. Edgington, Franklin Park, Illinois 60131, (312) 451-1000.

2. A prototyping system for the M6800 family.



CLOCK CIRCUIT WAVEFORMS (Using Circuit of Figure 7)

FIGURE 8a — MPU $\phi 1$ CLOCKFIGURE 8b — MPU $\phi 2$ CLOCKFIGURE 8c — MPU $\phi 1$ AND $\phi 2$ CLOCKSFIGURE 8d — MPU $\phi 2$ CLOCK AND BUS $\phi 2$ FIGURE 8e — MPU CLOCK NON-OVERLAP REGION
 $\phi 1$ to $\phi 2$ FIGURE 8f — MPU CLOCK NON-OVERLAP REGION
 $\phi 2$ to $\phi 1$ 

Both $\phi 1$ and $\phi 2$ clock high times were designed to be 15-30 ns wider than the minimum required by the MPU ($\phi 1 = 430$ ns, $\phi 2 = 450$ ns) to provide system margin. Rise and fall times were minimized to provide maximum clock high times consistent with non-critical circuit layout considerations. The overlap margin shown easily meets the MPU requirement of 0 ns at 0.5 volt but will decrease as the capacitive loading increases, as shown in Table 1. The MPU tested for this data had a typical clock input capacitance of 120 pF.

In many systems, especially in the breadboard and evaluation stage, it may be desirable to have the flexibility to vary the system clock to test the effects on data throughput, real time operation with interrupts, or to help diagnose a system timing problem. In these applications, or in those not requiring crystal oscillator stability, an even simpler clock circuit can be used. A pair of cross coupled monostable multivibrators with individual pulse width adjustments can be used as the clock oscillator with the previously described clock buffer circuit. This approach is shown in Figure 9. The non-overlapping clock is generated by the propagation delays through the monostable multivibrators. Figures 10a — 10d show waveforms resulting from this circuit. Table 2 shows test data taken of this

circuit over the voltage and temperature range. Note the small variations in the pulse widths.

The fast rise and fall times produced by this circuitry and the highly capacitive loads require some care in layout to avoid excessive ringing and/or pulse distortion. While no particular care was taken in the construction of the wire-wrap test boards other than placing all of the discrete devices into one header board, the following construction guidelines are recommended. Wide power and ground lines (50-100 mils) should be used to provide low impedance voltage and ground sources. The clock driver should be physically located as near the MPU as possible to avoid ringing down long lines. Close proximity of the clock circuitry to the MPU allows common power and ground connections so that any noise appears common mode rather than differential to the MPU and clock driver. Finally, it is recommended that the MPU $\phi 2$ clock signal not be used to clock any device other than the MPU so that it is not distributed all over the system with the possibility of picking up noise and causing reflections. The circuits shown provide an additional buffer for the other $\phi 2$ loads in the system to isolate MPU $\phi 2$ from all the other $\phi 2$ loads.

TABLE 1 — PERFORMANCE OF CIRCUIT OF FIGURE 7
($C_L = 120$ pF unless otherwise noted.)

Test Conditions	MPU $\phi 1$					MPU $\phi 2$					Non-Overlap Region	
	PW	t_r	t_f	"1" LL*	"0" LL*	PW	t_r	t_f	"1" LL*	"0" LL*	$\phi 1 \downarrow$ to $\phi 2 \uparrow$	$\phi 2 \downarrow$ to $\phi 1 \uparrow$
T = 20°C												
$V_{CC} = 4.75$ V	460 ns	15 ns	10 ns	4.75 V	0.1 V	465 ns	15 ns	10.5 ns	4.75 V	0 V	10.5 ns	12 ns
$V_{CC} = 5.00$ V	460	16	11	5.00	0.1	465	16	10	5.00	0	10	11
$V_{CC} = 5.25$ V	460	16	11	5.25	0.1	465	16	11	5.25	0	9.5	10.5
$V_{CC} = 5.00$ V, ($C_L = 210$ pF)	450	21	15.5	5.00	0.1	460	22	15	5.00	0	2	5.5
T = 70°C												
$V_{CC} = 4.75$ V	460	15	12	4.75	0.1	465	16	12	4.75	0	9	10.5
$V_{CC} = 5.00$ V	460	16	12	5.00	0.1	465	17	12	4.75	0	8.5	10
$V_{CC} = 5.25$ V	455	17	12.5	5.25	0.1	465	17	13	5.25	0	8	9
T = 0°C												
$V_{CC} = 4.75$ V	460	14	10	4.75	0.1	465	15	10.5	4.75	0	11	12
$V_{CC} = 5.00$ V	460	15	10	5.00	0.1	465	15	10	5.00	0	10.5	11.5
$V_{CC} = 5.25$ V	460	15	10.5	5.25	0.1	465	15	10	5.25	0	10	10.5

*Resolution of this measurement $\approx \pm 50$ mV

LEGEND:

PW: Pulse width measured at $V_{CC} - 0.3$ V
 t_r : Rise time measured from 0.3 V to $V_{CC} - 0.3$ V
 t_f : Fall time measured from $V_{CC} - 0.3$ V to 0.3 V

"0" LL: Zero logic level
 "1" LL: One logic level
 Non-Overlap: Measured from 0.5 volt levels



FIGURE 9 — MONOSTABLE CLOCK GENERATOR

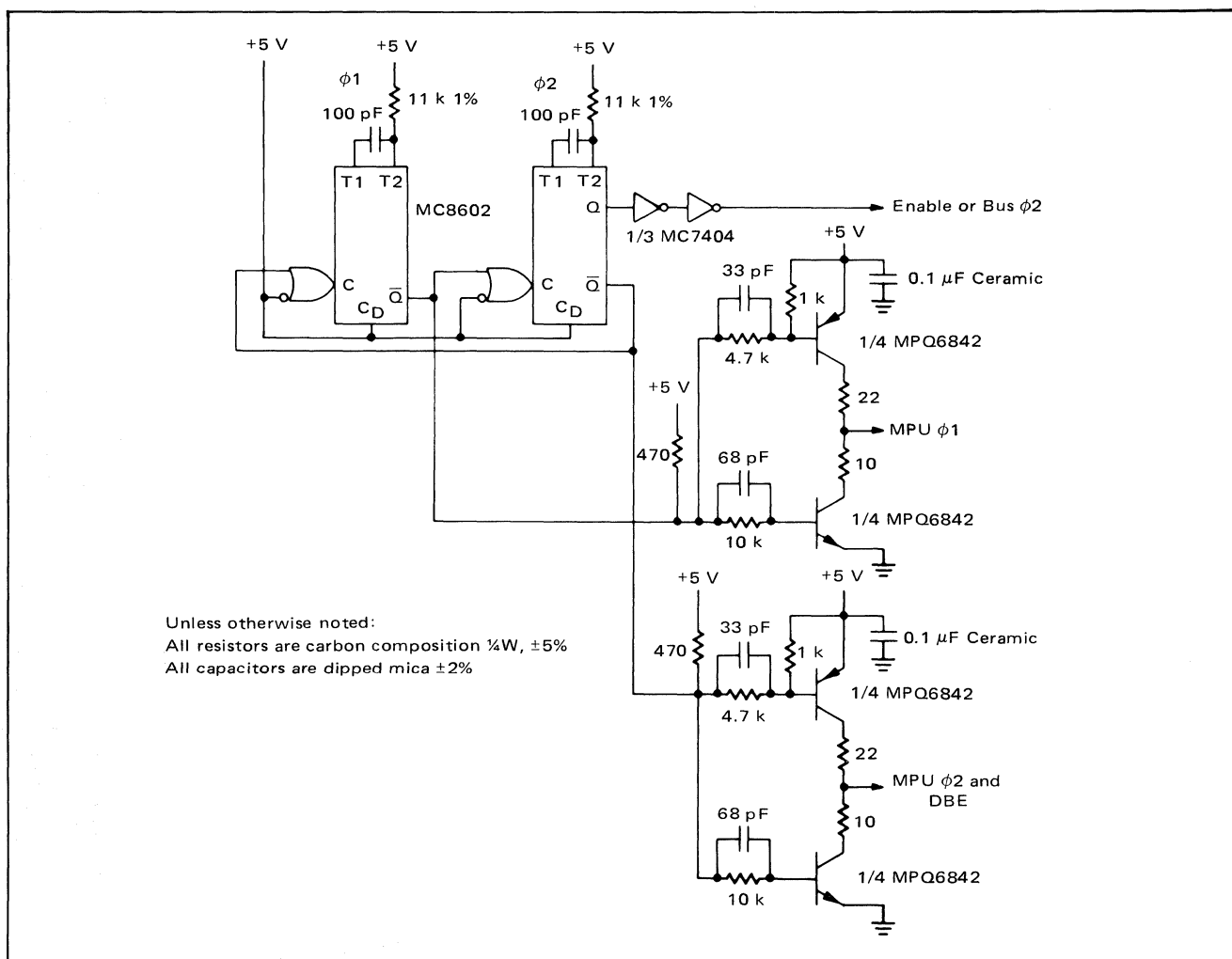


TABLE 2 — PERFORMANCE OF CIRCUIT OF FIGURE 9

Test Conditions	MPU φ1					MPU φ2					Non-Overlap Region	
	PW	t _r	t _f	"1" LL*	"0" LL*	PW	t _r	t _f	"1" LL*	"0" LL*	φ1↓ to φ2↑	φ2↓ to φ1↑
T = 20°C												
V _{CC} = 4.75 V	470 ns	11 ns	11.5 ns	4.75 V	0.1 V	450 ns	12 ns	12 ns	4.75 V	0 V	12 ns	11 ns
V _{CC} = 5.00 V	470	12.5	13	5.00	0.1	460	13	12.5	5.00	0	11	9.5
V _{CC} = 5.25 V	470	13	12	5.25	0.1	460	13.5	12.5	5.25	0	10	9
T = 70°C												
V _{CC} = 4.75 V	455	12.5	13.5	4.75	0.1	450	13	13	4.75	0	11	10
V _{CC} = 5.00 V	455	13	14	5.00	0.1	450	14	14	5.00	0	10	9
V _{CC} = 5.25 V	455	13	14.5	5.25	0.1	450	14	14	5.25	0	8.5	7
T = 0°C												
V _{CC} = 4.75 V	473	12	12	4.75	0.1	470	12	12	4.75	0	11	11
V _{CC} = 5.00 V	475	12	12	5.00	0.1	470	12.5	12	5.00	0	9	11
V _{CC} = 5.25 V	475	12.5	12.5	5.25	0.05	473	12.5	12	5.25	0	9	8

*Resolution of this measurement ≈ ±50 mV

LEGEND: PW: Pulse width measured at V_{CC} - 0.3 Vt_r: Rise time measured from 0.3 V to V_{CC} - 0.3 Vt_f: Fall time measured from V_{CC} - 0.3 V to 0.3 V

"0" LL: Zero logic level

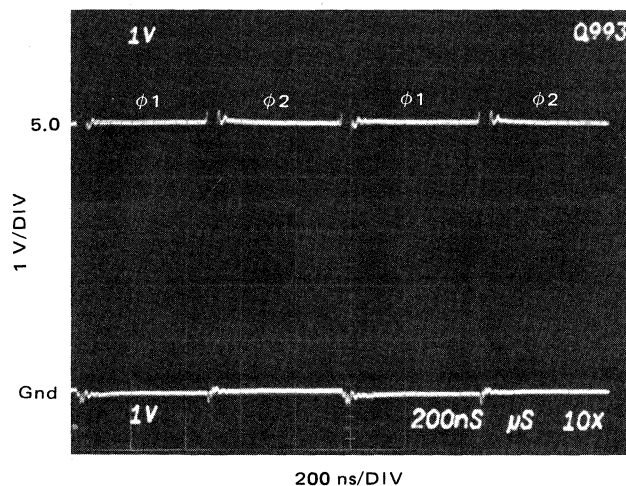
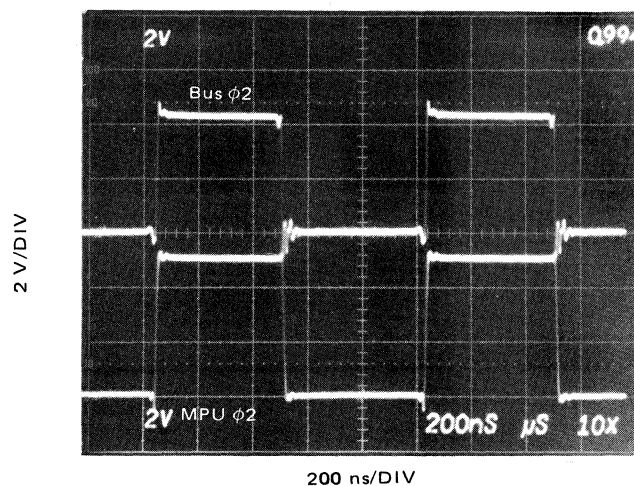
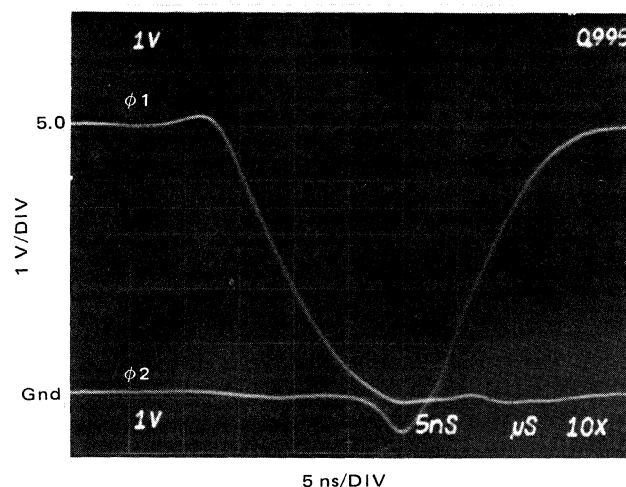
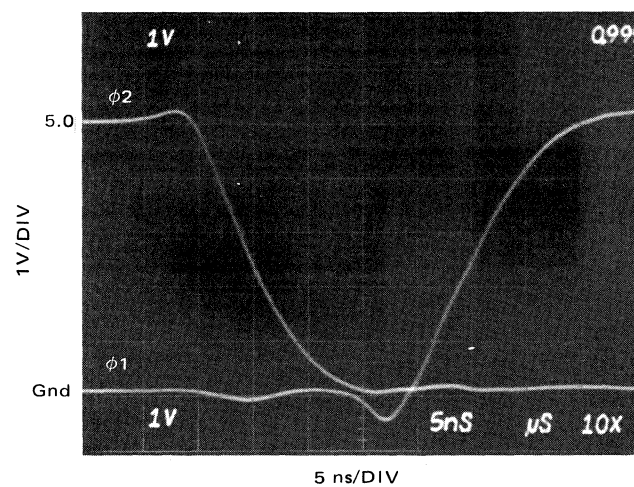
"1" LL: One logic level

Non-Overlap: Measured from 0.5 volt points



CLOCK CIRCUIT WAVEFORMS (Using Circuit of Figure 9)

FIGURE 10a — MPU CLOCK

FIGURE 10b — MPU $\phi 2$ CLOCK AND BUS $\phi 2$ FIGURE 10c — MPU CLOCK NON-OVERLAP REGION
 $\phi 1$ to $\phi 2$ FIGURE 10d — MPU CLOCK NON-OVERLAP REGION
 $\phi 2$ to $\phi 1$ 

CLOCK CIRCUITRY FOR SLOW AND DYNAMIC MEMORIES

The circuitry to modify the clock signals to interface the MC6800 with dynamic (e.g., MCM6605) and slow memories can be evolved from the clock circuitry described previously. Figure 11 expands the clock circuit of Figure 7 (which has a crystal stabilized source) to include interface signals for dynamic (Refresh Request and Refresh Grant) and slow memories (Memory Ready). Note that the only extra parts required are an MC7479 dual latch, an MC7404 hex inverter, and a pair of 10 k ohm pullup resistors. The state of Refresh Request is sampled during the leading edge of $\phi 1$ and, if it is low, the $\phi 1$ and $\phi 2$ clocks to the MPU are held in the high and low states respectively for at least one full clock cycle. A high Refresh Grant signal is issued to indicate to the dynamic memory system that this cycle is a refresh cycle. Upon receipt of the Refresh Grant signal, the memory system

controller sets Refresh Request back high; this is clocked through on the next leading edge of $\phi 1$, thereby restoring the system to normal operation. The Memory Ready line is sampled on the leading edge of $\phi 2$ and, if low, the MPU $\phi 1$ and $\phi 2$ clocks are held in the low and high states, respectively. The clocks will be held in these states until the Memory Ready line is brought high by the slow memory controller, allowing the slow memory controller to determine the amount by which $\phi 2$ is stretched. Figures 12a and 12b show the effect of Refresh Request and Memory Ready signals on the MPU clocks. Note that the Refresh Request signal is asynchronous with the MPU clocks as it is generated by the refresh oscillator in the dynamic memory controller. Figures 13a and 13b show the phase relationship between MPU $\phi 2$, Bus $\phi 2$ and Dynamic Memory Clock. Note that Bus $\phi 2$ and MPU $\phi 2$ are in phase and that the Dynamic Memory Clock leads MPU $\phi 2$ to help offset delays added by the memory



system controller in decoding and level shifting this signal onto the memory array.

The circuit in Figure 14 shows how the Memory Ready capability can be added to the cross-coupled monostable clock generator of Figure 9. The Memory Ready feature is

incorporated into this circuit by switching in or out of the $\phi 2$ pulse width generator an additional timing resistor. By selection of the timing resistors for $\phi 1$ and $\phi 2$, all combinations of $\phi 1$, $\phi 2$, and stretched $\phi 2$ pulse width can be generated.

FIGURE 11 – MPU CLOCK CIRCUITRY WITH INTERFACE FOR SLOW AND DYNAMIC MEMORY

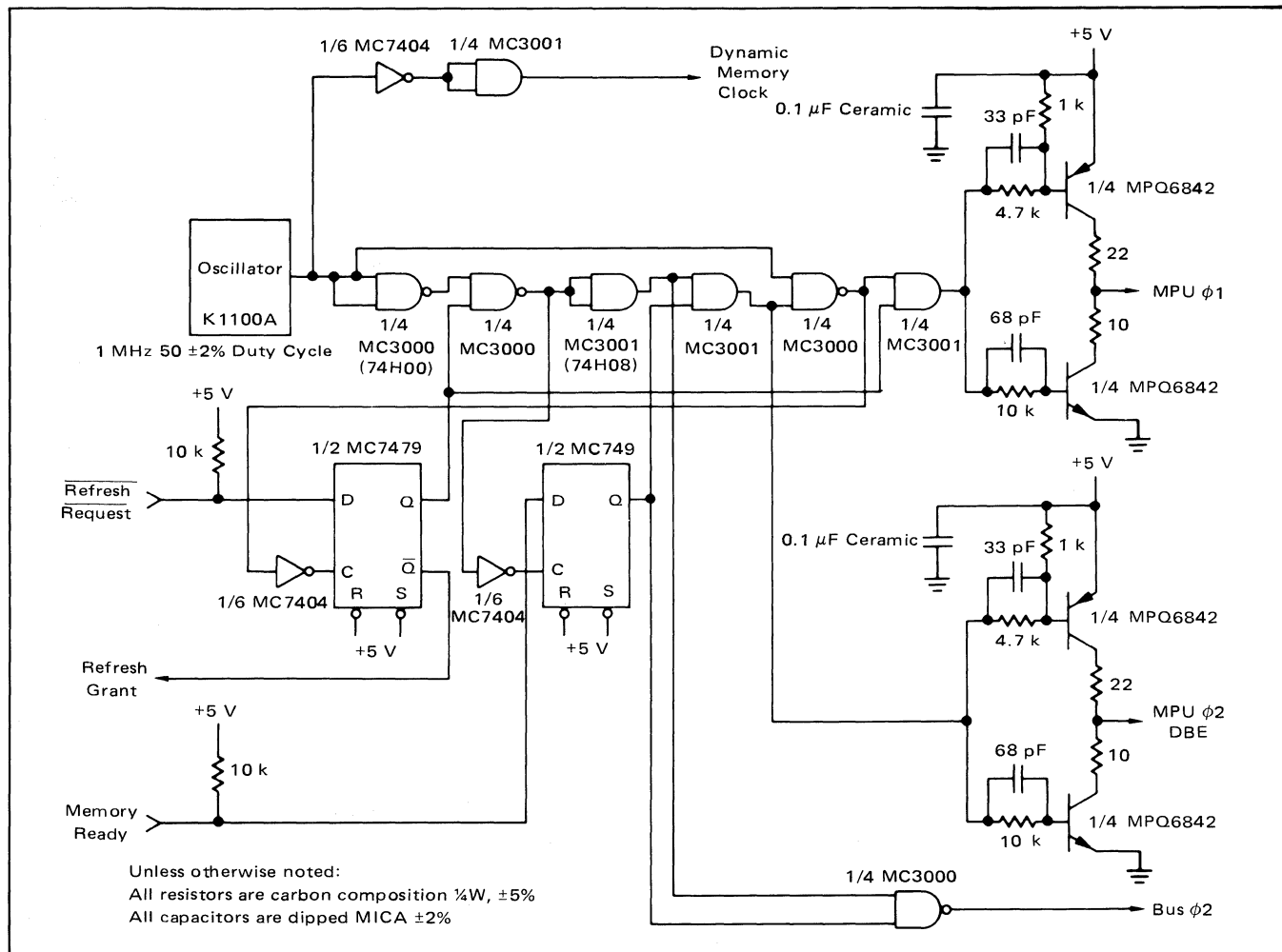


FIGURE 12a – MPU CLOCKS, REFRESH REQUEST, AND REFRESH GRANT

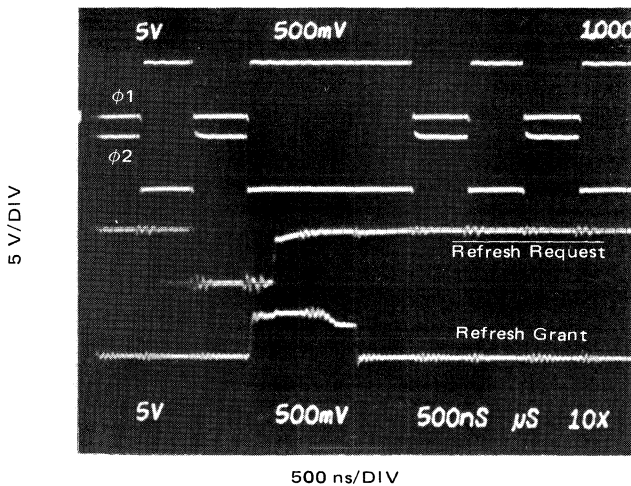


FIGURE 12b – MPU CLOCKS AND MEMORY READY

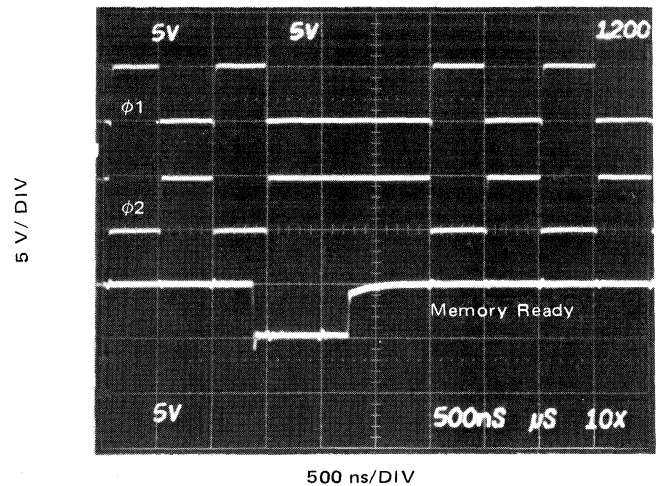


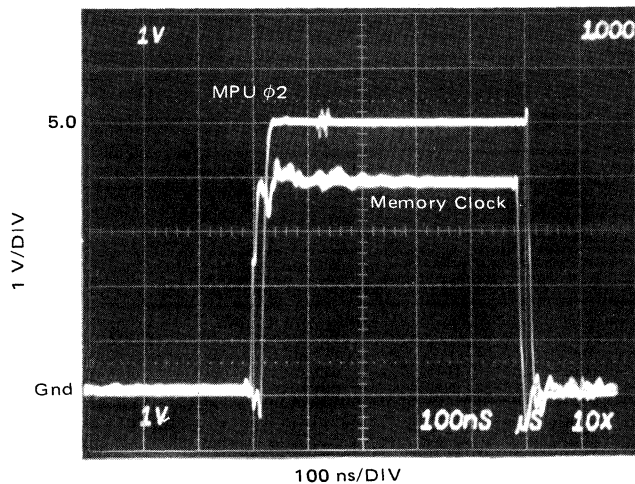
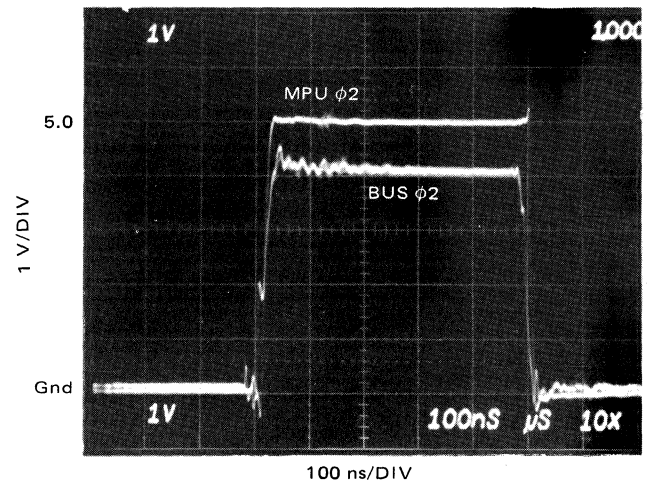
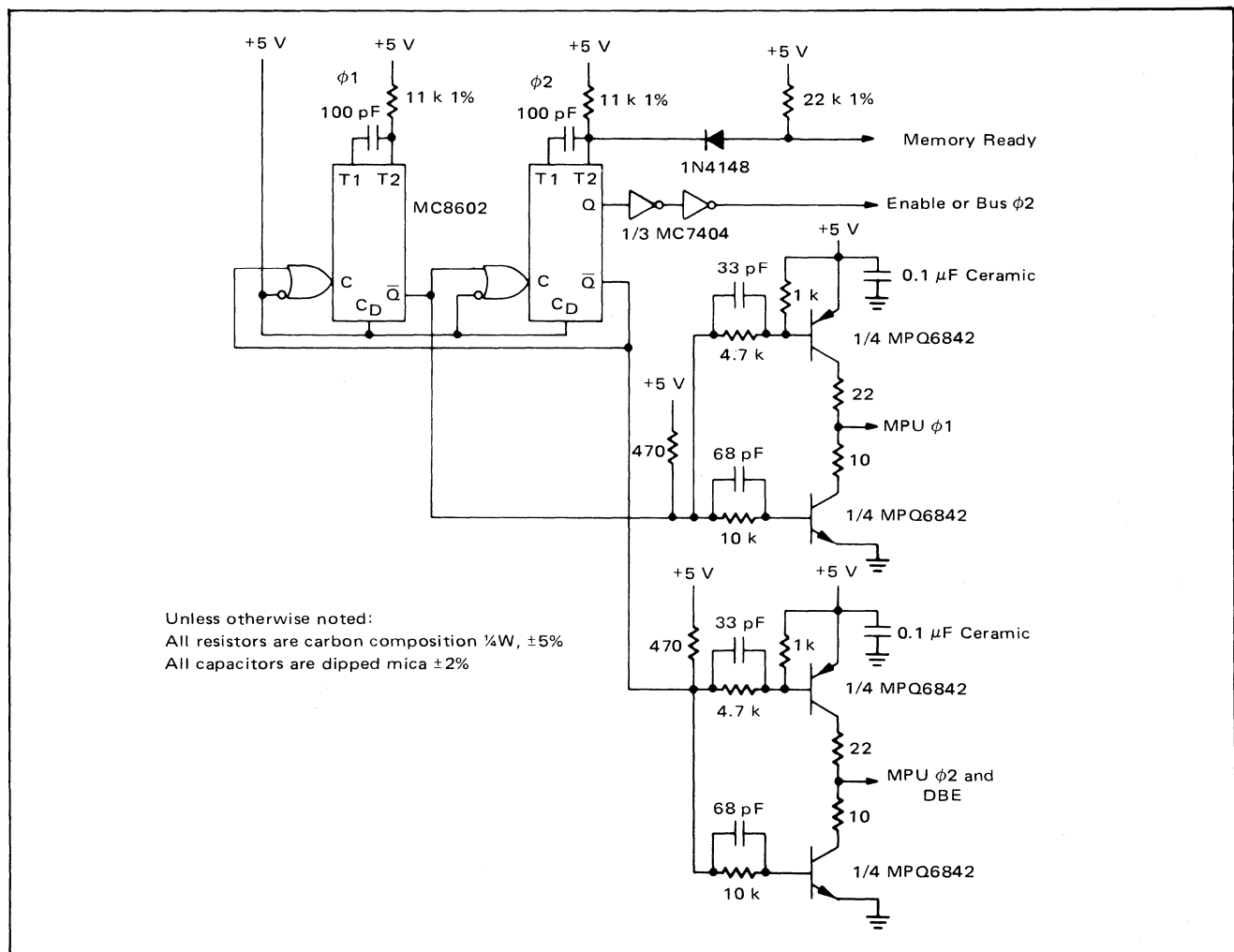
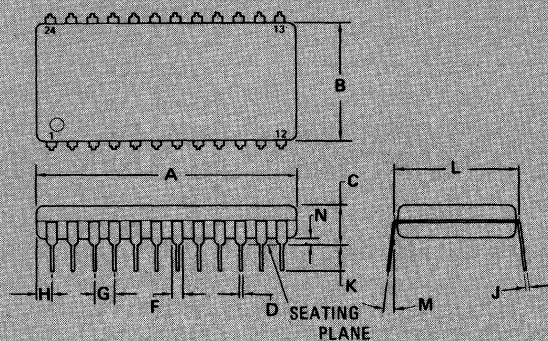
FIGURE 13a – DYNAMIC MEMORY
CLOCK AND MPU $\phi 2$ FIGURE 13b – BUS $\phi 2$ AND MPU $\phi 2$ 

FIGURE 14 – MONOSTABLE CLOCK GENERATOR WITH MEMORY READY



24 AND 40-LEAD PLASTIC PACKAGE DIMENSIONS

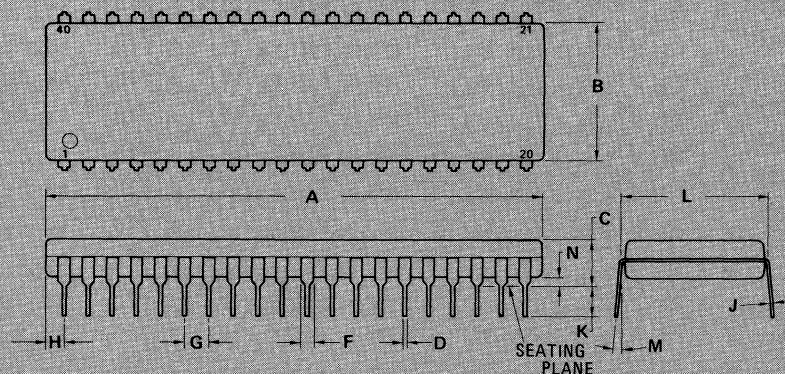
CASE 709-01



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	31.37	32.13	1.235	1.265
B	13.72	14.22	0.540	0.560
C	4.57	5.08	0.180	0.200
D	0.36	0.51	0.014	0.020
F	1.02	1.52	0.040	0.060
G	2.41	2.67	0.095	0.105
H	1.78	2.03	0.070	0.080
J	0.20	0.30	0.008	0.012
K	3.05	3.56	0.120	0.140
L	14.73	15.24	0.580	0.600
M	0°	10°	0°	10°
N	0.51	1.02	0.020	0.040

- NOTES:
1. LEADS, TRUE POSITIONED WITHIN 0.25 mm (0.010) DIA AT SEATING PLANE AT MAXIMUM MATERIAL CONDITION. (DIM. "D")
 2. DIM "L" TO CENTER OF LEADS WHEN FORMED PARALLEL.

CASE 711-01



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	51.82	52.32	2.040	2.060
B	13.72	14.22	0.540	0.560
C	4.57	5.08	0.180	0.200
D	0.36	0.51	0.014	0.020
F	1.02	1.52	0.040	0.060
G	2.41	2.67	0.095	0.105
H	1.65	2.16	0.065	0.085
J	0.20	0.30	0.008	0.012
K	3.68	4.19	0.145	0.165
L	14.99	15.49	0.590	0.610
M	0°	10°	0°	10°
N	0.51	1.02	0.020	0.040

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